

# Algebra, Geometry and Number Theory

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July 18, 2026

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# Chapter 1

## Introduction

The main purposes of these notes is to provide a detailed expositions of Galois Theory, Algebraic Number Theory, Algebraic Varieties over non-algebraically closed fields and Schemes, with particular interest in the Weil Conjectures. As such the section on Algebra, whilst broad, doesn't have huge depth, and often straightforward results are stated without proof. I have also tried to be rather explicit in dependence on earlier results, so much use is made of linked references. The section on Algebra largely follows Lang but with some I hope minor improvements in the exposition (e.g. Separability).

For the section on Algebraic Geometry I've tried to simultaneously develop the somewhat "elementary" approach (e.g. Hartshorne I, Kempf, JMilne) alongside the more technically challenging schemes approach (Stacks, Hartshorne II-III, Liu, EGA I) in order to motivate the constructions. I've also tried to adapt the elementary approach to work over non-algebraically closed fields so that it lends itself to talking about the Weil Conjectures at an early stage.

Finally I've included a very small amount of category theory, as it of course a useful language to talk about "universal properties" and helps frame some of the more technical results around schemes.

Some references I found useful

### Set Theory, Lattices

- Naive Set Theory - Halmos [[Hal17](#)]
- Lattice Theory - Birkhoff [[Bir40](#)]

### Algebra

- Algebra - Serge Lang [[Lan11](#)]
- Field Theory - Roman [[Rom05](#)]
- Introduction to Commutative Algebra - Atiyah, MacDonald [[AM69](#)]
- Local Rings - Nagata [[Nag75](#)]
- Commutative Algebra II - Zariski-Samuel [[ZS76](#)]

### Algebraic Geometry

- Algebraic Geometry - Robin Hartshorne [[Har13](#)]
- Algebraic Geometry - J.S. Milne [[Mil17](#)]
- Basic Algebraic Geometry - Shafarevich [[Sha94](#)]
- Introduction to Algebraic Geometry - Serge Lang [[Lan19](#)]
- Elements of Algebraic Geometry (EGA) - Alexander Grothendieck

### Analysis and Geometry

- General Topology - Kelley [[Kel71](#)]
- Foundations of Modern Analysis - Dieudonne [[Die11](#)]
- Differential Calculus - Henri Cartan [[Car71](#)]
- Real and Functional Analysis - Serge Lang [[Lan12](#)]



## Chapter 2

# Foundations

## 2.1 Set Theory

### 2.1.1 Relations

**Definition 2.1.1** (Binary Relation)

A **binary relation** (or just **relation**)  $R$  on a pair of sets  $(X, Y)$  is subset of the cartesian product  $X \times Y$ .

We write  $xRy$  to mean precisely  $(x, y) \in R$ .

**Definition 2.1.2** (Converse Relation)

Let  $R$  be a binary relation on  $(X, Y)$  then we the converse relation  $R^T$  on  $(Y, X)$  given by

$$yR^Tx \iff xRy$$

**Definition 2.1.3** (Domain and Range)

Let  $R$  be a relation on  $(X, Y)$ . We define the **domain** of  $R$  to be

$$\text{dom}(R) := \{x \in X \mid \exists y \in Y \text{ s.t. } xRy\}$$

and the **range** of  $R$

$$\text{range}(R) := \{y \in Y \mid \exists x \in X \text{ s.t. } xRy\}$$

**Definition 2.1.4** (Equivalence Relation)

Let  $R$  be a binary relation on  $(X, X)$ . It is said to be

- a) **reflexive** if  $xRx$  for all  $x \in X$
- b) **symmetric** if  $xRy \implies yRx$  for all  $x, y \in X$
- c) **transitive** if  $xRy \wedge yRz \implies xRz$  for all  $x, y, z \in X$

A relation which satisfies all these properties is called an **equivalence relation** on  $X$ . In this case we would write

$$x \sim y$$

instead of  $xRy$ . For an element  $x \in X$  denote the **equivalence class** of  $x$  by

$$[x]_R = \{y \mid xRy\}$$

Note that  $R^T = R$ .

**Definition 2.1.5** (Partition)

Let  $X$  be a set and  $\mathcal{F}$  a family of subsets of  $X$ . It is said to be a **partition** if

- a)  $X = \bigcup_{A \in \mathcal{F}} A$
- b)  $A, B \in \mathcal{F} \implies A = B \text{ or } A \cap B = \emptyset$

**Proposition 2.1.6** (Equivalence Classes form a Partition)

Let  $E$  be an equivalence relation on  $X$ . The family

$$\mathcal{F} = \{[x]_E \mid x \in X\}$$

forms a partition of  $X$ . Denote by  $X/E$  the family of equivalence classes, called the **quotient** of  $X$  with respect to  $E$ .

*Proof.* It's clear that

$$X = \bigcup_{A \in \mathcal{F}} A$$

because by reflexive-ness  $x \in [x]_E$  for all  $x \in X$ .

We claim that for any  $z \in [x]_E$  we have  $[z]_E = [x]_E$ . Suppose  $y \in [x]_E$  then  $xRz$  and  $xRy$ . By symmetry and transitivity we then have  $zRy$  which implies  $y \in [z]_E$ . In other words  $[x]_E \subseteq [z]_E$ . By symmetry of  $R$  we have  $x \in [z]_E$ , so by the same token  $[z]_E \subseteq [x]_E$ , which shows they are equal.

Therefore it's clear that  $[x]_E \cap [y]_E \neq \emptyset \implies [x]_E = [y]_E$  and thus  $\mathcal{F}$  forms a partition.  $\square$

**Definition 2.1.7** (Composition of Relation)

Suppose  $R$  is a relation on  $(X, Y)$  and  $S$  a relation on  $(Y, Z)$ . We define the composition  $S \circ R$  on  $(X, Z)$

$$S \circ R = \{(x, z) \mid \exists y \in Y \text{ s.t. } xRy \text{ and } yRz\}$$

## 2.1.2 Functions

**Definition 2.1.8** (Function)

A **function**  $f : X \rightarrow Y$  consists of a binary relation  $\Gamma(f)$  on  $(X, Y)$  such that

- $\text{dom}(f) = X$
- $\Gamma(f)$  is **single-valued** that is  $x\Gamma(f)y \wedge x\Gamma(f)y' \implies y = y'$

Equivalently for all  $x \in X$  there exists precisely one  $y \in Y$  such that  $x\Gamma(f)y$ .

We write  $f(x) = y$  for the unique element  $y \in Y$  such that  $x\Gamma(f)y$ .

**Proposition 2.1.9** (Equality of Functions)

Two functions  $f, g : X \rightarrow Y$  are equal if and only if  $f(x) = g(x)$  for all  $x \in X$ .

**Proposition 2.1.10** (Composition of Functions)

Let  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$  be functions then the **composition**  $\Gamma(g) \circ \Gamma(f)$  is still a function, which we write  $g \circ f$ , and

$$(g \circ f)(x) = g(f(x))$$

Furthermore composition is **associative** in the sense that

$$(h \circ g) \circ f = h \circ (g \circ f)$$

**Definition 2.1.11** (Injective, Surjective and Bijective)

Let  $f : X \rightarrow Y$  be a function then we say

- $f$  is **injective** if  $f(x) = f(x') \implies x = x'$
- $f$  is **surjective** if for all  $y \in Y$  there exists  $x$  such that  $f(x) = y$
- $f$  is **bijective** if it is both injective and surjective

**Definition 2.1.12** (Inverse Function)

Let  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$  be functions. We say

- $g$  is a **left inverse** for  $f$  if  $g \circ f = 1_X$
- $g$  is a **right inverse** for  $f$  if  $f \circ g = 1_Y$
- $g$  is a **two-sided inverse** for  $f$  if it is both a left and right inverse

**Proposition 2.1.13**

Let  $f : X \rightarrow Y$  be a function then

- $f$  is **injective** if and only if it has a **left inverse**
- $f$  is **surjective** if and only if it has a **right inverse**
- $f$  is **bijective** if and only if it has a **two-sided inverse**

**Definition 2.1.14** (Idempotent Function)

A function  $p : X \rightarrow X$  is **idempotent**  $p \circ p = p$ .

**Lemma 2.1.15** (Idempotent Criterion)

Let  $p : X \rightarrow X$  be a function. Then  $\text{Fix}(p) \subseteq \text{Im}(p)$  and these are equal if and only if  $p$  is idempotent.

### 2.1.3 Partial Orders

#### Definition 2.1.16 (Poset)

A binary relation  $\leq$  on  $(X, X)$  is a **partial order** if

- **reflexivity**  $x \leq x$
- **antisymmetry**  $x \leq y$  and  $y \leq x \implies y = x$
- **transitivity**  $x \leq y$  and  $y \leq z \implies x \leq z$

We may refer to  $(X, \leq)$  as a **partially ordered set** or **poset**.

#### Definition 2.1.17 (Dual Poset)

Given a poset  $(X, \leq)$  denote the set  $X$  with the converse relation by  $(X, \leq^d)$ . This is the **dual poset** to  $(X, \leq)$ .

#### Example 2.1.18

Let  $\mathcal{F}$  be a family of subsets of a fixed set  $E$ . Then  $(\mathcal{F}, \subseteq)$  is a poset ordered under inclusion.

#### Definition 2.1.19 (Top and Bottom)

Let  $(X, \leq)$  we say  $\top$  (resp.  $\perp$ ) is a **top element** (resp. **bottom element**) if it is greater than (resp. less than) every element of  $x$ . In this case it is unique.

#### Definition 2.1.20 (Monotone/Antitone Function)

Let  $(X, \leq)$  and  $(Y, \leq)$  be posets. A function  $f : X \rightarrow Y$  is

- **monotone / order-preserving** if  $x \leq y \implies f(x) \leq f(y)$
- **antitone / order-reversing** if  $x \leq y \implies f(y) \leq f(x)$
- a **monotone embedding** if  $x \leq y \iff f(x) \leq f(y)$
- an **order isomorphism** if it is bijective and monotone
- a **dual isomorphism** if it is bijective and antitone

#### Proposition 2.1.21

Let  $f : X \rightarrow Y$  be a monotone function. Then it is an embedding if and only if it is injective.

In what follows the notion of closure and kernel operator will be important.

#### Definition 2.1.22 (Closure operator)

Let  $(X, \leq)$  be a partially ordered set. A function  $c : X \rightarrow X$  is a **closure operator** if it is

- a) **extensive**  $x \leq c(x)$
- b) **monotone**  $x \leq y \implies c(x) \leq c(y)$
- c) **idempotent**  $c(c(x)) = c(x)$

#### Definition 2.1.23 (Kernel operator)

Let  $(X, \leq)$  be a partially ordered set. A function  $\kappa : X \rightarrow X$  is a **kernel operator** if it is

- **co-extensive**  $\kappa(x) \leq x$
- **monotone**  $x \leq y \implies \kappa(x) \leq \kappa(y)$
- **idempotent**  $\kappa(\kappa(x)) = \kappa(x)$

Note these definitions are “dual” with respect to the ordering on  $X$ .

### 2.1.4 Lattices

Certain families of subsets of algebraic structures (e.g. ideals, subgroups, normal subgroups, submodules) form a “sublattice” of the power set. Certain operations on, and results about, these subsets share common features regardless of the type of algebraic structure. Therefore we detail some elements of “Lattice Theory” (see Birkhoff) which may clarify the exposition.

#### Definition 2.1.24 (Upper and Lower Bounds)

Let  $(X, \leq)$  be a poset and  $S \subseteq X$ . Define the set of **upper bounds** for  $S$  by

$$S^\uparrow = \{x \in X \mid s \leq x \quad \forall s \in S\}$$

and the set of **lower bounds** for  $S$  by

$$S^\downarrow = \{x \in X \mid x \leq s \quad \forall s \in S\}$$

Note by convention  $\emptyset^\uparrow = \emptyset^\downarrow = X$ . Furthermore

$$X^\uparrow = \begin{cases} \{\top\} & X \text{ has a top element} \\ \emptyset & \text{otherwise} \end{cases}$$

and

$$X^\downarrow = \begin{cases} \{\perp\} & X \text{ has a bottom element} \\ \emptyset & \text{otherwise} \end{cases}$$

**Lemma 2.1.25** (Upper/Lower bounds are antitone maps)

Let  $(X, \leq)$  be a poset and  $S, T$  subsets of  $X$  then

- **antitone**  $S \subseteq T \implies T^\uparrow \subseteq S^\uparrow$  and  $T^\downarrow \subseteq S^\downarrow$
- **unit-counit relations**  $S \subseteq S^{\uparrow\downarrow}$  and  $T \subseteq T^{\downarrow\uparrow}$
- **triangular identities**  $S^\uparrow = S^{\uparrow\downarrow\uparrow}$  and  $T^\downarrow = T^{\downarrow\uparrow\downarrow}$

*Proof.* We prove only the first triangular identity as the others are straightforward consequences of the definitions. Firstly  $S \subseteq S^{\uparrow\downarrow} \implies S^{\uparrow\downarrow\uparrow} \subseteq S^\uparrow$  by the antitone property. Given the relation  $T \subseteq T^{\downarrow\uparrow}$  substitute  $T = S^\uparrow$  to get the reverse inclusion.  $\square$

**Lemma 2.1.26**

Let  $(X, \leq)$  be a poset and  $S, T$  subsets of  $X$ . Then the intersections  $S \cap S^\uparrow$  and  $T \cap T^\downarrow$  contain at most one element. **When they exist** write the elements as  $\top_S$  and  $\perp_T$  respectively, and are referred to as the **maximum** and **minimum** elements respectively.

*Proof.* Given  $x, y \in S \cap S^\uparrow$  then by definition  $x \leq y$  and  $y \leq x$ . By anti-symmetry we have  $x = y$  as required.  $\square$

**Definition 2.1.27** (Supremum and Infimum)

Let  $(X, \leq)$  be a poset and  $S \subseteq X$  a subset. We say a **supremum** of  $S$  is the minimal upper bound, i.e. the unique element of

$$S^\uparrow \cap S^{\uparrow\downarrow}$$

when it exists and write this as  $\sup S$ . Similarly an **infimum** of  $S$  is the maximal lower bound, i.e. the unique element of

$$S^\downarrow \cap S^{\downarrow\uparrow}$$

when it exists and write this as  $\inf S$ .

**Lemma 2.1.28** (Maximum = Supremum)

Let  $(X, \leq)$  be a poset and  $S \subseteq X$  a subset. Then  $\top_S$  exists if and only if  $\sup S$  exists and is a member of  $S$ . In this case  $\top_S = \sup S$ .

**Lemma 2.1.29**

Let  $(X, \leq)$  be a poset. Then  $\{\sup S\}^\uparrow = S^\uparrow$  and  $\{\inf T\}^\downarrow = T^\downarrow$  when these exist.

**Lemma 2.1.30** (Sup is monotone and Inf is antitone)

Let  $(X, \leq)$  be a poset and  $S, T$  subsets of  $X$ . Then  $S \subseteq T \implies \sup S \leq \sup T$  and  $\inf T \leq \inf S$  when these exist.

*Proof.* Note  $S \subseteq T \implies T^\uparrow \subseteq S^\uparrow$  so  $\sup T \in S^\uparrow$ . By definition  $\sup S \in S^{\uparrow\downarrow}$  therefore  $\sup S \leq \sup T$ .

Similarly  $S \subseteq T \implies T^\downarrow \subseteq S^\downarrow$ . By definition  $\inf T \in T^\downarrow \implies \inf T \in S^\downarrow$ . By definition  $\inf S \in S^{\downarrow\uparrow}$  therefore  $\inf T \leq \inf S$ .  $\square$

**Remark 2.1.31**

Note that  $\emptyset^\uparrow = X$  and therefore  $\sup \emptyset = \perp$  when it exists. Similarly  $\inf \emptyset = \top$  when it exists.

When  $\top$  exists  $\sup X = \top$ , otherwise it is not defined. Similarly when  $\perp$  exists  $\inf X = \perp$ , otherwise it is not defined.

**Definition 2.1.32** (Lattice)

A poset  $(X, \leq)$  is a **lattice** if every pair of elements  $x, y$  admits both a supremum and infimum. In this case we write

$$a \vee b := \sup\{a, b\}$$

and

$$a \wedge b := \inf\{a, b\}$$

These are called the **join** and **meet** operations. A subset  $Y$  is called a **sub-lattice** if

$$a, b \in Y \implies a \wedge b \in Y \text{ and } a \vee b \in Y.$$

Similarly it is a **complete lattice** if every subset  $S$  admits both a supremum and infimum. This is written

$$\bigvee S := \sup S$$

and

$$\bigwedge S := \inf S$$

Note a complete lattice has both a top and a bottom element (by considering  $\sup \emptyset$  and  $\inf \emptyset$ ), and a lattice admits **finite joins** and **meets**.

Trivially

$$\bigwedge \{x\} = \bigvee \{x\} = x$$

**Example 2.1.33** (Power Set)

For a fixed set  $E$  the collection of subsets  $\mathcal{P}(E)$  is a complete lattice under the union and intersection operator **with the convention that empty intersection is the whole set and empty union is the empty set**

In this case  $\top = E$  and  $\perp = \emptyset$ .

**Proposition 2.1.34** (Principal down-sets are lattices)

Let  $(X, \leq)$  be a lattice and  $x, y \in X$ . Then the subsets  $\{x\}^\uparrow$ ,  $\{x\}^\downarrow$  and  $\{x\}^\uparrow \cap \{y\}^\downarrow$  are sub-lattices.

Verifying a poset is a lattice is slightly easier than it may first appear.

**Lemma 2.1.35** (Supremum is Infimum of upper bounds)

Let  $(X, \leq)$  be a poset and  $S$  a subset of  $X$ . Then

$$\sup S = \inf S^\uparrow$$

when either exists. Dually

$$\inf S = \sup S^\downarrow$$

*Proof.* By definition  $\sup S$  is the unique element of  $S^\uparrow \cap S^{\uparrow\downarrow}$  and  $\inf S^\uparrow$  is the unique element of  $S^{\uparrow\downarrow} \cap S^{\uparrow\downarrow\uparrow}$ . By (2.1.25)  $S^{\uparrow\downarrow\uparrow} = S^\uparrow$  so they are equivalent.  $\square$

**Proposition 2.1.36** (Criteria to be a Complete Lattice)

Let  $(X, \leq)$  be a poset. Then the following are equivalent

- a)  $X$  is a complete lattice
- b)  $X$  admits arbitrary infimums (and in particular has  $\top = \inf \emptyset$ )
- c)  $X$  admits arbitrary supremums (and in particular has  $\perp = \sup \emptyset$ )

In this case we have the relationships

$$\begin{aligned}\bigvee S &= \bigwedge S^\uparrow \\ \bigwedge S &= \bigvee S^\downarrow\end{aligned}$$

*Proof.* 1  $\implies$  2,3 is clear.

2,3  $\implies$  1 follows from the previous Lemma.  $\square$

**Lemma 2.1.37**

Let  $(X, \leq)$  be a poset and  $(Y, \leq)$  a sub-poset. Let  $S \subseteq Y$  be a subset. Then  $\inf_Y S$  exists if and only if  $\inf_X S$  exists and belongs to  $Y$ . In this case they are equal.

*Proof.* Note in general that  $T^{\downarrow, Y} = T^{\downarrow, X} \cap Y$  and  $T^{\uparrow, Y} = T^{\uparrow, X} \cap Y$ . Therefore

$$S^{\downarrow, Y} \cap S^{\uparrow, Y} = S^{\downarrow, X} \cap S^{\uparrow, X} \cap Y$$

Recall  $\inf S$  is the unique element of  $S^\downarrow \cap S^\uparrow$  if it exists. Then the result follows easily.  $\square$

**Definition 2.1.38** (Moore Family)

Let  $(X, \leq)$  be a complete lattice. A sub-poset  $(Y, \leq)$  is a **Moore family** over  $X$  if it satisfies the following property

$$S \subseteq Y \implies \bigwedge_X S \in Y$$

In particular this includes the case  $S = \emptyset$  and so  $\top \in Y$ .

**Example 2.1.39** (Moore family of sets)

Given a fixed set  $E$ , then  $\mathcal{P}(E)$  is a complete lattice ordered under inclusion. Then a family of subsets  $\mathcal{F}$  is a Moore family precisely when

- $E \in \mathcal{F}$
- $A_{i \in I} \in \mathcal{F} \implies \bigcap_{i \in I} A_i \in \mathcal{F}$

**Proposition 2.1.40** (Equivalent Formulations of Complete Sub-lattice)

Let  $(X, \leq)$  be a complete lattice and  $(Y, \leq)$  a sub-poset. Then the following are equivalent

- a)  $(Y, \leq)$  is a Moore family
- b)  $(Y, \leq)$  is a complete lattice
- c)  $Y$  is the image of some closure operator  $c : X \rightarrow X$

In this case the closure operator is given by

$$c(x) = \bigwedge_X \{y \in Y \mid x \leq y\}$$

For  $S \subseteq Y$

$$\begin{aligned}\bigwedge_Y S &= \bigwedge_X S \\ \bigvee_Y S &= c\left(\bigvee_X S\right)\end{aligned}$$

and for  $S \subseteq X$  we have

$$c\left(\bigvee_X S\right) = \bigwedge_X (S^\uparrow \cap Y)$$

*Proof.*  $a) \implies b)$  By (2.1.37)  $S \subseteq Y \implies \bigwedge_Y S = \bigwedge_X S$ . By (2.1.36) then  $Y$  is a complete lattice.

$b) \implies c)$  Suppose that  $(Y, \leq)$  is a complete lattice then define the function  $c : X \rightarrow X$  by  $c(x) = \bigwedge_X \Gamma_x$  where  $\Gamma_x = \{y \in Y \mid x \leq y\}$ . We need to show that it is a closure operator. Evidently  $x \in \Gamma_x^\perp$  and  $c(x) \in \Gamma_x^{\perp\uparrow}$  by definition of infimum. Therefore  $x \leq c(x)$  and  $c$  is extensive. Note  $x \leq y \implies \Gamma_y \subseteq \Gamma_x$ . By (2.1.30) we have  $\inf \Gamma_x \leq \inf \Gamma_y$ , whence  $c(x) \leq c(y)$  and  $c$  is monotone.

$Y$  is a complete lattice, so by (2.1.37) we have  $c(x) \in Y$  so that  $\text{Im}(c) \subseteq Y$ . We claim that  $x \in Y \implies c(x) = x$ . In this case  $x \in \Gamma_x$  and  $c(x) \in \Gamma_x^\perp$  whence  $c(x) \leq x$  and therefore  $x = c(x)$  as required. Therefore  $Y \subseteq \text{Fix}(c) \subseteq \text{Im}(c) \subseteq Y$ , whence  $Y = \text{Im}(c) = \text{Fix}(c)$  and  $c$  is idempotent by (2.1.15). As  $c$  is extensive, monotone and idempotent it is by definition a closure operator.

$c) \implies a)$  In order for  $Y := \text{Im}(c)$  to be a Moore family, we need to show  $S \subseteq Y \implies \bigwedge_X S \in Y$ . We claim that by properties of  $c$  we have

$$S \subseteq Y \implies c(S^\perp) \subseteq S^\perp$$

$$T \subseteq X \implies c(T^\uparrow) \subseteq T^\uparrow$$

Therefore  $c$  maps the singleton set  $S^\perp \cap S^{\perp\uparrow} = \{\bigwedge_X S\}$  to itself. In otherwords  $\bigwedge_X S \in \text{Fix}(c) = \text{Im}(c) = Y$  as required.

Define  $\Gamma_x := \{y \in Y \mid x \leq y\}$ . We wish to show that  $c(x) = \bigwedge_X \Gamma_x$ . As  $x \leq c(x)$  we have  $c(x) \in \Gamma_x$ . Furthermore  $y \in \Gamma_x \implies c(x) \leq c(y) = y$ . So  $c(x) \in \Gamma_x^\perp$ . Therefore  $c(x) = \perp_{\Gamma_x} = \bigwedge_X \Gamma_x$  as required.

Finally by (2.1.29)  $\{\bigvee_X S\}^\uparrow = S^\uparrow$  for any  $S \subseteq X$ . Therefore, as  $c(x) = \bigwedge_X \Gamma_x$  we find

$$c\left(\bigvee_X S\right) = \bigwedge_X \left( \left\{ \bigvee_X S \right\}^\uparrow \cap Y \right) = \bigwedge_X (S^\uparrow \cap Y)$$

as required. In particular when  $S \subseteq Y$  we find by (2.1.40)

$$\bigvee_Y S = \bigwedge_Y S^{\uparrow,Y} = \bigwedge_X S^{\uparrow,Y} = \bigwedge_X (S^\uparrow \cap Y) = c\left(\bigvee_X S\right) \quad (2.1)$$

□

#### Remark 2.1.41

For a given complete lattice  $(X, \leq)$  we have established a correspondence between

$$\left\{ \text{closure operators } c : X \rightarrow X \right\} \longleftrightarrow \left\{ \text{complete sub-lattices } (Y, \leq) \right\}$$

#### Corollary 2.1.42 (Moore family admits a closure operator)

Let  $E$  be a fixed set and  $\mathcal{F}$  a *Moore family* over  $(\mathcal{P}(E), \subseteq)$ . Then there exists a surjective closure operator  $c : \mathcal{P}(E) \rightarrow \mathcal{F}$  given by

$$c(F) = \bigcap_{F \subseteq E_\alpha \in \mathcal{F}} E_\alpha$$

Any such closure operator  $c : \mathcal{P}(E) \rightarrow \mathcal{P}(E)$  gives rise to a Moore family  $\mathcal{F} := \text{Im}(c)$ .

#### Proposition 2.1.43 (Alternative expression for join)

Let  $(X, \leq)$  be a complete lattice and  $c : X \rightarrow X$  a closure operator with image  $Y$ . Then for any subset  $S \subset X$

$$c\left(\bigvee_X S\right) = c\left(\bigvee_X c(S)\right) = \bigvee_Y c(S) = \bigwedge_X (S^\uparrow \cap Y)$$

i.e. it's the smallest "closed" set containing each element of  $S$ .

*Proof.* By (2.1.40) the expression for  $c(\bigvee_X S)$  yields

$$c(\bigvee_X c(S)) = \bigwedge_X (c(S)^\uparrow \cap Y) = \bigwedge_X (S^\uparrow \cap Y) = c(\bigvee_X S)$$

where the middle equality follows because if  $y \in Y$  then  $c(s) \leq y \iff s \leq y$ . Furthermore  $c(S) \subseteq Y$  so the expression for  $\bigvee_Y$  yields

$$c(\bigvee_X c(S)) = \bigvee_Y c(S).$$

as required. □

### 2.1.5 Distributive Lattice

#### Proposition 2.1.44

Let  $(X, \leq)$  be a *lattice* then the following relations hold

- a)  $x \wedge (y \vee z) \geq (x \wedge y) \vee (x \wedge z)$
- b)  $x \vee (y \wedge z) \leq (x \vee y) \wedge (x \vee z)$

*Proof.* a) By definition  $x \wedge y \leq x$  and  $x \wedge y \leq y \leq y \vee z$ . Therefore  $x \wedge y \leq x \wedge (y \vee z)$ . By symmetry in  $y$  and  $z$  we have  $x \wedge z \leq x \wedge (y \vee z)$ . Whence  $(x \wedge y) \vee (x \wedge z) \leq x \wedge (y \vee z)$  as required.

b) follows by duality □

#### Definition 2.1.45 (Distributive Lattice)

We say a *lattice*  $(X, \leq)$  is **distributive** if it satisfies the following relations for all  $x, y, z \in X$

- $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$
- $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$

#### Proposition 2.1.46

Let  $(X, \leq)$  be a *lattice* then TFAE

- a)  $(X, \leq)$  is a *distributive lattice*
- b)  $x \wedge (y \vee z) \leq (x \wedge y) \vee (x \wedge z)$
- c)  $x \vee (y \wedge z) \geq (x \vee y) \wedge (x \vee z)$

#### Example 2.1.47

Any family of subsets closed under finite intersection and union is a distributive lattice.

### 2.1.6 Galois Connections

#### Definition 2.1.48 (Galois Connection)

Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets. A pair of functions  $(f_\star, f^\star)$

$$X \xrightleftharpoons[f_\star]{f^\star} Y$$

is called an **antitone Galois connection** if it satisfies the **adjoint property**

- $x \leq_X f^\star(y) \iff y \leq_Y f_\star(x) \quad \forall x \in X, y \in Y$

We say it is a **monotone Galois connection** if instead

- $x \leq_X f^\star(y) \iff f_\star(x) \leq_Y y \quad \forall x \in X, y \in Y$

We will assume that if not otherwise specified the connection is antitone.

**Proposition 2.1.49** (Equivalent Condition for Galois Connection)

Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be posets. Consider a pair of functions

$$X \xrightleftharpoons[f_*]{f^*} Y$$

Then this constitutes an **antitone Galois Connection** if and only if

- $f_*$  and  $f^*$  are both antitone
- $x \leq_X f^*(f_*(x))$  and  $y \leq_Y f_*(f^*(y))$  for all  $x \in X, y \in Y$  (i.e.  $f^* \circ f_*$  and  $f_* \circ f^*$  are extensive)

Similarly it constitutes a **monotone Galois Connection** if and only if

- $f_*$  and  $f^*$  are both monotone
- $x \leq_X f^*(f_*(x))$  and  $f_*(f^*(y)) \leq_Y y$  for all  $x \in X, y \in Y$

*Proof.* We consider only the antitone case, as the monotone follows from duality (flip  $\leq_Y$ ).

Suppose that  $(f_*, f^*)$  satisfies the adjoint property

$$\begin{aligned} f_*(x) = f_*(x) &\implies f_*(x) \leq_Y f_*(x) \implies x \leq_X f^*(f_*(x)) \\ f^*(y) = f^*(y) &\implies f^*(y) \leq_Y f^*(y) \implies y \leq_Y f_*(f^*(y)) \end{aligned}$$

whence the extensive property follows. Furthermore

$$\begin{aligned} x \leq_X x' &\implies x \leq_X f^*(f_*(x')) \implies f_*(x') \leq_Y f_*(x) \\ y \leq_Y y' &\implies y \leq_Y f_*(f^*(y')) \implies f^*(y) \leq_X f^*(y') \end{aligned}$$

which shows that the functions  $f_*$  and  $f^*$  are antitone.

Conversely suppose they satisfy the given conditions. Then by the antitone and extensive properties in turn

$$x \leq_X f^*(y) \implies f_*(f^*(y)) \leq_Y f_*(x) \implies y \leq_Y f_*(x)$$

and

$$y \leq_Y f_*(x) \implies f^*(f_*(x)) \leq_X f^*(y) \implies x \leq_X f^*(y).$$

which is the adjoint property as required. □

**Definition 2.1.50** (Closed sets)

Let  $(f_*, f^*)$  be a Galois connection. Then define the **closed sets** to be

$$\begin{aligned} X^* &:= f^*(Y) \\ Y^* &:= f_*(X) \end{aligned}$$

**Proposition 2.1.51** (Isomorphism on closed sets)

Consider an antitone (resp. monotone) Galois connection  $X \xrightleftharpoons[f_*]{f^*} Y$ . Then it restricts to a **dual isomorphism** (resp. order isomorphism) on **closed sets**

$$X^* \xrightleftharpoons[f_*]{f^*} Y^*$$

Furthermore the following properties hold

- $f_* \circ f^* \circ f_* = f_*$
- $f^* \circ f_* \circ f^* = f^*$
- In the antitone case  $f^* \circ f_*$  and  $f_* \circ f^*$  are **closure operators**.
- In the monotone case  $f^* \circ f_*$  is a closure operator and  $f_* \circ f^*$  is a **kernel operator**.

- $X^\star = \text{Fix}(f^\star \circ f_\star) = \text{Im}(f^\star \circ f_\star)$
- $Y^\star = \text{Fix}(f_\star \circ f^\star) = \text{Im}(f_\star \circ f^\star)$

*Proof.* We detail the antitone case as the monotone case follows by duality. We first prove so-called triangular identities, for by the extensive property (2.1.49)

$$x \leq f^\star(f_\star(x)) \implies f_\star(f^\star(f_\star(x))) \leq f_\star(x)$$

and by the other extensive property

$$f_\star(x) \leq f_\star(f^\star(f_\star(x)))$$

whence they are equal. The other case is similar.

It's immediate that  $f^\star \circ f_\star$  and  $f_\star \circ f^\star$  are idempotent, and they are extensive by (2.1.49). And the composition of two antitone functions is monotone so  $f^\star \circ f_\star$  and  $f_\star \circ f^\star$  are closure operators.

Observe

$$\text{Im}(f^\star \circ f_\star) \subseteq \text{Im}(f^\star) \subseteq \text{Fix}(f^\star \circ f_\star)$$

where the first inclusion is trivial and the second inclusion follows from the second triangular identity. However both sides are equal by (2.1.15) and the expression for  $X^\star$  follows. The expression for  $Y^\star$  follows similarly.

This shows that the maps are mutual inverses as required.  $\square$

In certain circumstances we may consider a smaller subset of  $X$ , by applying a suitable closure operator which is compatible with the Galois correspondence :

**Proposition 2.1.52** (Subordinated Closure Operator)

Let  $X \xrightleftharpoons[f_\star]{f^\star} Y$  be a Galois connection and  $c : X \rightarrow X$  be a closure operator with image  $X_c$ . Then

$$c(x) \leq (f^\star \circ f_\star)(x) \quad \forall x \in X \iff \text{Im}(f^\star) \subseteq \text{Fix}(c) \iff X^\star \subseteq X_c$$

In this case

$$f_\star(c(x)) = f_\star(x)$$

*Proof.* Suppose  $c(x) \leq (f^\star \circ f_\star)(x)$ . Substitute  $x = f^\star(y)$  then, because  $c$  is extensive,

$$f^\star(y) \leq c(f^\star(y)) \leq (f^\star \circ f_\star \circ f^\star)(y) = f^\star(y).$$

Therefore  $c(f^\star(y)) = f^\star(y)$  and  $\text{Im}(f^\star) \subseteq \text{Fix}(c)$  as required. Conversely suppose this holds, then by the monotone property of  $c$  and extensive property of  $f^\star \circ f_\star$

$$x \leq (f^\star \circ f_\star)(x) \implies c(x) \leq c((f^\star \circ f_\star)(x)) = (f^\star \circ f_\star)(x).$$

as required. Finally by the extensive property of  $c$

$$x \leq c(x) \leq (f^\star \circ f_\star)(x)$$

and by the antitone/monotone property of  $f_\star$  and triangular identity

$$f_\star(x) \leq f_\star(c(x)) \leq f_\star(x)$$

whence  $f_\star(c(x)) = f_\star(x)$  as required.  $\square$

The meaning of the “adjoint” criterion can be explained by the following rather generic situation

**Example 2.1.53** (Canonical example of an antitone Galois connection)

Suppose there is a predicate

$$\psi : X \times Y \rightarrow \{0, 1\}$$

Define a connection

$$\mathcal{P}(X) \xrightleftharpoons[f_*]{f^*} \mathcal{P}(Y)$$

by

$$\begin{aligned} f_*(S) &= \{y \in Y \mid \psi(x, y) = 1 \quad \forall x \in S\} \\ f^*(T) &= \{x \in X \mid \psi(x, y) = 1 \quad \forall y \in T\} \end{aligned}$$

Then

$$S \subseteq f^*(T) \iff \psi(s, t) = 1 \quad \forall s \in S \quad t \in T \iff T \subseteq f_*(S)$$

**Proposition 2.1.54** (Joins under Galois Correspondence)

Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be complete lattices with an antitone Galois connection  $X \xrightleftharpoons[f_*]{f^*} Y$ . Then for  $S \subseteq X$

$$f_*(\bigvee S) = \bigwedge f_*(S)$$

Similarly for  $T \subseteq Y$  we have

$$f^*(\bigvee T) = \bigwedge f^*(T)$$

*Proof.* Let  $a = \bigvee S$  and  $b = \bigwedge f_*(S)$ . Then  $s \leq a \implies f_*(a) \leq f_*(s)$  for all  $s \in S$ , which implies  $f_*(a) \leq b$ . Similarly  $b \leq f_*(s) \implies s \leq f^*(b)$  by the adjoint criterion. Therefore  $a \leq f^*(b)$  by definition of join, which implies  $b \leq f_*(a)$  by the adjoint criterion again. Whence  $f_*(a) = b$  as required.

The second statement follows from duality. □

### 2.1.7 Axiom of Choice

**Theorem 2.1.55** (Axiom of choice)

There are a number of essentially equivalent formulations of the axiom of choice

- a) The Cartesian product of a non-empty family of sets is non-empty
- b) For any set  $X$  of non-empty sets there exists a function  $f : X \rightarrow \bigcup X$  such that  $A \in X \implies f(A) \in A$ .
- c) **Zorn's Lemma** Suppose a partially ordered set  $(X, \leq)$  is such that every *chain* in  $X$  has an upper bound in  $X$ . Then  $X$  contains at least one maximal element.
- d) Every surjective function has a right inverse.

**Corollary 2.1.56** (Choose representatives)

Let  $\pi : X \rightarrow Y$  be a surjective function and  $T \subseteq Y$  a subset. Then there exists a subset  $S \subseteq X$  such that  $\pi|_S$  is bijective.

When  $T$  is finite  $\#S = \#T$ .

**Definition 2.1.57** (Finite Character)

A family of sets  $\mathcal{F}$  has **finite character** if it satisfies the following property

$$A \in \mathcal{F} \iff (B \subseteq A \text{ and finite} \implies B \in \mathcal{F})$$

**Corollary 2.1.58** (Tukey's Lemma)

Let  $\mathcal{F}$  be a family of sets of finite character. Then it is chain-complete when ordered by inclusion.

In particular every set is contained in a maximal set.

### 2.1.8 Chain Conditions

**Definition 2.1.59** (Totally ordered / chains)

A poset  $(\mathcal{F}, \leq)$  is **totally ordered** if  $x \leq y$  or  $y \leq x$  for all  $x, y \in \mathcal{F}$ .

**Definition 2.1.60** (Chain)

A non-empty subset  $C$  of  $\mathcal{F}$  is a **chain** if it is totally ordered under  $\leq$ .

The length of the chain is simply  $\ell(C) := |C| - 1$ .

A chain  $C$  is

- **saturated** if  $x \leq z \leq y$  and  $x, y \in C \implies z \in C$ .
- **maximal** if it's not contained properly in another chain.

**Definition 2.1.61** (Chain-Complete)

A poset  $(\mathcal{F}, \leq)$  is **chain complete** if every chain  $C$  has a supremum in  $\mathcal{F}$ . It is **co-chain complete** if every chain  $C$  has an infimum in  $\mathcal{F}$ .

**Proposition 2.1.62** (Noetherian / Artinian Poset)

Let  $(X, \leq)$  be a poset then the following conditions are equivalent

- a) Any ascending chain

$$x_1 \leq x_2 \leq \dots \leq x_n \leq \dots$$

eventually stabilizes

- b) Any non-empty subset  $Y \subseteq X$  has a maximal element

Such a poset is called **Noetherian**. If it satisfies the dual condition then it is called **Artinian**.

*Proof.* a)  $\implies$  b) If  $Y$  has no maximal elements then we may (by axiom of dependent choice) construct a strictly increasing sequence, which by definition does not stabilize.

b)  $\implies$  a) Clear. □

## 2.2 Numbers

Informally we consider the set of integers

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

and the subset of natural numbers

$$\mathbb{N} = \{0, 1, 2, \dots\}$$

Although it's possible to construct the integers painstakingly from a small set of axioms (see ...) we instead for brevity simply state the most commonly used results as axioms.

### 2.2.1 Integers

We suppose the existence of a set  $\mathbb{Z}$  with distinguished elements  $0 \neq 1$  together with

- A binary operation

$$+ : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$$

and an involution

$$(-) : \mathbb{Z} \rightarrow \mathbb{Z}$$

satisfying

- $-0 = 0$
- $-(-x) = x$
- $x = -x \iff x = 0$
- $x + 0 = 0 = 0 + x$
- $x + y = y + x$
- $(x + y) + z = x + (y + z)$
- $x + (-x) = 0 = (-x) + x$
- $-(x + y) = (-x) + (-y)$

- A subset  $\mathbb{N}$  such that

- $0, 1 \in \mathbb{N}$
- $x, y \in \mathbb{N} \implies x + y \in \mathbb{N}$
- $x \in \mathbb{Z} \implies (x \in \mathbb{N}) \vee (-x \in \mathbb{N})$
- $x \in \mathbb{N} \wedge -x \in \mathbb{N} \implies x = 0$

which also satisfies the principle of induction

- Let  $S \subseteq \mathbb{N}$  be a set such that

- $0 \in S$
- $x \in S \implies x + 1 \in S$

then  $S = \mathbb{N}$

It's possible to use these to show the existence of multiplication

**Proposition 2.2.1** (Multiplication exists)

*There exists a binary operation*

$$\cdot : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$$

*such that*

- $x \cdot 0 = 0 = 0 \cdot x$
- $x \cdot 1 = x = 1 \cdot x$
- $xy = yx$
- $(xy)z = x(yz)$

- $x(y + z) = xy + xz$
- $(y + z)x = yx + zx$
- $(-x)(y) = -(xy) = x(-y)$

We may also show the existence a partial ordering

**Proposition 2.2.2** (Order exists)

*There exists a relation  $\leq$  on  $\mathbb{Z}$  given by*

$$x \leq y \iff y - x \in \mathbb{N}$$

*which satisfies*

$$x \leq y \vee y \leq x$$

$$x \leq y \wedge y \leq x \implies x = y$$

*Define  $x < y$  in the obvious way then it satisfies the usual trichotomy law, namely precisely one of the following holds*

$$x < y, x = y, y < x$$

*and further*

- $z > 0$  then  $x < y \iff xz < yz$
- $z < 0$  then  $x < y \iff yz < xz$
- $y > 1$  and  $x > 0$  then  $x < xy$

Finally we can construct an absolute value function

**Proposition 2.2.3**

*There exists an **absolute value** function*

$$|\cdot| : \mathbb{Z} \rightarrow \mathbb{N}$$

*such that*

$$|x| = \begin{cases} x & 0 < x \\ 0 & x = 0 \\ -x & x < 0 \end{cases}$$

*It satisfies*

- $|x| = 0 \iff x = 0$
- $|x| = |y| \iff x = \pm y$
- $|xy| = |x| |y|$
- $|x + y| \leq |x| + |y|$

In many cases it may be more convenient to use the following form of induction

**Proposition 2.2.4** (Well-Ordering Principle)

*Let  $S \subset \mathbb{N}$  be a non-empty subset. Then it contains a minimal element  $d$ .*

## 2.2.2 Arithmetic

**Proposition 2.2.5** (Division Algorithm)

*Let  $x, y \in \mathbb{Z}$  be non-zero integers then there exists  $q, r$  such that*

$$x = yq + r$$

*and*

$$0 \leq r < |y|$$

Furthermore  $(q, r)$  is the unique such pair.

*Proof.* Suppose first that  $x, y > 0$ . Let  $S = \{x - yn \mid n \in \mathbb{Z}\} \cap \mathbb{N}$ . Then  $x \in S$  so it is non-empty. By the Well-Ordering principle it has a minimal element  $r$ . By assumption

$$x = yq + r$$

for some  $q \in \mathbb{Z}$  and  $r \geq 0$ . Suppose  $r \geq y$ , then  $0 \leq x - y(q + 1) < r$  contradicting minimality.

The case  $x > 0, y < 0$  is then straightforward, as is the case  $x < 0$ .

For uniqueness suppose  $yq' + r' = yq + r$  then  $|y||q - q'| = |r' - r| < |y|$  from which it follows  $|q - q'| = 0 \implies q = q' \implies r = r'$ .  $\square$

**Corollary 2.2.6** (Ideals are Principal)

Let  $S \subseteq \mathbb{Z}$  be a non-empty set such that

$$x, y \in S \implies x \pm y \in S$$

Then  $S = d\mathbb{Z}$  for a unique  $d \geq 0$ .

*Proof.* First we claim that  $0 \in S$ . For if  $x \in S$  then  $0 = x - x \in S$  by assumption. Furthermore  $x \in S \implies -x = 0 - x \in S$ .

Consider the set  $S' = (S \cap \mathbb{N}) \setminus \{0\}$ . If it's empty then  $S = \{0\}$  (for  $x \in S \implies -x \in S$ ) and  $d = 0$ .

Otherwise it has a minimal element  $d > 0$  by the well-ordering principle. Then by induction  $d\mathbb{Z} \subseteq S$ . Conversely suppose  $y \in S$  then by the division algorithm  $y = qd + r$  with  $0 \leq r < d$ . By assumption  $r = y - qd \in S$  and by minimality must be equal to 0. Therefore  $y \in d\mathbb{Z}$  and  $d\mathbb{Z} = S$  as required.  $\square$

**Definition 2.2.7** (Divisibility)

Let  $x, y \in \mathbb{Z}$  be two integers. We say that  $x$  divides  $y$  if there exists  $a$  such that  $ax = y$ . In this case we write

$$x \mid y$$

and

$$\frac{y}{x}$$

for the unique integer  $a$  such that  $ax = y$ .

**Lemma 2.2.8**

Let  $x, y \in \mathbb{Z}$  be two integers then

$$x \mid y \implies |x| \leq |y|$$

In particular  $x \mid y \wedge y \mid x \implies x = \pm y$ .

**Proposition 2.2.9** (Bezout's Theorem)

Let  $x, y$  be non-zero integers. Then there exists a unique positive integer  $d$  such that

- $d$  is a common divisor of  $x, y$
- For any other common divisor  $e$  we have  $e \mid d$ .

Further there exists integers  $a, b$  such that  $ax + by = d$ . We write this as  $(x, y)$ .

*Proof.* Let  $S = \{ax + by \mid a, b \in \mathbb{Z}\}$ . Then by (2.2.6) we have  $S = d\mathbb{Z}$  for a unique  $d > 0$ . As  $x, y \in S$  by definition  $d$  is a common divisor, and by definition  $d = d \cdot 1 = ax + by$  for some integers  $a, b$ . Suppose  $e$  is a common divisor then  $d = ax + by = e(ap + bq)$  and  $e \mid d$  as required.

Any two such common divisors have  $d = \pm d'$  by the previous Lemma. Since they are positive and non-zero we have  $d = d'$ .  $\square$

**Proposition 2.2.10**

Let  $a, x, y$  be non-zero integers then

$$|a|(x, y) = (ax, ay)$$

In particular

$$\left( \frac{x}{(x, y)}, \frac{y}{(x, y)} \right) = 1$$

*Proof.* This follows from the characterization of  $(x, y)$  as the minimal positive integer in the set  $\{mx + ny\}$ .  $\square$

**2.2.3 Prime Factorization****Definition 2.2.11**

Let  $x \in \mathbb{Z}$  be a non-zero integer. We say that  $x$

- is a **unit** if it's equal to 1 or  $-1$ .
- is **prime** if it's not a unit and  $x \mid p$  implies  $x = \pm 1$  or  $x = \pm p$
- **composite** otherwise

**Lemma 2.2.12**

Let  $p$  be a positive prime and  $a$  a non-zero integer. Then precisely one of the following holds

- $(p, a) = 1$
- $(p, a) = p$  and  $p \mid a$

*Proof.* Note that  $(p, a)$  is positive and divides both  $p$  and  $a$  so the result follows by definition of prime.  $\square$

**Proposition 2.2.13** (Euclid's Lemma)

Suppose  $x \mid ab$  then  $\frac{x}{(x, a)} \mid b$ .

In particular if  $p$  is a prime and  $p \mid ab$ , then  $p \mid a$  or  $p \mid b$ .

*Proof.* First suppose that  $(x, a) = 1$ . Then by assumption  $zx = ab$  and by [Bezout's Theorem](#)  $mx + na = 1$  for some integers  $m, n$ . Multiply by  $z$  to find that  $abm + na = a(bm + n) = z$ . Therefore  $a(bm + n)x = ab$  and cancel  $a$  to find  $x \mid b$  as required.

For the general case define  $x' = x/(x, a)$  and  $a' = a/(x, a)$ . Then by [\(2.2.10\)](#)  $(x', a') = 1$ . Furthermore it's clear that  $x' \mid a'b$  so we have  $x' \mid b$  by the special case just proven.

Finally suppose  $p \mid ab$ . If  $(p, a) = 1$  then  $p \mid b$  by the first result. By [\(2.2.12\)](#) if this does not hold then  $p \mid a$  as required.  $\square$

Using these results we may show that there exists a unique factorization into primes, unique up to multiplication by a unit.

## 2.3 Matroids

The theory of bases of vector spaces (Section ??) and transcendence bases of field extensions (Section ??) have some formal similarities, as noted in [vdW91]. Here we use the theory of Matroids to formalise this precisely so that the proofs need not be repeated in each case.

### Definition 2.3.1 (Matroid)

Consider a set  $X$  together with a *closure operator*  $c : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$  (“span” operator) such that  $c(\emptyset) = \emptyset$ . We say

- $S \subset X$  is *independent* if  $x \in S \implies x \notin c(S \setminus \{x\})$
- $\Gamma \subset X$  is *spanning* if  $c(\Gamma) = X$ .

Note by definition  $X$  is spanning and  $\emptyset$  is independent. Moreover all singletons  $\{x\}$  are independent.

We call the pair  $(X, c)$  a *matroid* if it also satisfies the following properties

- **Finitary**  $x \in c(\Gamma) \implies x \in c(\Gamma')$  for some finite subset  $\Gamma'$  of  $\Gamma$ .
- **Exchange Property** For all  $x, y \in X$  and  $Y \subseteq X$  we have

$$x \in c(Y \cup \{y\}) \setminus c(Y) \implies y \in c(Y \cup \{x\})$$

We say  $(X, c)$  has *finite rank* if it has a finite spanning set.

Finally we say  $\mathcal{B} \subseteq X$  is a *basis* if it is both *spanning* and *independent*.

We begin with some elementary characterizations of independent sets

### Lemma 2.3.2

Suppose  $S \subset X$  is a subset

- $A \subseteq c(S) \implies c(S \cup A) = c(S)$
- $S$  is independent if and only if no proper subset has the same span.

*Proof.* We prove each in turn

- By monotonicity

$$c(S) \subseteq c(S \cup A) \subseteq c(c(S) \cup A) = c(c(S)) = c(S)$$

- Suppose  $S$  is independent and  $S' \subsetneq S$  is a proper subset such that  $c(S') = c(S)$ . Choose  $x \in S \setminus S'$  then by definition  $x \in S \implies x \in c(S) = c(S') \subseteq c(S' \cup \{x\})$  contradicting independence.

Conversely suppose for some  $x \in S$  we have  $x \in c(S \setminus \{x\})$ . Define  $S' := S \setminus \{x\}$ . Then  $x \in c(S')$  implies  $c(S) = c(S')$  by a). As  $S'$  is a proper subset this contradicts the hypothesis.

□

### Lemma 2.3.3

Every subset of an independent set is independent. Furthermore the family of independent sets has *finite character*.

*Proof.* The first statement is straightforward. Suppose  $S$  a dependent set such that every finite subset is independent. Then there exists  $x \in S$  such that  $x \in c(S \setminus \{x\})$ . Then by the finitary property there exists a finite subset  $S' \subseteq S \setminus \{x\}$  such that  $x \in c(S')$ . Therefore by definition  $S' \cup \{x\}$  is not independent, a contradiction. □

The finitary condition ensures that  $\mathcal{E}$  is “inductively ordered”

### Corollary 2.3.4

Let  $\{S_i\}_{i \in I}$  be a chain of independent subsets. Then  $S = \bigcup_{i \in I} S_i$  is also independent.

### Lemma 2.3.5 (Extension Property)

Suppose  $S$  is an independent set and  $x \notin c(S)$ , then  $S \cup \{x\}$  is independent.

*Proof.* We require to prove that for all  $y \in S$  we have  $y \notin c(S \cup \{x\} \setminus \{y\})$ . By independence of  $S$  we have  $y \notin c(S \setminus \{y\})$ , so by the Exchange Property  $y \in c(S \cup \{x\} \setminus \{y\})$  would imply  $x \in c(S)$ , contradicting the hypothesis. □

### Proposition 2.3.6

Let  $\mathcal{F}$  be a family of independent sets which satisfies the following properties

- $\emptyset \in \mathcal{F}$

- **extension property**  $S \in \mathcal{F}$  and  $x \notin c(S) \implies S \cup \{x\} \in \mathcal{F}$
- $\mathcal{F}$  has *finite character*

then  $\mathcal{F}$  consists of all independent sets.

*Proof.* It's enough to show that  $\mathcal{F}$  contains all finite independent sets, which follows by induction on  $\#S$ . For given an independent set  $S$  and  $x \in S$ , then by definition  $x \notin c(S \setminus \{x\})$ . By the induction hypothesis  $S \setminus \{x\} \in \mathcal{F}$  whence by the extension property  $S \in \mathcal{F}$  as required.  $\square$

**Proposition 2.3.7** (Basis exists)

Let  $(X, c)$  be a matroid,  $S$  independent and  $\Gamma$  a subset such that  $S \subseteq \Gamma$ . Then there exists an independent set  $\mathcal{B}$  such that  $S \subseteq \mathcal{B} \subseteq \Gamma$  and  $c(\mathcal{B}) = c(\Gamma)$ .

In particular if  $\Gamma$  is spanning then  $\mathcal{B}$  is a basis.

*Proof.* Consider the collection

$$\mathcal{I} = \{T \text{ independent} \mid S \subseteq T \subseteq \Gamma\}$$

By (2.3.4) is chain-complete. Therefore it has a maximal element  $\mathcal{B}$  by Zorn's Lemma. Suppose  $x \in \Gamma \setminus c(\mathcal{B})$  then  $\mathcal{B} \cup \{x\}$  is independent by (2.3.5), contradicting maximality. Therefore  $\Gamma \subseteq c(\mathcal{B}) \implies c(\Gamma) \subseteq c(\mathcal{B})$ . The reverse inequality is clear so that  $c(\Gamma) = c(\mathcal{B})$ .  $\square$

**Corollary 2.3.8** (Criteria for bases)

Let  $(X, c)$  be a matroid. Then the following are equivalent

- $\mathcal{B}$  is a basis
- $\mathcal{B}$  is a minimal spanning set
- $\mathcal{B}$  is maximally independent (possibly in some spanning set  $\Gamma$ )

*Proof.* a)  $\implies$  b). Let  $\mathcal{B}$  be a basis and  $\Gamma \subseteq \mathcal{B}$  a spanning set. Then by (2.3.2).b)  $\Gamma = \mathcal{B}$ .

b)  $\implies$  a). Let  $\mathcal{B}$  be a minimal spanning set, then by (2.3.7) there exists a subset  $\mathcal{B}'$  which is a basis, and in particular spanning. By minimality  $\mathcal{B} = \mathcal{B}'$ .

c)  $\implies$  a). By (2.3.7) there exists a basis  $\mathcal{B}'$  containing  $\mathcal{B}$ . By maximality  $\mathcal{B}' = \mathcal{B}$ .

a)  $\implies$  c). Suppose  $\mathcal{B} \subseteq S$  where  $S$  is independent. Then  $S$  is spanning too, and so by (2.3.2)  $\mathcal{B} = S$ .  $\square$

**Lemma 2.3.9** (Mini Exchange Lemma)

Let  $S \subseteq \Gamma$  be finite sets and  $x \in X \setminus S$  such that

- $S$  is independent
- $x \in c(\Gamma) \setminus c(S)$

Then there exists  $y \in \Gamma \setminus S$  such that  $c(\Gamma \setminus \{y\} \cup \{x\}) = c(\Gamma)$ .

*Proof.* We may assume without loss of generality that  $x \notin \Gamma$ .

Consider  $\tilde{\Gamma} \subseteq \Gamma$  minimal subject to  $S \subseteq \tilde{\Gamma}$  and  $x \in c(\tilde{\Gamma})$ . If  $S = \tilde{\Gamma}$  then  $x \in c(S)$  a contradiction. Therefore we may choose  $y \in \tilde{\Gamma} \setminus S$ . By minimality we have  $x \in c(\tilde{\Gamma}) \setminus c(\tilde{\Gamma} \setminus \{y\})$ . Therefore by the Exchange Property we have  $y \in c(\tilde{\Gamma} \setminus \{y\} \cup \{x\})$ . Then by (2.3.2) applied twice

$$c(\Gamma \setminus \{y\} \cup \{x\}) = c(\Gamma \cup \{x\}) = c(\Gamma)$$

as required.  $\square$

**Proposition 2.3.10** (Exchange Lemma)

Let  $S$  be an independent set and  $\Gamma$  be a finite set such that  $S \subseteq c(\Gamma)$ . Then there exists a subset  $T \subseteq \Gamma$  such that

- $\#T = \#S$
- $c(\Gamma \setminus T \cup S) = c(\Gamma)$ .

In particular  $\#S \leq \#\Gamma$ .

*Proof.* By considering the sub-matroid  $(c(\Gamma), c)$  we may assume without loss of generality that  $X = c(\Gamma)$ .

Let  $n = \#\Gamma$  and consider the set of pairs

$$\mathcal{F} := \{(A, B) \mid A \subseteq S, \quad B \subseteq \Gamma, \quad \#A = \#B, \quad c(\Gamma \setminus B \cup A) = X\}$$

Essentially  $\mathcal{F}$  is the set of swaps we may perform from  $S$  to  $\Gamma$  whilst preserving the span. It is non-empty because  $(\emptyset, \emptyset) \in \mathcal{F}$  and we wish to show that  $(S, B) \in \mathcal{F}$  for some  $B$ .

Observe that for all  $(A, B) \in \mathcal{F}$  we have  $\#B \leq n$  so choose a pair such that  $\#B$  is maximal. We wish to show that in this case  $A = S$ , so we suppose to the contrary that  $A \subsetneq S$ .

We claim that  $B \subsetneq \Gamma$ , for  $B = \Gamma$  implies by construction  $c(A) = X$  which would imply  $c(A) = c(S)$ , contradicting the criteria for independence of  $S$  given by Lemma (2.3.2).

Define  $\Gamma' := \Gamma \setminus B \cup A$ . Then by assumption  $c(\Gamma') = X$  and  $A \subseteq \Gamma'$  is independent. Choose  $x \in S \setminus A$  then by definition of independence  $x \notin c(A)$ . By (2.3.9) there exists  $y \in \Gamma' \setminus A$  such that

$$c(\Gamma' \setminus \{y\} \cup \{x\}) = c(\Gamma') = X$$

Note by construction that  $y \notin B$ , so we see that  $(A \cup \{x\}, B \cup \{y\}) \in \mathcal{F}$  has greater length, which contradicts maximality.  $\square$

**Corollary 2.3.11** (Bases have the same cardinality)

*Every base of a finite rank matroid is finite and of the same size. Denote this by  $r(X)$ .*

*Proof.* There is a finite basis by (2.3.7). Then apply (2.3.10).  $\square$

**Corollary 2.3.12**

*Let  $(X, c)$  be a finite-rank matroid and  $S \subseteq X$  an independent subset, then  $\#S \leq r(X)$ . Similarly a spanning subset  $\Gamma$  satisfies  $\#\Gamma \geq r(X)$ .*

*Proof.* Apply (2.3.7) and (2.3.11).  $\square$

**Corollary 2.3.13** (Basis Criteria)

*Let  $\mathcal{B}$  be a subset of a finite-rank matroid  $(X, c)$ . Then the following are equivalent*

- a)  $\mathcal{B}$  is a basis
- b)  $\mathcal{B}$  is independent and  $\#\mathcal{B} \geq r(X)$
- c)  $\mathcal{B}$  is spanning and  $\#\mathcal{B} \leq r(X)$

*Proof.* Apply (2.3.7) and (2.3.11).  $\square$

**Definition 2.3.14** (Submatroid)

*A subset  $Y \subseteq X$  is a **sub-matroid** if  $c(Y) = Y$ . In this case  $S \subseteq Y \implies c(S) \subseteq Y$  and so we have an induced matroid structure  $(Y, c)$ .*

**Proposition 2.3.15**

*Let  $Y \subseteq X$  be a sub-matroid of a finite-rank matroid. Then  $Y = X \iff r(Y) = r(X)$ .*

*Proof.* Let  $\mathcal{B}$  be a basis for  $Y$  then  $\#\mathcal{B} = r(Y) = r(X)$  and is a-fortiori independent in  $X$ . Therefore by (2.3.13)  $\mathcal{B}$  is a basis for  $X$  and hence  $X = c(\mathcal{B}) = Y$ .  $\square$

## 2.4 Decomposition in Noetherian and Distributive Lattices

An analogue of irreducible factorization in rings (see ??) applies to Noetherian Lattices. Furthermore uniqueness holds when the lattice is distributive. For a canonical reference see [Bir40].

**Definition 2.4.1** (Meet-Prime and Meet-Irreducible)

Let  $(X, \leq)$  be a lattice and  $x \in X$ . Then we say that  $x$  is

- **meet-irreducible** if  $y \wedge z = x \implies y = x$  or  $z = x$
- **meet-prime** if  $y \wedge z \leq x \implies y \leq x$  or  $z \leq x$
- **join-irreducible** if  $y \vee z = x \implies y = x$  or  $z = x$
- **join-prime** if  $x \leq y \vee z \implies x \leq y$  or  $x \leq z$

The following result is proven in [Bir40, Ch. IX Lemma 4.1].

**Proposition 2.4.2** (Prime = Irreducible)

Let  $(X, \leq)$  be a lattice. In general  $\text{meet-prime} \implies \text{meet-irreducible}$  and  $\text{join-prime} \implies \text{join-irreducible}$ . If  $X$  is a distributive lattice, then the converse holds.

In the distributive case we denote by  $\mathcal{M}(X)$  and  $\mathcal{J}(X)$  the sub-poset of meet-prime and join-prime elements respectively.

*Proof.* The first statement is straightforward. Conversely suppose  $X$  is a distributive lattice and  $x$  is meet-irreducible. If  $y \wedge z \leq x$  then  $x = x \vee (y \wedge z) = (y \vee x) \wedge (z \vee x) \implies x = y \vee x$  or  $x = z \vee x$ , whence the result follows.  $\square$

**Proposition 2.4.3**

Let  $(X, \leq)$  be a distributive lattice and  $Y$  a sub-lattice, then

$$\mathcal{M}(Y) = \mathcal{M}(X) \cap Y$$

$$\mathcal{J}(Y) = \mathcal{J}(X) \cap Y$$

In particular this holds when  $Y = \{x\}^\uparrow, \{y\}^\downarrow, \{x\}^\uparrow \cap \{y\}^\downarrow$ .

**Proposition 2.4.4**

Let  $(X, \leq)$  be a distributive lattice. If it is chain-complete (resp. co-chain-complete) then so is  $\mathcal{J}(X)$  (resp.  $\mathcal{M}(X)$ ).

Every join-prime element is bounded above by a maximal join-prime element, and every meet-prime element is bounded below by a minimal meet-prime element

*Proof.* Let  $C$  be a chain of join-prime elements and  $x := \bigvee C$ . Suppose that  $y \vee z \geq x$  then  $y \vee z \geq w$  for all  $w \in C$ . Then  $y \geq w$  or  $z \geq w$  for all  $w \in C$ . Let  $C_1 := \{w \in C \mid y \geq w\}$ . If  $C_1 = C$  then we are done as  $x \leq y$ . Otherwise suppose  $w_0 \notin C_1$  then by prime-ness  $z \geq w_0$ . Clearly  $w \leq w_0 \implies w \leq z$ . Further  $w \geq w_0 \implies w \not\leq y$  (as otherwise  $w_0 \leq y$ ) whence  $w \leq z$ . Therefore  $x \leq z$ .

The last statement follows from Zorn's Lemma by considering the sub-lattices  $\{x\}^\uparrow$  and  $\{y\}^\downarrow$  which inherit the chain complete properties.  $\square$

**Definition 2.4.5**

Let  $(X, \leq)$  be a lattice and  $Y \subseteq X$  a finite subset. Then we say that  $Y$  is

- **(meet-)irredundant** if no proper subset has the same meet
- **incomparable** (or an **antichain**) if no two elements are comparable

**Lemma 2.4.6** (incomparable  $\iff$  irredundant)

Let  $(X, \leq)$  be a lattice and  $Y \subseteq X$  then  $Y$  meet-irredundant  $\implies Y$  incomparable. Conversely if  $Y$  is a finite incomparable subset of meet-prime elements then  $Y$  is meet-irredundant.

*Proof.* The first part is straightforward, for if  $y_1 \leq y_2$  are elements of  $Y$  then  $\bigwedge Y = \bigwedge Y \setminus \{y_2\}$ .

Conversely suppose  $Y' \subsetneq Y$  is such that  $\bigwedge Y' = \bigwedge Y$ . Choose  $y_2 \in Y \setminus Y'$ , then  $\bigwedge Y' \leq y_2$ , whence by definition of meet-prime (and induction)  $y_1 \leq y_2$  for some  $y_1 \in Y'$ .  $\square$

The following is [Bir40, Chapter IX Theorem 9]

**Proposition 2.4.7** (Decomposition in Noetherian Lattice)

Let  $(X, \leq)$  be a *Noetherian distributive lattice*. Then every element  $x \in X$  has a unique decomposition

$$x = x_1 \wedge \dots \wedge x_n$$

where  $x_i$  are meet-prime and meet-irredundant (equivalently incomparable). These are precisely the meet-primes minimal over  $x$ .

Dually, if  $(X, \leq)$  is an *Artinian distributive lattice*, then every element  $x \in X$  has a unique decomposition

$$x = x_1 \vee \dots \vee x_n$$

where  $x_i$  are join-prime and join-irredundant (equivalently incomparable). These are precisely the join-primes maximal below  $x$ .

*Proof.* Let  $Y$  be the subset of elements which are not finite meets of meet-prime elements, and suppose it is non-empty. Then by (2.1.62)  $Y$  has a maximal element  $x_0$ . It cannot be meet-prime, and therefore not meet-irreducible (2.4.2), so there must exist elements  $y_0, z_0 \in X$  such that  $x_0 = y_0 \wedge z_0$  but  $x_0 \not\leq y_0$  and  $x_0 \not\leq z_0$ . By maximality  $y_0, z_0$  are finite meets of prime elements, and therefore so is  $x_0$  a contradiction.

Therefore we have a decomposition into distinct primes

$$x = x_1 \wedge \dots \wedge x_n$$

Consider the family of subsets of  $\{x_1, \dots, x_n\}$  which have the same meet. Then there exists a minimal subset which by definition is meet-irredundant and by (2.4.6) incomparable. Suppose there is another such decomposition  $x = x'_1 \wedge \dots \wedge x'_m$ . Then for every  $i = 1 \dots n$  we have  $x'_{\sigma(i)} \leq x_i$  and for every  $j = 1 \dots m$  we have  $x_{\tau(j)} \leq x'_j$  whence  $x_{\tau(\sigma(i))} \leq x'_{\sigma(i)} \leq x_i$ . As the decomposition is incomparable we have  $\tau(\sigma(i)) = i$  and  $x'_{\sigma(i)} = x_i$ . Therefore  $\sigma$  is injective and  $n \leq m$ . By symmetry  $m \leq n$  and  $\sigma$  is a bijection. In otherwords the decomposition is unique.

Note  $x \leq z$  and  $z$  meet-prime implies  $x_j \leq z$  for some  $j$ . Therefore if  $z$  is a minimal prime then  $x_j = z$  and all the meet-primes minimal over  $x$  must occur in the decomposition. Similarly if  $z \leq x_i$  then by incomparability  $x_j = z = x_i$ . Therefore each  $x_i$  is also minimal.  $\square$

## 2.5 Krull Dimension

The purpose of this section is to abstract the notions of Krull Dimension in commutative ring theory (??) and topology (Section 3.1.12). A more standard approach (eg EGA IV) would be to develop the topological notion first, and then link to commutative ring case using the prime spectrum (??). Generally the concept is not well-behaved, so stronger conditions are defined which generally hold in geometric cases. Principle references are (EGA0 IV 14.3, Heinrich).

**Definition 2.5.1** (Finite-Dimensional Poset)

Let  $(\mathcal{G}, \leq)$  be a poset, we say that it is **finite-dimensional** if

$$\dim(\mathcal{G}) := \sup\{\ell(C) \mid C \subseteq \mathcal{G} \text{ a chain}\} < \infty$$

In this case we define

$$\begin{aligned} \dim(x) &:= \dim(\{x\}^\downarrow) \\ \text{codim}(y) &:= \dim(\{y\}^\uparrow) \\ \text{codim}(y, x) &:= \dim(\{x\}^\downarrow \cap \{y\}^\uparrow) \end{aligned}$$

Note  $\mathcal{G}$  is both Noetherian and Artinian, but finite-dimensionality is a stronger condition. Note also that  $\{x\}^\downarrow, \{y\}^\uparrow$  and  $\{x\}^\downarrow \cap \{y\}^\uparrow$  are finite-dimensional posets

**Definition 2.5.2** (Krull Lattice)

Let  $(\mathcal{F}, \leq)$  be an **Artinian** distributive lattice. We say it is a **Krull Lattice** if the poset of **join-prime** elements  $\mathcal{J}(\mathcal{F})$  is finite-dimensional and define

$$\dim(\mathcal{F}) := \dim(\mathcal{J}(\mathcal{F}))$$

By (2.4.3) we have  $\mathcal{H} = \{x\}^\downarrow, \{y\}^\uparrow, \{x\}^\downarrow \cap \{y\}^\uparrow$  are Krull Lattices such that

$$\mathcal{J}(\mathcal{H}) = \mathcal{J}(\mathcal{F}) \cap \mathcal{H}$$

For  $x, y \in \mathcal{F}$  we have a unique decomposition into maximal join-prime elements  $x_i, y_j \in \mathcal{J}(\mathcal{F})$  (2.4.7)

$$x = x_1 \vee \dots \vee x_n$$

$$y = y_1 \vee \dots \vee y_m$$

Note that  $y \leq x \iff$  for all  $j$  we have  $y_j \leq x_i$  for some  $i$  and we may define

$$\dim(x) := \max_i \dim(x_i) = \dim(\{x\}^\downarrow) \quad (2.2)$$

$$\text{codim}(y) := \min_j \text{codim}(y_j) \quad (2.3)$$

$$\text{codim}(y, x) := \min_j \max_i \{\text{codim}(y_j, x_i) \mid y_j \leq x_i\} \quad (2.4)$$

$$= \min_j \text{codim}(y_j, x) \quad (2.5)$$

note it's required to be careful in definition of co-dimension in order to have a sensible co-dimension formula. Note also that

$$\dim(y; \{x\}^\downarrow) = \dim(y) \quad (2.6)$$

**Remark 2.5.3**

For the topological case, we would define  $\mathcal{F}$  to be the closed subsets of  $X$  and  $\mathcal{J}(\mathcal{F})$  would be the collection of irreducible closed subsets, see (3.1.60).

**Proposition 2.5.4** (Extending chains)

Let  $(\mathcal{G}, \leq)$  be a finite-dimensional poset

- Every chain is contained in a saturated chain
- Every chain is contained in a maximal chain
- Every maximal chain is of the form

$$x_0 \leq x_1 \leq \dots \leq x_n$$

for  $x_0$  minimal and  $x_n$  maximal in  $\mathcal{G}$ .

**Definition 2.5.5** (Properties)

Let  $\mathcal{G}$  be a finite-dimensional poset. Then we say it is

- **Irreducible** if it has a top element
- **Equidimensional** if every maximal element has the same dimension
- **Equicodimensional** if every minimal element has the same codimension
- **Biequidimensional** if every maximal chain has the same length
- **Quasi-Biequidimensional** if  $\{x\}^\downarrow$  is biequidimensional for all  $x$  (equivalently all maximal  $x$ )
- **Catenary** if for every pair  $y \leq x$ , every saturated chain in  $[y, x] := \{y\}^\uparrow \cap \{x\}^\downarrow$  has the same length, namely  $\text{codim}(y, x)$ .

If  $\mathcal{F}$  is a Krull Lattice then we say it inherits these properties from  $\mathcal{J}(\mathcal{F})$ . Note if  $\mathcal{F}$  is irreducible then it also has (the same) top element.

Trivially irreducible implies equidimensional. Similarly biequidimensional implies both equidimensional and equicodimensional, but not conversely.

Finally **quasi-biequidimensional + equidimensional  $\iff$  biequidimensional**.

**Proposition 2.5.6** (Simple Properties)

Let  $\mathcal{G}$  be a finite-dimensional poset then

- a) If  $x$  is maximal then  $\text{codim}(x) = 0$
- b) If  $x$  is minimal then  $\dim(x) = 0$
- c)  $\dim(\mathcal{G}) = \sup\{\dim(x) \mid x \text{ maximal}\} = \sup\{\text{codim}(x) \mid x \text{ minimal}\}$
- d) For all  $z \leq y \leq x$  we have  $\text{codim}(z, y) + \text{codim}(y, x) \leq \text{codim}(z, x)$

If  $\mathcal{F}$  is a Krull Lattice then

- e) For all  $y \leq x$  we have  $\dim(y) + \text{codim}(y, x) \leq \dim(x)$
- f) For all  $y \leq x$  we have  $\text{codim}(y, x) = 0 \iff y_j = x_i$  some  $i, j$

Alternatively  $\text{codim}(y, x) > 0$  if and only if  $(y_j \leq x_i \implies y_j \neq x_i)$ .

*Proof.* e) The case of a finite-dimensional poset is (relatively) clear. In the general case then we have

$$\dim(y_j) + \text{codim}(y, x) \leq \dim(y_j) + \text{codim}(y_j, x) = \max_i (\dim(y_j) + \text{codim}(y_j, x_i)) \leq \max_i \dim(x_i) = \dim(x)$$

and taking max over  $j$  yields the result.

f) The case  $x, y \in \mathcal{J}(\mathcal{F})$  is clear by (2.5.4). For the general case  $\text{codim}(y, x) = 0 \iff \text{codim}(y_j, x) = 0$  for some  $j \iff (y_j \leq x_i \implies \text{codim}(y_j, x_i) = 0) \iff y_j = x_i$  for some  $i, j$ .  $\square$

**Corollary 2.5.7** (Codimension 1 formula)

Let  $\mathcal{F}$  be a Krull Lattice with  $y \in \mathcal{F}$ ,  $x \in \mathcal{J}(\mathcal{F})$  and  $y \leq x$ . Then

$$\dim(y) = \dim(x) - 1 \implies \text{codim}(y, x) = 1$$

Suppose further that  $\mathcal{F}$  is irreducible then

$$\dim(y) = \dim(\mathcal{F}) - 1 \implies \text{codim}(y) = 1$$

*Proof.* By (2.5.6).e)  $\text{codim}(y, x) \leq 1$ . If  $\text{codim}(y, x) = 0$  then by f) we see that  $y_j = x$  for some  $j$ , whence  $y = x$  which contradicts  $\dim(y) = \dim(x) - 1$ . Therefore  $\text{codim}(y, x) = 1$  as required.

If  $\mathcal{F}$  is irreducible the result follows with  $x = \top$ .  $\square$

**Remark 2.5.8** (Duality)

We note that the concepts of **dimension** (of a poset), **biequidimensional** and **catenary** are self-dual, in the sense that they are preserved when considering the **dual poset**  $(\mathcal{G}, \leq^d)$ .

Similarly the concepts of **equidimensional** and **equicodimensional** are dual to each other.

**Proposition 2.5.9**

Let  $\mathcal{G}$  be a finite-dimensional poset. Then the following are equivalent

- $\mathcal{G}$  is catenary
- For every triplet  $z < y < x$  in  $\mathcal{G}$  we have

$$\text{codim}(z, x) = \text{codim}(z, y) + \text{codim}(y, x)$$

When  $\mathcal{G}$  is irreducible this is equivalent to

$$\text{codim}(z) = \text{codim}(z, y) + \text{codim}(y)$$

for all  $z < y$ .

*Proof.* Suppose  $\mathcal{G}$  is catenary. Choose saturated chains  $C_1$  in  $[z, y]$  and  $C_2$  in  $[y, x]$ . One may show that  $C_1 \cap C_2 = \{y\}$  and  $C_1 \cup C_2$  is a saturated chain in  $[x, z]$ . The result follows by definition of catenary.

Conversely, consider a saturated chain between  $x$  and  $y$  of length  $n$

$$x = x_0 \leq x_1 \leq \dots \leq x_n = y$$

Then clearly  $\text{codim}(x_i, x_{i+1}) = 1$  for all  $i = 0 \dots n - 1$ . Therefore by repeatedly applying the relation we may show that

$$\text{codim}(x, y) = n$$

and therefore every saturated chain between  $x$  and  $y$  has the same length.

When  $\mathcal{G}$  is irreducible we may deduce the second formula by setting  $x = \top$ . Conversely we may see that

$$\text{codim}(z, x) = \text{codim}(z) - \text{codim}(x) = \text{codim}(z) - \text{codim}(y) + \text{codim}(y) - \text{codim}(x) = \text{codim}(z, y) + \text{codim}(y, x)$$

□

**Lemma 2.5.10**

Let  $\mathcal{G}$  be a biequidimensional finite-dimensional poset. Then for  $x \in \mathcal{G}$  we have

- a)  $\{x\}^\downarrow$  and  $\{x\}^\uparrow$  are biequidimensional
- b)  $\dim(\mathcal{G}) = \dim(x) + \text{codim}(x)$
- c) If  $x$  is maximal then  $\dim(x) = \dim(\mathcal{G})$  and in particular  $\mathcal{G}$  is equidimensional
- d) If  $x$  is minimal then  $\text{codim}(x) = \dim(\mathcal{G})$  and in particular  $\mathcal{G}$  is equicodimensional

In particular  $\mathcal{G}$  is quasi-biequidimensional.

*Proof.* Consider a fixed maximal chain  $C$  of  $\{x\}^\uparrow$ , necessarily containing  $x$ . Any maximal chain  $C'$  of  $\{x\}^\downarrow$  also contains  $x$  and combines with  $C$  to yield a maximal chain of  $\mathcal{F}$ . Whence  $\ell(C') + \ell(C) = \dim(\mathcal{F})$  and  $\{x\}^\downarrow$  is biequidimensional. By duality  $\{x\}^\uparrow$  is biequidimensional and  $\ell(C) = \text{codim}(x)$  from which the formula follows.

If  $x$  is maximal then clearly  $\text{codim}(x) = 0$ , and similarly if  $x$  is minimal then  $\dim(x) = 0$ , so the last two statements follow immediately.

□

**Proposition 2.5.11** (Equivalent Characterizations of Biequidimensionality)

Let  $\mathcal{G}$  be a finite-dimensional poset. Then the following are equivalent

- a)  $\mathcal{G}$  is quasi-biequidimensional
- b)  $\mathcal{G}$  is catenary and for every maximal  $x$  we have  $\{x\}^\downarrow$  is equicodimensional
- c)  $\mathcal{G}$  satisfies  $\dim(x) = \dim(y) + \text{codim}(y, x)$  for  $y \leq x$
- d)  $\mathcal{G}$  satisfies c) whenever  $\text{codim}(y, x) = 1$

Furthermore the following relationship holds

$$\text{codim}(y) = \text{codim}(y, x) + \text{codim}(x) \quad y \leq x$$

*Proof.* **a)**  $\implies$  **c)**. By assumption  $\{x\}^\downarrow$  is biequidimensional. Then

$$\dim(x) = \dim(\{x\}^\downarrow) \stackrel{(2.5.10)}{=} \dim(y; \{x\}^\downarrow) + \text{codim}(y; \{x\}^\downarrow) = \dim(y) + \text{codim}(y, x)$$

For **c)**  $\implies$  **b)**. Suppose  $z < y < x$  in  $\mathcal{G}$  then by the codimension formula applied twice

$$\text{codim}(z, x) = \dim(x) - \dim(z) = \dim(x) - \dim(y) + \dim(y) - \dim(z) = \text{codim}(y, x) + \text{codim}(z, y)$$

so by (2.5.9)  $\mathcal{G}$  is catenary. Let  $x$  be a maximal element and  $z$  a minimal element of  $\{x\}^\downarrow$ , then by the codimension formula

$$\text{codim}(z, x) = \dim(x) - \dim(z) = \dim(x)$$

whence  $\{x\}^\downarrow$  is equicodimensional.

**b)**  $\implies$  **a)**. Let  $x$  be a maximal element and  $C$  a maximal chain in  $\{x\}^\downarrow$  with minimum element  $y$  then, as  $\mathcal{G}$  is catenary, we have  $\ell(C) = \text{codim}(y, x)$ . As  $\{x\}^\downarrow$  is equicodimensional we have  $\text{codim}(y, x) = \dim(x)$  which shows that  $\{x\}^\downarrow$  is biequidimensional.

Clearly **c)**  $\implies$  **d)**. Conversely for **d)**  $\implies$  **a)** consider a maximal chain in  $\{x\}^\downarrow$

$$x_0 \leq \dots \leq x_n = x$$

Then clearly  $\text{codim}(x_i, x_{i+1}) = 1$  whence by induction  $\ell(C) = \dim(x)$ .

The final statement follows from (2.5.9) because  $\{x\}^\downarrow$  is irreducible and catenary.  $\square$

### Remark 2.5.12

*This is a corrected version of EGA IV 14.3.3, as noted by Heinrich.*

### Corollary 2.5.13

*When  $\mathcal{F}$  is a quasi-biequidimensional Krull Lattice then we have the following codimension formulas for all  $x \in \mathcal{J}(\mathcal{F})$  and  $y \in \mathcal{F}$  such that  $y \leq x$*

$$\dim(x) = \dim(y) + \text{codim}(y, x) \tag{2.7}$$

$$\text{codim}(y) = \text{codim}(y, x) + \text{codim}(x) \tag{2.8}$$

Furthermore if  $\mathcal{F}$  is biequidimensional then for all  $y \in \mathcal{F}$

$$\dim(\mathcal{F}) = \dim(y) + \text{codim}(y)$$

*Proof.* The first two relations hold when  $x, y \in \mathcal{J}(\mathcal{F})$  by (2.5.11). We may generalise this as follows.

For the first relation and fixed  $x \in \mathcal{J}(\mathcal{F})$  we see  $\dim(y_j)$  is maximal precisely when  $\text{codim}(y_j, x)$  is minimal. Therefore it holds for all  $y \in \mathcal{F}$ .

Similarly for the second relation we see that  $\text{codim}(y_j, x)$  is minimal precisely when  $\text{codim}(y_j)$  is minimal.

The final statement follows from the first relation by taking the supremum over all maximal elements  $x$ .  $\square$

### Corollary 2.5.14

*Let  $\mathcal{G}$  be an irreducible finite-dimensional poset. Then the following are equivalent*

- a)  $\mathcal{G}$  is biequidimensional
- b)  $\mathcal{G}$  is quasi-biequidimensional
- c)  $\mathcal{G}$  is catenary and equicodimensional
- d)  $\dim(x) = \dim(y) + \text{codim}(y, x) \quad \forall y \leq x$
- e)  $\mathcal{G}$  satisfies d) when  $\text{codim}(y, x) = 1$

*Proof.* The equivalence follows from (2.5.11) by noting that an irreducible poset has only one maximal element and is in particular equidimensional.  $\square$

**Proposition 2.5.15**

*Let  $\mathcal{F}$  be a quasi-biequidimensional Krull Lattice. Then for all  $x \in \mathcal{F}$  we have  $\{x\}^\downarrow$  is a quasi-biequidimensional Krull Lattice.*

*Proof.* Consider  $y \in \mathcal{J}(\{x\}^\downarrow) \stackrel{(2.4.3)}{=} J(\mathcal{F}) \cap \{x\}^\downarrow$ . Then by definition  $\{y\}^\downarrow$  is biequidimensional, and so  $\{x\}^\downarrow$  is quasi-biequidimensional.  $\square$

## 2.6 Category Theory

### 2.6.1 Categories

**Definition 2.6.1** (Category)

A (locally small) **category**  $\mathcal{C}$  consists of

- a class  $\text{ob}(\mathcal{C})$  of objects
- for every pair of objects  $a, b \in \text{ob}(\mathcal{C})$  a set of **morphisms**  $\text{Mor}(a, b)$
- for every three objects  $a, b, c$  a **law of composition**

$$\begin{aligned} \text{Mor}(a, b) \times \text{Mor}(b, c) &\rightarrow \text{Mor}(a, c) \\ (g, f) &\rightarrow g \circ f \end{aligned}$$

such that the following conditions hold

- $h \circ (g \circ f) = (h \circ g) \circ f$  **associativity**
- There exists  $1_a \in \text{Mor}(a, a)$  such that  $1_a \circ f = f$  and  $g \circ 1_a = g$ .

**Example 2.6.2**

The category of sets is **Set** with maps in the usual way. Note associativity is automatically satisfied.

**Example 2.6.3** ( $n$ -pointed category)

Given a category  $\mathcal{C}$  where objects are sets, we may consider the pointed category  $(\mathcal{C}, \star^n)$  consisting of pairs  $(A, a)$  where  $A \in \text{ob}(\mathcal{C})$  and  $a \in A^n$ . We consider only morphisms  $f : A \rightarrow B$  such that  $f(a_i) = b_i$ .

**Definition 2.6.4** (Opposite Category)

Given a category  $\mathcal{C}$  the **opposite category** is denoted  $\mathcal{C}^{op}$  and given by

- The same class of objects  $\text{ob } \mathcal{C}^{op} = \text{ob}(\mathcal{C})$
- For every pair of objects  $a, b \in \text{ob } \mathcal{C}$  the morphisms are reversed

$$\text{Mor}^{op}(a, b) := \text{Mor}(b, a)$$

- The law of composition is reversed

$$\begin{aligned} \text{Mor}^{op}(a, b) \times \text{Mor}^{op}(b, c) &\rightarrow \text{Mor}^{op}(a, c) \\ (g, f) &\rightarrow f \circ g \end{aligned}$$

**Example 2.6.5**

The polynomial ring  $A[X]$  is an initial object in the category of pointed  $A$ -algebras.

**Definition 2.6.6** (Isomorphism)

A morphism  $f : a \rightarrow b$  is an **isomorphism** if there exists  $g : b \rightarrow a$  such that

$$g \circ f = 1_a$$

and

$$f \circ g = 1_b$$

**Proposition 2.6.7** (Initial objects are unique)

An initial object is unique up to isomorphism

*Proof.* First observe by uniqueness  $\eta_a^a = 1_a$ . Let  $a, a'$  be two initial objects with morphisms  $\eta_a^a$  and  $\eta_{a'}^{a'}$  respectively. Then by definition

$$\eta_a^{a'} \circ \eta_{a'}^a = \eta_a^a = 1_a$$

and vice-versa.

□

**Definition 2.6.8** (Functor)

A **covariant functor**  $F : \mathcal{C} \rightarrow \mathcal{D}$  consists of a mapping of objects

$$F : \text{ob}(\mathcal{C}) \rightarrow \text{ob}(\mathcal{D})$$

together with a mapping of morphisms

$$F(-) : \text{Mor}(a, b) \rightarrow \text{Mor}(F(a), F(b))$$

which satisfies

- $F(1_a) = 1_{F(a)}$
- $F(f \circ g) = F(f) \circ F(g)$

A **contravariant functor**  $\mathcal{C} \rightarrow \mathcal{D}$  is equivalent to a covariant functor on the opposite category  $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ , namely where arrows are reversed.

**Definition 2.6.9** (Full and faithful)

A functor  $F$  is said to be

- **Faithful** if  $F(-)$  is injective.
- **Full** if  $F(-)$  is surjective.

**Definition 2.6.10** (Concrete Category)

A **concrete category** is a pair  $(\mathcal{C}, U)$  where  $\mathcal{C}$  and a “**forgetful functor**”  $U : \mathcal{C} \rightarrow \mathbf{Set}$  which is faithful

**Example 2.6.11** (Forgetful Functor)

The category of groups (resp. rings, modules, ...) is a concrete category in the obvious way.

**Definition 2.6.12** (Mor functor)

For any objects  $a, b, c \in \text{ob}(\mathcal{C})$ , there is a canonical covariant functor

$$\begin{array}{ccc} \text{Mor}(a, -) : \mathcal{C} & \longrightarrow & \mathbf{Set} \\ b & \longrightarrow & \text{Mor}(a, b) \end{array}$$

which acts on a morphism  $f : b \rightarrow c$  by

$$\begin{array}{ccc} \text{Mor}(a, f) : \text{Mor}(a, b) & \rightarrow & \text{Mor}(a, c) \\ g & \rightarrow & f \circ g \end{array}$$

It's a functor precisely because composition of functions is associative. Similarly there's a canonical contravariant functor  $\text{Mor}(-, b)$ .

**Definition 2.6.13** (Natural Transformation)

Let  $F, G : \mathcal{C} \rightarrow \mathcal{D}$  be covariant functors. A **natural transformation**  $\eta : F \Rightarrow G$  consists of a family of morphisms

$$\eta_c : F(c) \rightarrow G(c) \quad c \in \text{ob}(\mathcal{C})$$

such that the following diagram commutes holds for all  $f : c \rightarrow c'$

$$\begin{array}{ccc} F(c) & \xrightarrow{\eta_c} & G(c) \\ \downarrow F(f) & & \downarrow G(f) \\ F(c') & \xrightarrow{\eta_{c'}} & G(c') \end{array}$$

for all  $f : c \rightarrow c'$ .

**Definition 2.6.14** (Natural isomorphism)

A natural transformation  $\eta : F \Rightarrow G$  is a natural isomorphism if  $\eta_c$  is an isomorphism for all  $c \in \mathcal{C}$ .

**Definition 2.6.15**

We say  $\mathcal{C}' \subset \mathcal{C}$  is a **subcategory** if

- $\text{ob}(\mathcal{C}') \subseteq \text{ob}(\mathcal{C})$
- $\text{Mor}_{\mathcal{C}'}(c, d) \subseteq \text{Mor}_{\mathcal{C}}(c, d)$
- *Composition agrees when it is well-defined*

We say  $\mathcal{C}'$  is a **full subcategory** if additionally  $\text{Mor}_{\mathcal{C}'}(c, d) = \text{Mor}_{\mathcal{C}}(c, d)$ .

## 2.6.2 Pointed Categories

**Definition 2.6.16** (Initial object)

An **initial object** of a category  $\mathcal{C}$  is an object  $\emptyset$  such that for all objects  $c$

$$\text{Mor}(\emptyset, c) = \{\eta_{\emptyset}^c\}$$

consists of a single element.

In this case an initial object is unique up to unique isomorphism.

**Definition 2.6.17** (Terminal Object)

A **terminal object** of a category  $\mathcal{C}$  is an object  $\star$  such that for all objects  $c \in \mathcal{C}$  we have

$$\text{Mor}(c, \star) = \{!_c\}$$

is a singleton.

In this case a terminal object is unique up to unique isomorphism.

**Definition 2.6.18** (Zero Object / Pointed Category)

A **zero object** of a category  $\mathcal{C}$  is both an initial and terminal object. We denote this by  $0$  or  $0_{\mathcal{C}}$

For two objects  $c, c' \in \mathcal{C}$  we denote by  $0_{c, c'}$  the composite morphism  $c \rightarrow 0 \rightarrow c'$ . This is independent of the choice of zero object.

A category with a distinguished zero object is **pointed**.

**Proposition 2.6.19**

Let  $\mathcal{C}$  be a pointed category and  $c \in \mathcal{C}$ . Then the following are equivalent

- $c$  is an initial object
- $c$  is a terminal object
- $c$  is a zero object

**Proposition 2.6.20** (Functors Preserve Zeros)

Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a functor of pointed categories for which  $F(0_{\mathcal{C}})$  is a zero object of  $\mathcal{D}$ . Then for all objects  $c, c'$  we have

$$F(0_{c, c'}) = 0_{F(c), F(c')}$$

## 2.6.3 Product Categories and Bifunctors

**Definition 2.6.21** (Product Category)

Given two categories  $\mathcal{C}, \mathcal{D}$  we may construct the product category  $\mathcal{C} \times \mathcal{D}$  as follows

- The objects are given by pairs

$$\text{ob}(\mathcal{C} \times \mathcal{D}) := \text{ob}(\mathcal{C}) \times \text{ob}(\mathcal{D})$$

- The morphisms are given by pairs

$$\text{Mor}((c, c'), (d, d')) := \text{Mor}(c, c') \times \text{Mor}(d, d')$$

Concretely given  $f : c \rightarrow c'$  and  $g : d \rightarrow d'$  write  $(f, g) : (c, d) \rightarrow (c', d')$

- The law of composition is determined by

$$(f, g) \circ (h, k) := (f \circ h, g \circ k)$$

It's clear that the law of composition is associative and the identity morphism for  $(c, d)$  is  $(1_c, 1_d)$ . Furthermore we observe the following property

$$(f, 1_d) \circ (1_c, g) = (f, g) = (1_c, g) \circ (f, 1_d) \quad (2.9)$$

**Definition 2.6.22** (Bifunctor)

A functor on a product category,  $F : \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{E}$  is termed a **bifunctor**.  $F$  induces a family of functors

- $F(c, -) : \mathcal{D} \rightarrow \mathcal{E}$  given by

$$F(c, g) := F(1_c, g)$$

- $F(-, d) : \mathcal{C} \rightarrow \mathcal{E}$  given by

$$F(f, d) := F(f, 1_d)$$

which satisfy the compatibility conditions

$$\begin{array}{ccc} F(c, d) & \xrightarrow{F(f, d)} & F(c', d) \\ F(c, g) \downarrow & \searrow F(f, g) & \downarrow F(c', g) \\ F(c, d') & \xrightarrow{F(f, d')} & F(c', d') \end{array}$$

by applying  $F$  to Eq. (2.9).

**Proposition 2.6.23** (Reconstruct Bifunctor)

Consider a family of functors  $\{F_L(c) : \mathcal{D} \rightarrow \mathcal{E}\}_{c \in \text{ob}(\mathcal{C})}$  and  $\{F_R(d) : \mathcal{C} \rightarrow \mathcal{E}\}_{d \in \text{ob}(\mathcal{D})}$ . Suppose the following conditions hold

- $F_L(c)(d) = F_R(d)(c)$
- $F_L(c')(g) \circ F_R(d)(f) = F_R(d')(f) \circ F_L(c)(g)$

then these determine a well-defined bifunctor given by

$$F(f \times g) := F_L(c')(g) \circ F_R(d)(f)$$

**Proposition 2.6.24**

Let  $F : \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{E}$  be a bifunctor. For  $f : c \rightarrow c'$  then there is a natural transformation

$$F(f, -) : F(c, -) \Rightarrow F(c', -)$$

given by  $F(f, -)_d := F(f, 1_d)$ . Similarly for  $g : d \rightarrow d'$  then there is a natural transformation

$$F(-, g) : F(-, d) \Rightarrow F(-, d')$$

*Proof.* The naturality condition is immediate from Equation (2.9) and the commutative diagram in (2.6.22).  $\square$

**Proposition 2.6.25** (Criteria for Natural Transformation)

Let  $F, G : \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{E}$  be two bifunctors and suppose we have a family of natural transformations

$$\eta_c : F(c, -) \Rightarrow G(c, -)$$

for all  $c \in \mathcal{C}$ . Then the following are equivalent

- $\eta : F(-, -) \Rightarrow G(-, -)$  is a natural transformation
- The following diagram commutes for all  $d \in \mathcal{D}$  and  $f : c \rightarrow c'$

$$\begin{array}{ccc} F(c, d) & \xrightarrow{\eta_{c, d}} & G(c, d) \\ F(f, 1_d) \downarrow & & \downarrow G(f, 1_d) \\ F(c', d) & \xrightarrow{\eta_{c', d}} & G(c', d) \end{array}$$

*Proof.* a)  $\implies$  b) This diagram is a special case of the naturality condition.

b)  $\implies$  a) Suppose  $f : c \rightarrow c'$  and  $g : d \rightarrow d'$  then we require to show that

$$G(f, g)(\eta_{c, d}(\phi)) = \eta_{c', d'}(F(f, g)(\phi))$$

However

$$\begin{aligned} G(f, g)(\eta_{c, d}(\phi)) &= G(f, 1_{d'})(G(1_c, g)(\eta_{c, d}(\phi))) \\ &= G(f, 1_{d'})\eta_{c, d'}(F(1_c, g)(\phi)) \\ &= \eta_{c', d'}(F(f, 1_{d'})(F(1_c, g)(\phi))) \\ &= \eta_{c', d'}(F(f, g)(\phi)) \end{aligned}$$

where we have used that  $\eta_{c, d}$  is natural in  $d$  and  $c$  individually. □

**Example 2.6.26** (Mor is a bifunctor)

The canonical example is the following, given any locally small category  $\mathcal{C}$ , we have a bifunctor

$$\text{Mor} : \mathcal{C}^{op} \times \mathcal{C} \rightarrow \mathbf{Set}$$

given by the set of morphisms. The action on morphisms is given by

$$\text{Mor}(f, g)(\phi) = g \circ \phi \circ f$$

and we verify by associativity of  $\mathcal{C}$  that

$$\text{Mor}((f, g) \circ (h, k)) = \text{Mor}(f, g) \circ \text{Mor}(h, k)$$

and it's clear that the commutativity condition in Eq. (2.9) is satisfied.

**Example 2.6.27** (Concrete bifunctors)

Let  $\mathcal{C}, \mathcal{D}$  be concrete categories and consider the following bifunctor

$$\begin{aligned} F & : \mathcal{D}^{op} \times \mathcal{C} \rightarrow \mathbf{Set} \\ F(d, c) & := \{\phi : d \rightarrow c \mid P(\phi)\} \end{aligned}$$

where  $P(-)$  is some predicate such that  $P(\phi) \implies P(g \circ \phi \circ f)$ .

## 2.6.4 Equivalence of categories

**Definition 2.6.28** (Equivalence of categories)

Let  $\mathcal{C}, \mathcal{D}$  be categories. An **equivalence of categories** consists of a pair of functors (either both covariant or both contravariant)

$$\mathcal{C} \xrightleftharpoons[F]{G} \mathcal{D}$$

together with natural isomorphisms

$$\begin{aligned} \eta : \mathbf{1} &\Rightarrow GF \\ \epsilon : FG &\Rightarrow \mathbf{1} \end{aligned}$$

We say  $F$  is an equivalence of categories if there exists some  $G$  satisfying these conditions.

**Definition 2.6.29**

We say a functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  is **essentially surjective** if for all  $d \in \mathcal{D}$  there exists  $c \in \mathcal{C}$  such that  $F(c)$  is isomorphic to  $d$ .

**Lemma 2.6.30**

Let  $F : \mathcal{C} \rightarrow \mathcal{D}$ ,  $G : \mathcal{D} \rightarrow \mathcal{C}$  be functors.

If there exists a natural isomorphism  $\eta : \mathbf{1} \Rightarrow GF$  then  $F$  is faithful. Explicitly  $F(-)$  has a left-inverse given by

$$g \rightarrow \eta_{c'}^{-1} \circ G(g) \circ \eta_c$$

Furthermore  $GF(\eta_c) = \eta_{GF(c)}$ .

*Proof.* Consider the sequence of maps

$$\text{Mor}(c, c') \xrightarrow{F(-)} \text{Mor}(F(c), F(c')) \xrightarrow{G(-)} \text{Mor}(GF(c), GF(c')) \xrightarrow{\text{Mor}(\eta_c, \eta_{c'}^{-1})} \text{Mor}(c, c')$$

Note that the composite of this map is given by

$$f \rightarrow \eta_{c'}^{-1} \circ GF(f) \circ \eta_c = \eta_{c'}^{-1} \circ \eta_{c'} \circ f = f$$

in other words  $F(-)$  has a left inverse and therefore  $F$  is faithful.

Note by naturality we have  $GF(\eta_c) \circ \eta_c = \eta_{GF(c)} \circ \eta_c$ . Since  $\eta_c$  is an isomorphism we may cancel to find  $GF(\eta_c) = \eta_{GF(c)}$ .  $\square$

**Proposition 2.6.31** (Equivalence is full and faithful)

Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a functor then the following are equivalent

- $F$  is full, faithful and *essentially surjective*
- $F$  is an *equivalence of categories*

In other words  $F(-)$  is bijective and hence has a two-sided inverse. Explicitly it is given by

$$\begin{array}{ccc} \text{Mor}(c, c') & \longleftrightarrow & \text{Mor}(F(c), F(c')) \\ f & \rightarrow & F(f) \\ \eta_{c'}^{-1} \circ G(g) \circ \eta_c & \leftarrow & g \end{array}$$

*Proof.* We prove only the second implies the first. By assumption there is an equivalence with  $G$  and by the previous Lemma both  $F$  and  $G$  are faithful by considering  $\eta$  and  $\epsilon^{-1}$  in turn. Further the given map is already shown to be a left inverse. We claim it's also a right inverse, for consider

$$g' := F(\eta_{c'}^{-1}) \circ FG(g) \circ F(\eta_c).$$

We claim that  $G(g') = G(g)$ . As  $G$  is faithful this would imply  $g' = g$  and the given map is a right inverse as required. Observe

$$G(g') = GF(\eta_{c'}^{-1}) \circ GFG(g) \circ GF(\eta_c) = \eta_{GF(c')}^{-1} \circ GFG(g) \circ \eta_{GF(c)} = \eta_{GF(c')}^{-1} \circ \eta_{GF(c')} \circ G(g) = G(g)$$

where we have used the result that  $GF(\eta_c) = \eta_{GF(c)}$ . Since the maps are mutual inverses we see that  $F$  is full and faithful as required.

Given  $d \in \mathcal{D}$  then  $F(G(d))$  is isomorphic to  $d$  via  $\epsilon$  so  $F$  is essentially surjective.  $\square$

**Proposition 2.6.32** (Duality)

Let  $(-)^* : \mathcal{C} \rightarrow \mathcal{C}$  be a contravariant functor such that there is a natural isomorphism

$$\eta : \mathbf{1}_{\mathcal{C}} \Rightarrow (-)^{**}$$

then  $(-)^*$  is an equivalence of categories and in particular full and faithful.

*Proof.* Define  $\epsilon = \eta^{-1}$  to determine the equivalence of categories. By the previous result then  $(-)^*$  is full and faithful.  $\square$

### 2.6.5 Properties of Morphisms

**Definition 2.6.33** (Injective, Surjective and Bijective)

Let  $(\mathcal{C}, U)$  be a *concrete category* and  $f : a \rightarrow c$  a morphism. Then we say

- $f$  is **injective** if  $U(f)$  is injective
- $f$  is **surjective** if  $U(f)$  is surjective

**Remark 2.6.34**

Note if  $f$  is both surjective and injective it need not be an isomorphism.

The concepts of monic/split-monic, epic/split-epic, iso generalize the notion of injective, surjective and bijective to general categories as we shall see.

**Definition 2.6.35** (Monomorphism)

A morphism  $f : a \rightarrow b$  is said to be a **monomorphism** (or **monic**) if

$$f \circ g_1 = f \circ g_2 \implies g_1 = g_2$$

for all  $g_1, g_2 : c \rightarrow a$ .

**Definition 2.6.36** (Epimorphism)

A morphism  $f : a \rightarrow b$  is said to be an **epimorphism** (or **epic**) if

$$g_1 \circ f = g_2 \circ f \implies g_1 = g_2$$

for all  $g_1, g_2 : b \rightarrow c$ .

**Definition 2.6.37** (Split-monic / Section)

A morphism  $f : a \rightarrow b$  is **split-monic** if it has a left inverse,  $g : b \rightarrow a$

$$g \circ f = 1_a$$

We say  $g$  is a **section** of  $f$ .

**Definition 2.6.38** (Split-epic / Retraction)

A morphism  $f : a \rightarrow b$  is **split-epic** if it has a right inverse,  $g : b \rightarrow a$

$$f \circ g = 1_b$$

We say  $g$  is a **retraction** of  $f$ .

**Proposition 2.6.39** (Split Monic  $\implies$  Monic)

For a general category  $\mathcal{C}$  we have

- $\text{split-monic} \implies \text{monic}$
- $\text{split-epic} \implies \text{epic}$

We say an **isomorphism** is a morphism with a two-sided inverse. We can refine the criteria for  $f$  to be an isomorphism using the notions just defined

**Proposition 2.6.40** (Isomorphism Criteria)

Let  $f : a \rightarrow b$  be a morphism. Then the following are equivalent

- a)  $f$  is an **isomorphism**
- b)  $f$  is both **split-epic** and **split-monic**
- c)  $f$  is **split-epic** and **monic**
- d)  $f$  is **split-monic** and **epic**

In this case a morphism  $g$  is a retraction if and only if it is a section. And such a  $g$  is unique, so we denote it by  $f^{-1}$

*Proof.* This is mostly formal

1  $\implies$  2) Clear.

2  $\implies$  3, 4) This follows from (2.6.39).

3  $\implies$  2) Suppose  $g$  is a retraction of  $f$ , that is  $fg = 1_b$ . Then  $f(gf) = (fg)f = 1_b \circ f = f = f \circ 1_a$ . As  $f$  is monic we conclude that  $gf = 1_a$  and  $g$  is a section of  $f$ .

4  $\implies$  2) Analogous

2  $\implies$  1) We've shown that any retraction is a section and vice-versa. Furthermore by monic/epic-ness a retraction or section is unique.  $\square$

### Proposition 2.6.41

For the category **Set** we have

- $\text{split-monic} \iff \text{monic} \iff \text{injective}$
- $\text{split-epic} \iff \text{epic} \iff \text{surjective}$
- $\text{isomorphism} \iff \text{bijective}$

### Definition 2.6.42 (Preserves/Reflects)

Let  $\mathcal{P}$  be a property of morphisms and  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a functor then we say

- $F$  **preserves**  $\mathcal{P}$  if ( $f$  satisfies  $\mathcal{P} \implies F(f)$  satisfies  $\mathcal{P}$ )
- $F$  **reflects**  $\mathcal{P}$  if ( $F(f)$  satisfies  $\mathcal{P} \implies f$  satisfies  $\mathcal{P}$ )

### Proposition 2.6.43

Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  be a covariant functor then

- $F$  preserves split-monic, split-epic and iso morphisms.

If in addition  $F$  is faithful then

- $F$  reflects monic and epic morphisms.

and if  $F$  is full and faithful then

- $F$  reflects split-epic, split-monic and isomorphisms.

Similar statements apply when  $F$  is contravariant.

*Proof.* The first statement is easy, for example if  $gf = 1_a$  then  $F(g) \circ F(f) = 1_{F(a)}$ .

Suppose  $F$  is faithful,  $F(f)$  is monic and  $fg_1 = fg_2$ . Then  $F(f)F(g_1) = F(f)F(g_2) \implies F(g_1) = F(g_2)$  by assumption. As  $F$  is faithful  $g_1 = g_2$  as required. The other statement is similar.

Suppose  $F$  is full and faithful and  $F(f)$  is split-monic. Then  $hF(f) = 1_{F(a)}$ . As  $F$  is full  $h = F(g)$  and  $1_{F(a)} = F(gf)$ . As  $F$  is faithful then  $gf = 1_a$ . the other statements are similar.  $\square$

### Proposition 2.6.44

Let  $(\mathcal{C}, U)$  be a *concrete category* then

- $f$  split-monic  $\implies f$  injective  $\implies f$  monic
- $f$  split-epic  $\implies f$  surjective  $\implies f$  epic
- $f$  isomorphism  $\implies f$  bijective

*Proof.* Suppose  $f$  is split-monic, then  $U(f)$  is split-monic by (2.6.43) and so by (2.6.41)  $U(f)$  is injective.

Suppose  $U(f)$  injective, then by (2.6.41)  $U(f)$  is monic. By (2.6.43)  $U$  reflects monics and so  $f$  is monic.

The other statements are similar.  $\square$

We can restate the criteria for split-epic/split-monic

### Proposition 2.6.45

Let  $f : a \rightarrow b$  be a morphism then

- $f$  is split-monic if and only if  $\text{Mor}(f, c)$  is surjective for all  $c \in \mathcal{C}$
- $f$  is epic if and only if  $\text{Mor}(f, c)$  is injective for all  $c \in \mathcal{C}$

dually

- $f$  is split-epic if and only if  $\text{Mor}(c, f)$  is surjective for all  $c \in \mathcal{C}$
- $f$  is monic if and only if  $\text{Mor}(c, f)$  is injective for all  $c \in \mathcal{C}$

*Proof.*  $f$  is epic (resp. monic) iff  $\text{Mor}(f, c)$  (resp.  $\text{Mor}(c, f)$ ) is injective precisely by the definitions.

Suppose  $f$  is split-monic, then  $gf = 1_a \implies (hg)f = h$  for any  $h$ . That is  $\text{Mor}(f, c)$  is surjective. Conversely if it's surjective then let  $c = b$  and choose  $g$  such that  $gf = \text{Mor}(f, b)(g) = 1_a$ .

A similar statement follows dually for  $f$  split-epic, and  $f$  monic.  $\square$

**Corollary 2.6.46** (Isomorphism Criteria)

Let  $f : a \rightarrow b$  be a morphism then TFAE

- $f$  is an isomorphism
- $\text{Mor}(f, c)$  is bijective for all  $c \in \mathcal{C}$
- $\text{Mor}(c, f)$  is bijective for all  $c \in \mathcal{C}$

*Proof.* This follows from combining (2.6.45) with (2.6.40).  $\square$

**Definition 2.6.47** (Conservative)

We say a functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  is **conservative** if it reflects isomorphisms.

We say a concrete category  $(\mathcal{C}, U)$  is **conservative** if the functor  $U : \mathcal{C} \rightarrow \mathbf{Set}$  is conservative.

**Proposition 2.6.48**

Let  $(\mathcal{C}, U)$  be a conservative concrete category which admits directed limits. Then the following are equivalent

- $U$  preserves directed colimits
- $U$  reflects directed colimits

**Definition 2.6.49** (Algebraic Category)

We say a concrete category  $(\mathcal{C}, U)$  is an **algebraic category** if

- $U$  reflects (and preserves) isomorphisms
- $\mathcal{C}$  has directed limits and  $U$  commutes with them

## 2.6.6 Yoneda Lemma

The motivation for the following result becomes more clear in the next section. Roughly speaking properties of an object,  $c$ , are encoded in the functor  $\text{Mor}(c, -)$ .

**Proposition 2.6.50** (Yoneda Lemma)

Suppose  $c, c' \in \mathcal{C}$  then there is a bijection between morphisms and natural transformation of functors

$$\begin{array}{ccc} \text{Mor}(c', c) & \xrightarrow{\sim} & \text{Nat}(\text{Mor}(c, -), \text{Mor}(c', -)) \\ f & \longrightarrow & \text{Mor}(f, -) : (\phi \rightarrow \phi \circ f) \\ \alpha_c(1_c) & \longleftarrow & \alpha \end{array}$$

Observe that under this correspondence

- $f = \text{Mor}(f, c)(1_c)$
- $\text{Mor}(f \circ g, -) = \text{Mor}(g, -) \circ \text{Mor}(f, -)$
- $f$  is an isomorphism  $\iff \text{Mor}(f, -)$  is a natural isomorphism.

In the latter case  $\text{Mor}(f, -)^{-1} = \text{Mor}(f^{-1}, -)$

In many cases the set of morphisms  $\text{Mor}(c, c')$  has an additional structure (typically an abelian group or module). We encode this with the following definition

**Definition 2.6.51** (Enriched Hom Functor)

Let  $(\mathcal{S}, U)$  be a **concrete category** for which  $U$  reflects isomorphisms. We say a bifunctor

$$\text{Hom} : \mathcal{C}^{op} \times \mathcal{C} \rightarrow \mathcal{S}$$

is an **enriched hom functor** if we have

- $U(\text{Hom}(c, d)) = \text{Mor}(c, d)$
- $U(\text{Hom}(f, d)) = \text{Mor}(f, d)$
- $U(\text{Hom}(c, g)) = \text{Mor}(c, g)$

and bijections

$$\begin{array}{ccc}
 & \text{Mor}(c', c) & \\
 \text{Hom}(f, -) \swarrow & & \searrow \text{Mor}(f, -) \\
 \text{Nat}(\text{Hom}(c, -), \text{Hom}(c', -)) & \xrightarrow[U]{\sim} & \text{Nat}(\text{Mor}(c, -), \text{Mor}(c', -))
 \end{array}$$

Write the inverse of  $f \rightarrow \text{Hom}(f, -)$  as  $\mathcal{Y}$ . Then for  $\alpha \in \text{Nat}(\text{Hom}(c, -), \text{Hom}(c', -))$

- a)  $\alpha$  is a natural isomorphism  $\iff U\alpha$  is a natural isomorphism  $\iff \mathcal{Y}(\alpha)$  is an isomorphism.
- b)  $\mathcal{Y}(\alpha) = U(\alpha_c)(1_c)$

Note the right hand arrow is always a bijection by Yoneda's Lemma (2.6.50) and the diagram always commutes by assumption.

*Proof.* Suppose  $\alpha$  is a natural isomorphism then  $U\alpha^{-1} \circ U\alpha = U(1_{\text{Hom}(c, -)}) = 1_{\text{Mor}(c, -)}$  and similarly  $U\alpha \circ U\alpha^{-1} = 1_{\text{Mor}(c', -)}$ , so  $U\alpha$  is a natural isomorphism. □

Note this trivially includes the usual case  $\mathcal{S} = \mathbf{Set}$ .

### 2.6.7 Representable Functors

Many algebraic constructions (e.g. tensor product) may be formalised in terms of “representable” functors (2.6.52). The usual definition involves the set-valued functor  $\text{Mor}(c, -)$  as defined in (2.6.12).

As a generalisation follows we consider an **enriched hom functor**  $\text{Hom}(-, -)$  taking values in the concrete category  $(\mathcal{S}, U)$  where  $U$  reflects isomorphisms. This clearly generalises the usual case.

**Definition 2.6.52** (Representable Functor)

Let  $F : \mathcal{C} \rightarrow \mathcal{S}$  a covariant functor. We say  $F$  is **representable** if there is a pair  $(x, \Phi)$  where  $x \in \mathcal{C}$  and  $\Phi$  is a natural isomorphism

$$\text{Hom}(x, -) \xrightarrow[\sim]{\Phi} F(-)$$

We say  $F$  is **represented** by the pair  $(x, \Phi)$ . Note this implies  $U \circ F$  is represented by  $U\Phi$ .

It is often useful to have another conception of representable functor in terms of universal properties as this is more intuitive for applications.

**Definition 2.6.53** (Universal Element)

Let  $F : \mathcal{C} \rightarrow \mathcal{S}$  be a covariant functor, then we say that a pair  $(x, i)$ , where  $x \in \mathcal{C}$ ,  $i \in (U \circ F)(x)$ , is a **universal element** for  $F$  if for all  $c \in \mathcal{C}$  there are bijections

$$\begin{array}{ccc}
 \text{Mor}(x, c) & \longrightarrow & (U \circ F)(c) \\
 f & \longrightarrow & (U \circ F)(f)(i)
 \end{array} \tag{2.10}$$

In this case we may also say  $F$  is **represented** by the universal element  $(x, i)$ .

We show the equivalence of the two notions by some abstract nonsense

**Proposition 2.6.54** (Representable Set-Valued Functor = Universal Element)

Let  $F : \mathcal{C} \rightarrow \mathbf{Set}$  be a covariant functor then there is a natural correspondence between representations and universal elements

$$\begin{array}{ccc}
 \{ \text{representations of } F \} & \longleftrightarrow & \{ \text{universal elements of } F \} \\
 (x, \Phi) & \longrightarrow & (x, \Phi(1_x)) \\
 (x, f \rightarrow F(f)(i)) & \longleftarrow & (x, i)
 \end{array}$$

The following may in particular be used to show that representations are unique up to isomorphism.

**Proposition 2.6.55** (Morphisms Between Representations)

Suppose  $(x, \Phi)$  and  $(x', \Phi')$  represent the covariant functors  $F, F' : \mathcal{C}^{op} \rightarrow \mathcal{S}$  respectively. Then there are bijections

$$\begin{array}{ccccc} \text{Mor}(x, x') & \xrightarrow{\sim} & \text{Nat}(\text{Hom}(x', -), \text{Hom}(x, -)) & \xrightarrow{\sim} & \text{Nat}(F'(-), F(-)) \\ f & \rightarrow & \text{Hom}(f, -) & \rightarrow & \widehat{f} \\ U(\eta_{x'})(1_{x'}) & \leftarrow & \eta & \leftarrow & \alpha \end{array}$$

where we define the mutual inverses

$$\begin{aligned} \widehat{f} &= \Phi \circ \text{Hom}(f, -) \circ (\Phi')^{-1} \\ \alpha^* &= U(\Phi_{x'}^{-1} \circ \alpha_{x'} \circ \Phi'_{x'})(1_{x'}) \end{aligned}$$

Note this has the following properties

- $\widehat{1_x} = \text{id}_F$
- $\widehat{g \circ f} = \widehat{f} \circ \widehat{g}$  where  $g : x' \rightarrow x''$  is morphism and  $(x'', \Phi'')$  represents  $F''$
- $f$  is an isomorphism  $\iff \widehat{f}$  is a natural isomorphism and in this case  $\widehat{f^{-1}} = \widehat{f}^{-1}$ .

Further  $\alpha^* \in \text{Mor}(x, x')$  is the unique morphism such that the following diagram of natural transformations commutes

$$\begin{array}{ccc} \text{Hom}(x', -) & \xrightarrow{\Phi'} & F'(-) \\ \text{Hom}(\alpha^*, -) \downarrow & & \downarrow \alpha \\ \text{Hom}(x, -) & \xrightarrow{\Phi} & F(-) \end{array}$$

and satisfies

- $\text{id}_F^* = 1_x$
- $(\beta \circ \alpha)^* = \alpha^* \circ \beta^*$
- $\alpha^*$  is an isomorphism  $\iff \alpha$  is a natural isomorphism

*Proof.* The first bijection is simply the Yoneda Lemma and the second bijection is obvious.  $\square$

**Corollary 2.6.56** (Representation is Unique)

Let  $F : \mathcal{C} \rightarrow \mathcal{S}$  be a covariant functor which is represented by pairs  $(x, \Phi)$  and  $(x', \Phi')$ . Then they are isomorphic with two-sided inverses

$$x \xrightleftharpoons[U(\Phi'^{-1})(i)]{U(\Phi^{-1})(i')} x'$$

where  $i := (U\Phi)(1_x)$  and  $i' := (U\Phi')(1_{x'})$ .

*Proof.* We may apply the previous result with  $F = F'$  and the identity natural transformation  $\text{id}_F$ .  $\square$

In practice we typically have a family of representations, and we want to show the construction is functorial

**Definition 2.6.57** (Representable Bifunctor)

Let  $F : \mathcal{D}^{op} \times \mathcal{C} \rightarrow \mathcal{S}$  be a bifunctor. We say that it is **representable** by  $(G, \Phi)$  where  $G : \mathcal{D} \rightarrow \mathcal{C}$  is a functor, if  $\Phi$  is a natural isomorphism of bifunctors

$$\Phi : \text{Hom}(G(-), -) \xrightarrow{\sim} F(-, -)$$

The following shows we only need to construct the representation pointwise.

**Proposition 2.6.58**

Let  $F : \mathcal{D}^{op} \times \mathcal{C} \rightarrow \mathcal{S}$  be a bifunctor such that  $F(d, -)$  is representable for all  $d \in \mathcal{D}$ , by  $(G(d), \Phi_d)$ . Then  $G$  can be made into a covariant functor uniquely such that  $\Phi$  constitutes a natural isomorphism of bifunctors

$$\Phi_{d,c} : \text{Hom}(G(d), c) \xrightarrow{\sim} F(d, c)$$

Explicitly for any morphism  $h : d \rightarrow d'$  we have the identity

$$G(h) = \Phi_{d, G(d')}^{-1} (\Phi_{d', G(d')} (1_{G(d')}) \circ h)$$

*Proof.* It is straightforward to verify that for any covariant functor  $G$ , the mapping  $\text{Mor}(G(-), -)$  is a bifunctor (contravariant in the first “slot”).

Let  $h : d \rightarrow d'$  be a morphism then by (2.6.24)  $F(h, -) : F(d', -) \rightarrow F(d, -)$  is a natural transformation. Therefore we may define

$$G(h) := F(h, -)^*$$

in the notation of (2.6.55); it is the unique morphism making the following diagram commute

$$\begin{array}{ccc} \text{Hom}(G(d'), -) & \xrightarrow{\Phi_{d'}} & F(d', -) \\ \text{Hom}(G(h), -) \downarrow \Downarrow & & \downarrow F(h, -) \\ \text{Hom}(G(d), -) & \xrightarrow{\Phi_d} & F(d, -) \end{array} \quad (2.11)$$

Chasing  $1_{G(d')}$  gives the identity for  $G(h)$ .

Furthermore by uniqueness  $G(h \circ k) = F(h \circ k, -)^* = (F(k, -) \circ F(h, -))^* = F(h, -)^* \circ F(k, -)^* = G(h) \circ G(k)$ . We conclude by (2.6.25) that  $\Phi$  is a natural transformation of bifunctors.  $\square$

**Example 2.6.59** (Concrete Interpretation)

Let  $\mathcal{C}, \mathcal{D}$  be concrete categories and consider the following bifunctor

$$\begin{aligned} F & : \quad \mathcal{D}^{op} \times \mathcal{C} \rightarrow \mathbf{Set} \\ F(d, c) & := \{ \phi : d \rightarrow c \mid P(\phi) \} \end{aligned}$$

where  $P(-)$  is some predicate such that  $P(\phi) \implies P(g \circ \phi \circ f)$ . Then the universal element is simply an object  $G(d) \in \mathcal{C}$  together with a mapping

$$i_d : d \rightarrow G(d)$$

satisfying  $P$  such that there is a bijection

$$\begin{aligned} \text{Mor}(G(d), c) & \xrightarrow{\sim} F(d, c) := \{ f : d \rightarrow c \mid P(f) \} \\ \phi & \longrightarrow \phi \circ i_d \end{aligned}$$

The functoriality of  $G$  corresponds to a commutative diagram

$$\begin{array}{ccc} d & \xrightarrow{f} & d' \\ \downarrow i_d & & \downarrow i'_d \\ G(d) & \xrightarrow{G(f)} & G(d') \end{array}$$

where  $G(f)$  is the unique morphism making the diagram commute.

**Proposition 2.6.60** (Functorial Yoneda Lemma)

Let  $G, G' : \mathcal{D} \rightarrow \mathcal{C}$  be functors. Then there is a bijection

$$\begin{array}{ccc} \text{Nat}(G(-), G'(-)) & \xrightarrow{\sim} & \text{Nat}(\text{Hom}(G'(-), -) \text{Hom}(G(-), -)) \\ f_d & \rightarrow & \text{Hom}(f_d, 1_c) \\ U(\eta_{d, G'(d)})(1_{G'(d)}) & \leftarrow & \eta_{d, c} \end{array}$$

We have the following properties

- $\widehat{\text{id}_G} = \text{id}$
- $\widehat{g \circ f} = \widehat{f} \circ \widehat{g}$
- $f$  is a natural isomorphism iff  $\text{Hom}(f, -)$  is a natural isomorphism

*Proof.* In order to demonstrate that  $\text{Hom}(f_d, 1_c)$  is a natural transformation of bifunctors it's enough by (2.6.24) to show the following diagram commutes for  $h : d' \rightarrow d$

$$\begin{array}{ccc}
\mathrm{Hom}(G'(d), c) & \xrightarrow{\circ f_d} & \mathrm{Hom}(G(d), c) \\
\downarrow \circ G'(h) & & \downarrow \circ G(h) \\
\mathrm{Hom}(G'(d'), c) & \xrightarrow{\circ f_{d'}} & \mathrm{Hom}(G(d'), c)
\end{array}$$

However  $f_d \circ G(h) = G'(h) \circ f_{d'}$  by definition of natural transformation, from which it follows immediately. The map is a pointwise bijection (for every  $d \in D$ ) with the inverse given by the usual Yoneda Lemma.

The first two properties are straightforward. From these it follows that if  $f$  is a natural isomorphism then so is  $\mathrm{Hom}(f, -)$ . Conversely if  $\mathrm{Hom}(f, -)$  is a natural isomorphism then it has a two-sided inverse which is of the form  $\mathrm{Hom}(g, -)$ . It's then straightforward to argue that  $g$  is a two-sided inverse of  $f$ . □

### 2.6.8 Adjoint Functors

Some universal constructions may be expressed as an adjoint pair of functors. Using this concept we can simplify the verification of universal properties by appealing to general criteria for adjoint functors as below.

**Definition 2.6.61** (Adjoint Pair)

Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  and  $G : \mathcal{D} \rightarrow \mathcal{C}$  be functors. We say that  $F$  is **left adjoint** to  $G$  if there is a bijection

$$\psi_{c,d} : \mathrm{Mor}(F(c), d) \longrightarrow \mathrm{Mor}(c, G(d))$$

which is natural in  $c$  and  $d$  in the following sense. Let  $\alpha : c' \rightarrow c$ ,  $\beta : d \rightarrow d'$ , then for all  $f : F(c) \rightarrow d$  we have

$$\psi_{c',d'}(\beta \circ f \circ F(\alpha)) = G(\beta) \circ \psi_{c,d}(f) \circ \alpha \quad (2.12)$$

or equivalently for all  $g : c \rightarrow G(d)$

$$\beta \circ \psi_{c,d}^{-1}(g) \circ F(\alpha) = \psi_{c',d'}^{-1}(G(\beta) \circ g \circ \alpha) \quad (2.13)$$

Equivalently  $F(-)$  represents the bifunctor  $\mathrm{Mor}(-, G(-))$  (2.6.57).

**Proposition 2.6.62** (Adjoint  $\implies$  unit, counit)

Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  and  $G : \mathcal{D} \rightarrow \mathcal{C}$  be adjoint functors with relationship

$$\psi_{c,d} : \mathrm{Mor}(F(c), d) \xrightarrow{\sim} \mathrm{Mor}(c, G(d))$$

Then we have two natural transformations (**unit** and **counit** respectively)

$$\begin{aligned}
\eta : \mathbf{1} &\Rightarrow G \circ F \\
\epsilon : F \circ G &\Rightarrow \mathbf{1}
\end{aligned}$$

defined by

$$\begin{aligned}
\eta_c &= \psi_{c,F(c)}(1_{F(c)}) \\
\epsilon_d &= \psi_{G(d),d}^{-1}(1_{G(d)})
\end{aligned}$$

Furthermore we may recover the adjoint relationship via

$$\begin{aligned}
\psi_{c,d}(f) &= G(f) \circ \eta_c \\
\psi_{c,d}^{-1}(g) &= \epsilon_d \circ F(g)
\end{aligned}$$

*Proof.* We show that the transformations given are natural. Suppose  $\alpha : c \rightarrow c'$  then

$$\begin{aligned}
G(F(\alpha)) \circ \eta_c &= G(F(\alpha)) \circ \psi_{c,F(c)}(1_{F(c)}) \\
&= \psi_{c,F(c')} (F(\alpha) \circ 1_{F(c)}) \quad (2.12) \\
&= \psi_{c,F(c')} (1_{F(c')} \circ F(\alpha)) \\
&= \psi_{c',F(c')} (1_{F(c')}) \circ \alpha \quad (2.12) \\
&= \eta_{c'} \circ \alpha
\end{aligned}$$

so  $\eta$  is a natural transformation. Furthermore

$$\psi_{c,d}(f) = \psi_{c,d}(f \circ 1_{F(c)}) = G(f) \circ \psi_{c,F(c)}(1) = G(f) \circ \eta_c$$

as required. Similarly for  $\beta : d \rightarrow d'$

$$\begin{aligned}
 \beta \circ \epsilon_d &= \beta \circ \psi_{G(d),d}^{-1}(1_{G(d)}) \\
 &= \psi_{G(d),d'}^{-1}(G(\beta) \circ 1_{G(d)}) \quad (2.13) \\
 &= \psi_{G(d),d'}^{-1}(1_{G(d')} \circ G(\beta)) \\
 &= \psi_{G(d'),d'}^{-1}(1_{G(d')} \circ F(G(\beta))) \quad (2.13) \\
 &= \epsilon_{d'} \circ F(G(\beta))
 \end{aligned}$$

□

Given two natural transformations we may recover a corresponding adjoint

**Proposition 2.6.63** (Adjoint from unit and counit)

Let  $F : \mathcal{C} \rightarrow \mathcal{D}$  and  $G : \mathcal{D} \rightarrow \mathcal{C}$  be functors with two natural transformations

$$\begin{aligned}
 \eta : \mathbf{1} &\Rightarrow G \circ F \\
 \epsilon : F \circ G &\Rightarrow \mathbf{1}
 \end{aligned}$$

Then the following are equivalent

- a)  $F$  is left adjoint to  $G$  with unit and counit  $\eta$ ,  $\epsilon$ , and
- b) the so-called **triangular identities** are satisfied

$$1_{G(d)} : G(d) \xrightarrow{\eta_{G(d)}} GFG(d) \xrightarrow{G(\epsilon_d)} G(d) \quad (2.14)$$

$$1_{F(c)} : F(c) \xrightarrow{F(\eta_c)} FGF(c) \xrightarrow{\epsilon_{F(c)}} F(c) \quad (2.15)$$

More precisely the adjunction is given by

$$\begin{array}{ccc}
 \text{Mor}(F(c), d) & \xleftarrow[\psi]{\phi} & \text{Mor}(c, G(d)) \\
 f & \longrightarrow & G(f) \circ \eta_c \\
 \epsilon_d \circ F(g) & \longleftarrow & g
 \end{array}$$

*Proof.* Let  $\psi, \phi$  denote the proposed adjunction maps. We will use the triangular identities to show that these are mutually inverse. First observe by naturality of  $\eta$  and  $\epsilon$  that

$$\psi\phi(g) = G(\epsilon_d \circ F(g)) \circ \eta_c = G(\epsilon_d) \circ \eta_{G(d)} \circ g \quad (2.16)$$

$$\phi\psi(f) = \epsilon_d \circ F(G(f) \circ \eta_c) = f \circ \epsilon_{F(c)} \circ F(\eta_c) \quad (2.17)$$

It's then immediate that these are mutually inverse maps if and only if the triangular identities are satisfied (one way is obvious, the other way consider  $f = 1_{F(c)}$  and  $g = 1_{G(d)}$ ).

Further one may easily verify that  $\psi, \phi$  so-defined are natural in  $c$  and  $d$

$$\begin{aligned}
 \psi(\beta \circ f \circ F(\alpha)) &= G(\beta) \circ G(f) \circ GF(\alpha) \circ \eta_c \\
 &= G(\beta) \circ G(f) \circ \eta_{c'} \circ \alpha \\
 &= G(\beta) \circ \psi(f) \circ \alpha
 \end{aligned}$$

□

**Proposition 2.6.64** (Criteria for right adjoint to be full and faithful)

Let  $\mathcal{C} \xrightleftharpoons[G]{F} \mathcal{D}$  be adjoint functors with  $\eta, \epsilon$  unit and counit transformations. Then

- $G$  is faithful if and only if  $\epsilon$  is pointwise epic
- $G$  is full if and only if  $\epsilon$  is pointwise split-monic
- $G$  is full and faithful if and only if  $\epsilon$  is a pointwise isomorphism

*Proof.* Consider the composite map

$$\text{Mor}(d', d) \xrightarrow{\text{Mor}(\epsilon_{d'}, d)} \text{Mor}(F(G(d')), d) \xrightarrow{\psi_{G(d'), d}} \text{Mor}(G(d'), G(d))$$

which is natural in  $d$  and  $d'$ . Note that image of  $\alpha \in \text{Mor}(d', d)$  is

$$\psi_{G(d'), d}(\alpha \circ \epsilon_{d'}) = G(\alpha) \circ \psi_{G(d'), d'}(\epsilon_{d'}) = G(\alpha)$$

so the composite is just  $G(-)$ . The second map is bijective by the adjoint assumption. Therefore the first map is injective (resp. surjective) if and only if  $G$  is faithful (resp. full).

By (2.6.45)  $\text{Mor}(\epsilon_{d'}, d)$  is injective (resp. surjective) for all  $d, d'$  if and only if  $\epsilon_{d'}$  is epic (resp. split-monic) for all  $d'$ .

Then the first two statements follow easily. The last statement follows from the previous two, combined with (2.6.40).  $\square$

**Proposition 2.6.65** (Reflective Sub-Category)

Let  $\mathcal{C} \xrightleftharpoons[F]{G} \mathcal{D}$  be adjoint functors such that  $G$  is fully faithful. Then for  $c \in \mathcal{C}$  we have  $\eta_c$  is an isomorphism iff  $c$  is in the essential image of  $G$  (i.e. is isomorphic to  $G(d)$  for some  $d$ ).

In particular let  $\mathcal{C}'$  denote the **essential image** of  $\mathcal{D}$  under  $G$ . Then this restricts to an equivalence of categories

$$\mathcal{C}' \xrightleftharpoons[F]{G} \mathcal{D}$$

*Proof.* If  $\eta_c : c \rightarrow GF(c)$  is an isomorphism define  $d := F(c)$ , and we are done.

Conversely we first show that  $\eta_{G(d)}$  is an isomorphism. By the triangular identities (2.6.63) we have

$$G(\epsilon_d) \circ \eta_{G(d)} = 1_{G(d)}$$

so  $\eta_{G(d)}$  has a left inverse. Recall by (2.6.64)  $\epsilon_d$  is an isomorphism, whence so is  $G(\epsilon_d)$  (2.6.43). Therefore we may show directly that it is also a right inverse, and therefore  $\eta_{G(d)}$  is an isomorphism.

Now assume that we have an isomorphism  $\psi : c \xrightarrow{\sim} G(d)$  for some  $d \in \mathcal{D}$ . Then by naturality we have a commutative diagram

$$\begin{array}{ccc} c & \xrightarrow[\sim]{\psi} & G(d) \\ \eta_c \downarrow & & \sim \downarrow \eta_{G(d)} \\ GF(c) & \xrightarrow[\sim]{GF(\psi)} & GF(G(d)) \end{array}$$

observing that  $GF(\psi)$  is also an isomorphism. Then we conclude  $\eta_c$  is an isomorphism as required.  $\square$

The following criteria will be useful

**Proposition 2.6.66** (Adjoint can be constructed pointwise)

Let  $G : \mathcal{D} \rightarrow \mathcal{C}$  be a functor. Suppose that for all  $c \in \text{ob}(\mathcal{C})$  there exists  $F(c) \in \text{ob}(\mathcal{D})$  such that there is a bijection natural in  $d$

$$\psi_{c,d} : \text{Mor}(F(c), d) \xrightarrow{\sim} \text{Mor}(c, G(d))$$

Then there exists a unique functor  $F$  which ensures this is natural in  $c$ , more precisely for  $g : c \rightarrow c'$  this is given by

$$\begin{aligned} F(g) &= \psi_{c,F(c')}^{-1}(\eta_{c'} \circ g) \\ \eta_c &:= \psi_{c,F(c)}(1_{F(c)}) \end{aligned}$$

This is also the unique functor which ensures  $\eta_c$  is a natural transformation.

*Proof.* The hypothesis is precisely that  $(\psi_{c,-}, F(c))$  represents the functor  $d \rightarrow \text{Mor}(c, G(d))$ . Therefore the first uniqueness follows from representability of bifunctors (2.6.58).

We have already shown that if  $\psi_{c,d}$  is natural in  $c$  then  $\eta_c$  is a natural transformation (...). Conversely suppose that  $\eta_c$  is a natural transformation, since we have assumed that  $\psi_{c,d}$  is natural in  $d$ , this means  $\psi_{c,F(c')}(F(g)) = GF(g) \circ \eta_c = \eta_{c'} \circ g \implies F(g) = \psi_{c,F(c')}^{-1}(\eta_{c'} \circ g)$ .  $\square$

**Proposition 2.6.67**

Let  $\mathcal{C} \xrightleftharpoons[F]{G} \mathcal{D}$  be a pair of adjoint functors. Then  $F$  preserves initial objects and  $G$  preserves terminal objects.

If  $\mathcal{C}, \mathcal{D}$  are pointed categories then  $F$  and  $G$  also preserve zero objects and zero morphisms.

*Proof.* We only prove that  $F$  preserves initial objects, as the second statement follows dually. By hypothesis there is a bijection natural in  $d$

$$\text{Mor}(F(\emptyset), d) \xrightarrow{\sim} \text{Mor}(\emptyset, G(d))$$

Therefore if  $\emptyset$  is initial, then so is  $F(\emptyset)$ .

The last statement follows from (2.6.19) and (2.6.20). □

**2.6.9 Limits****Definition 2.6.68**

A functor  $F : I \rightarrow \mathcal{C}$  where  $I$  is a small category is referred to as a **diagram**

**Definition 2.6.69** (Cones)

Let  $F : I \rightarrow \mathcal{C}$  be a diagram. We say that the pair  $(L, \{\lambda_i : L \rightarrow F(i)\}_{i \in I})$  is a **cone** over  $F$  if

$$F(\alpha) \circ \lambda_i = \lambda_j \text{ for all morphisms } \alpha : i \rightarrow j$$

The cones for  $F$  constitute a **cone category**  $(\Delta \downarrow F)$  where a morphism  $\alpha : (L, \{\lambda_i\}_{i \in I}) \rightarrow (L', \{\lambda'_i\}_{i \in I})$  is just a morphism  $\alpha : L \rightarrow L'$  such that  $\lambda'_i \circ \alpha = \lambda_i$  for all  $i \in I$

**Definition 2.6.70** (Small Limit)

Let  $F : I \rightarrow \mathcal{C}$  be a diagram. A cone  $(L, \{\lambda_i\}_{i \in I})$  over  $F$  is a **limit** for  $F$  if it is a **terminal object** (2.6.16) in the cone category.

Explicitly for every cone  $(L', \{\lambda'_i\}_{i \in I})$  there is a unique morphism  $\alpha : L' \rightarrow L$  such that  $\lambda_i \circ \alpha = \lambda'_i$ .

We may denote this as  $\lim F$ .

**Proposition 2.6.71** (Product)

Let  $I$  be a discrete small category and  $\{c_i\}_{i \in I}$  be a family of objects in  $\mathcal{C}$ . Then there is a unique functor  $F : I \rightarrow \mathcal{C}$  such that  $F(i) = c_i$ .

Consider an object  $c \in \mathcal{C}$  and family of morphisms  $\pi_i : c \rightarrow c_i$ , then the following are equivalent

- a) The cone  $(c, \{\pi_i\}_{i \in I})$  is a limit for  $F$
- b) For any family of morphisms  $\lambda_i : c' \rightarrow c_i$  there exists a unique  $\lambda : c' \rightarrow c$  such that  $\pi_i \circ \lambda = \lambda_i$
- c) For all objects  $c'$  the map

$$\begin{aligned} \Phi : \text{Mor}(c', c) &\longrightarrow \prod_{i \in I} \text{Mor}(c', c_i) \\ f &\longmapsto \pi_i \circ f \end{aligned}$$

is a bijection, i.e.  $(c, \Phi)$  represents the contravariant functor  $\prod_{i \in I} \text{Mor}(-, c_i)$ .

In this case we say that  $(c, \{\pi_i\}_{i \in I})$  is a **product** for  $\{c_i\}_{i \in I}$  and denote this by  $\prod_{i \in I} c_i$ . Furthermore we may write the induced map  $\lambda$  as  $(\lambda_i)_{i \in I} : c' \rightarrow \prod_{i \in I} c_i$ .

**Proposition 2.6.72** (Functoriality of Product)

Let  $I$  be a discrete small category and suppose that  $\mathcal{C}$  has  $I$ -shaped limits. Then we may define a map

$$\prod_{i \in I} (-) : \mathcal{C}^I \rightarrow \mathcal{C}$$

Recall that a morphism  $\prod_{i \in I} c_i \rightarrow \prod_{i \in I} d_i$  in  $\mathcal{C}^I$  is simply a family of morphisms  $(f_i : c_i \rightarrow d_i)_{i \in I}$  in  $\mathcal{C}$ . Then we define

$$\prod_{i \in I} f_i := (f_i \circ \pi_i)_{i \in I}$$

which is the unique morphism satisfying

$$\pi_i \circ \prod_{i \in I} f_i = f_i \circ \pi_i$$

This makes  $\prod_{i \in I} (-)$  into a functor.

**Proposition 2.6.73** (Functoriality of Binary Product)

Let  $c, c', d, d' \in \mathcal{C}$  be such that  $c \times d$  and  $c' \times d'$  exists. For  $f : c \rightarrow c'$  and  $g : d \rightarrow d'$  there exists a unique morphism

$$f \times g : c \times d \rightarrow c' \times d'$$

such that

$$\pi_{c'} \circ (f \times g) = f \circ \pi_c$$

and

$$\pi_{d'} \circ (f \times g) = g \circ \pi_d$$

If  $\mathcal{C}$  has all binary products then this determines a bifunctor

$$\times : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$$

*Proof.* We have

$$(f \circ \pi_c, g \circ \pi_d) \in \text{Mor}(c \times d, c') \times \text{Mor}(c \times d, d')$$

By the universal property of  $c' \times d'$  we have a morphism

$$(f \times g) \in \text{Mor}(c \times d, c' \times d')$$

satisfying the required properties.

The uniqueness shows that we have functoriality, namely

$$(h \times k) \circ (f \times g) = (h \circ f) \times (k \circ g)$$

□

**Proposition 2.6.74** (Limits exist in Set)

Limits exist in **Set** more precisely for a diagram  $F : I \rightarrow \mathbf{Set}$

$$\lim F = \{(x_i)_{i \in I} \in \prod_{i \in I} F(i) \mid F(\alpha)(x_i) = x_j \text{ for all } \alpha : i \rightarrow j\}$$

**Definition 2.6.75**

Let  $U : \mathcal{C} \rightarrow \mathcal{D}$  be a functor and  $I$  a small category.

We say that  $U$  **creates limits** of shape  $I$  if for all diagrams  $F : I \rightarrow \mathcal{C}$  such that  $\lim U \circ F$  exists then  $\lim F$  exists and is the unique cone such that  $U(\lim F) = \lim(U \circ F)$ .

We say that  $U$  **preserves limits** of shape  $I$  if whenever  $\lim F$  exists in  $\mathcal{C}$  we have  $\lim U \circ F$  exists in  $\mathcal{D}$  and is equal to  $U(\lim F)$ .

If we restrict to the case  $I$  is discrete (resp.  $I = \mathbf{2}$ ) then we say that  $U$  creates/preserves products (resp. equalisers).

**Proposition 2.6.76** (Creates  $\implies$  Preserves)

Let  $U : \mathcal{C} \rightarrow \mathcal{D}$  which creates limits of shape  $I$  and  $\mathcal{D}$  has all limits of shape  $I$ . Then  $U$  preserves limits of shape  $I$ .

In particular this holds unconditionally for  $\mathcal{D} = \mathbf{Set}$  since this has all limits.

**2.6.10 Direct Limit / Filtered Colimit**

Here we consider only a certain type of colimit, which is referred to variously as **direct limit** [AM69], **limite inductive** [Gro64] or **filtered colimit** [ML98]. We also restrict to the case where the indexing category is “thin”, which is more properly referred to as a directed colimit.

**Definition 2.6.77** (Directed Poset)

We say a category  $I$  is a **directed poset** if it is small, thin and filtered, that is for any  $i, j \in I$

- there exists at most one morphism  $i \rightarrow j$ , and
- there exists  $k \in I$  and morphisms  $i \rightarrow k$  and  $j \rightarrow k$

If there exists a morphism  $i \rightarrow j$  then we write  $i \prec j$ .

**Example 2.6.78**

A typical example is any family of subsets of a set  $X$  ordered by reversion inclusion which are closed under finite intersection.

**Definition 2.6.79** (Direct limit / Directed colimit)

Let  $I$  be a directed poset and  $F : I \rightarrow \mathcal{C}$  a diagram.

A **cocone** over  $F$  is a pair  $(L, \{\mu : F(i) \rightarrow L\}_{i \in I})$  such that

$$i \prec j \implies \mu_j \circ F(i \rightarrow j) = \mu_i$$

The cocones over  $F$  constitute a category  $(F \downarrow \Delta)$  where morphisms  $\alpha : (L, \mu_i) \rightarrow (L', \mu'_i)$  are simply morphisms  $\alpha : L \rightarrow L'$  such that  $\mu'_i \circ \alpha = \mu_i$  for all  $i \in I$ .

A cone  $(L, \{\mu_i\}_{i \in I})$  over  $F$  is a **direct limit** for  $F$  if it is an **initial object** (2.6.16) in the cocone category  $(F \downarrow \Delta)$ .

Explicitly for every cocone  $(L', \{\mu'_i\}_{i \in I})$  there is a unique morphism  $\alpha : L \rightarrow L'$  such that  $\mu'_i \circ \alpha = \mu_i$  for all  $i \in I$ .

We denote this by  $\varinjlim_{i \in I} F(i)$ .

**Definition 2.6.80** (Cofinal subposet)

Let  $I$  be a directed poset. Then a full subcategory  $I' \subset I$  is said to be **cofinal** if

$$\forall i \in I \quad \exists i' \in I' \text{ s.t. } i \leq i'$$

In this case  $I'$  is also directed.

**Proposition 2.6.81** (Cofinal Direct Limit)

Let  $I$  be a directed poset and  $I'$  a full subcategory which is also directed. Suppose  $F : I \rightarrow \mathcal{C}$  is a diagram then there is a unique morphism of cocones

$$\varinjlim_{i \in I'} F(i) \longrightarrow \varinjlim_{i \in I} F(i)$$

provided both these exist.

It is an isomorphism if and only if  $I'$  is cofinal in  $I$ .

**Proposition 2.6.82** (Direct Limits of Sets)

Let  $I$  be a directed poset and  $F : I \rightarrow \mathbf{Set}$  a diagram. Then we may construct

$$\varinjlim_{i \in I} F_i := \{(i, x) \mid i \in I, x \in F(i)\} / \sim$$

where we consider the equivalence relation

$$(i, x) \sim (j, y)$$

if for some  $k \in I$  we have  $i \prec k, j \prec k$  and  $F(i \rightarrow k)(x) = F(j \rightarrow k)(y)$ .

**2.6.11 Equaliser and Coequaliser****Definition 2.6.83** (Equaliser)

Consider a commutative diagram in a category  $\mathcal{C}$

$$a \xrightarrow{e} b \xrightleftharpoons[\beta]{\alpha} c$$

such that  $\alpha \circ e = \beta \circ e$ . We say that  $e$  is an **equaliser** for  $(\alpha, \beta)$  if for all  $e' : a' \rightarrow b$  such that  $\alpha \circ e' = \beta \circ e'$  there is a unique factorisation

$$\begin{array}{ccc} a & \xrightarrow{e} & b \xrightleftharpoons[\beta]{\alpha} c \\ \uparrow & \nearrow e' & \\ a' & & \end{array}$$

This is equivalent to  $(a, e)$  being a limit for the obvious functor  $F : \mathbf{2} \rightarrow \mathcal{C}$  where  $\mathbf{2}$  denotes the category  $\bullet \rightrightarrows \bullet$ .

**Proposition 2.6.84**

Every equaliser is a monomorphism.

*Proof.* Suppose that  $(a, e)$  is an equaliser for  $\alpha, \beta : b \rightrightarrows c$ . Suppose that  $e \circ f = e \circ g = h$  then evidently  $\alpha \circ h = \beta \circ h$  so there is a unique morphism  $k$  such that  $e \circ k = h$ , whence  $k = f = g$ .  $\square$

**Proposition 2.6.85**

A diagram in  $\mathbf{Set}$

$$X \xrightarrow{i} Y \xrightleftharpoons[\beta]{\alpha} Z$$

is an equaliser if and only if  $i$  is injective and

$$\text{Im}(i) = \{y \in Y \mid \alpha(y) = \beta(y)\}$$

**Definition 2.6.86** (Coequaliser)

Consider a commutative diagram in a category  $\mathcal{C}$

$$b \xrightarrow[\beta]{\alpha} c \xrightarrow{\pi} a$$

such that  $\pi \circ \alpha = \pi \circ \beta$ . We say that  $\pi$  is a **co-equaliser** for  $(\alpha, \beta)$  if for all  $\pi' : c \rightarrow a'$  such that  $\pi' \circ \alpha = \pi' \circ \beta$  there is a unique factorisation

$$\begin{array}{ccccc} b & \xrightarrow[\beta]{\alpha} & c & \xrightarrow{\pi} & a \\ & & & \searrow \pi' & \downarrow \text{---} \\ & & & & a' \end{array}$$

This is equivalent to  $(a, \pi)$  being a colimit for the obvious functor  $F : \mathbf{2} \rightarrow \mathcal{C}$  where  $\mathbf{2}$  denotes the category  $\bullet \rightrightarrows \bullet$ .

**Proposition 2.6.87**

The following diagram is a coequaliser

$$b \xrightarrow[\beta]{\alpha} c \xrightarrow{\pi} a$$

if and only if the following is an equaliser in **Set** for all  $z \in \mathcal{C}$

$$\text{Mor}(a, z) \xrightarrow{\text{Mor}(\pi, z)} \text{Mor}(c, z) \xrightarrow[\text{Mor}(\beta, z)]{\text{Mor}(\alpha, z)} \text{Mor}(b, z)$$

*Proof.* We see from the definition that  $\pi$  is a coequaliser for  $(\alpha, \beta)$  if and only if  $\text{Mor}(\pi, z)$  is injective and

$$\text{Im}(\text{Mor}(\pi, z)) = \{\pi' : c \rightarrow z \mid \pi' \circ \alpha = \pi' \circ \beta\}$$

which is exactly the criteria (2.6.85) for an equaliser in **Set**. □

**Proposition 2.6.88** (Left Adjoints Preserve Coequalisers)

Let  $\mathcal{C} \xrightleftharpoons[G]{F} \mathcal{D}$  a pair of adjoint functors. Then  $F$  preserves coequalisers and  $G$  preserves equalisers.

*Proof.* Consider a coequaliser  $b \xrightarrow[\beta]{\alpha} c \xrightarrow{\pi} a$ . By the adjoint relationship we have a commutative diagram for all  $d \in \mathcal{D}$

$$\begin{array}{ccccc} \text{Mor}(F(c), d) & \xrightarrow{\text{Mor}(F(\pi), d)} & \text{Mor}(F(b), d) & \xrightarrow[\text{Mor}(F(\beta), d)]{\text{Mor}(F(\alpha), d)} & \text{Mor}(F(a), d) \\ \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\ \text{Mor}(c, G(d)) & \xrightarrow{\text{Mor}(\pi, G(d))} & \text{Mor}(b, G(d)) & \xrightarrow[\text{Mor}(\beta, G(d))]{\text{Mor}(\alpha, G(d))} & \text{Mor}(a, G(d)) \end{array}$$

By (2.6.87) the bottom row is an equaliser, and so by the isomorphisms so is the top row. Therefore we conclude by the same result that the following is an equaliser as required

$$F(b) \xrightarrow[F(\beta)]{F(\alpha)} F(c) \xrightarrow{F(\pi)} F(a)$$

□

## 2.6.12 Kernels and Cokernels

**Proposition 2.6.89** (Kernel)

Let  $\mathcal{C}$  be a pointed category (2.6.18). Suppose we have a sequence

$$\ker(\alpha) \xrightarrow{i} c \xrightarrow{\alpha} d$$

then the following are equivalent

- a)  $i$  is the equaliser of  $(\alpha, 0)$

b)  $\alpha \circ i = 0$  and  $\alpha \circ i' = 0 \implies \exists! e \text{ s.t. } i \circ e = i'$

In this case we say that  $(i, \ker(\alpha))$  is the **kernel** of  $\alpha$

**Proposition 2.6.90** (Cokernel)

Let  $\mathcal{C}$  be a pointed category (2.6.18). Suppose we have a sequence

$$c \xrightarrow{\alpha} d \xrightarrow{\pi} \text{coker}(\alpha)$$

then the following are equivalent

a)  $\pi$  is the coequaliser of  $(\alpha, 0)$ ,

b)  $\pi \circ \alpha = 0$  and  $\pi' \circ \alpha = 0 \implies \exists! h \text{ s.t. } h \circ \pi = \pi'$ ,

In this case we say that  $(\pi, \text{coker}(\alpha))$  is the **cokernel** of  $f$

**Proposition 2.6.91** (Left Adjoints Preserve Cokernels)

Let  $\mathcal{C}, \mathcal{D}$  be pointed categories and  $\mathcal{C} \xrightleftharpoons[G]{F} \mathcal{D}$  a pair of adjoint functors.

Then  $F$  preserves cokernels and  $G$  preserves kernels.

*Proof.* By (2.6.67) we see that  $F$  and  $G$  preserve zero morphisms. Therefore we are done by (2.6.88) since a cokernel of  $f$  is just the coequaliser of  $(f, 0)$ .  $\square$

### 2.6.13 Functorial Limits

**Proposition 2.6.92** (Pointwise Limits are Functorial)

Let  $J$  be a small category and  $F : \mathcal{C} \times J \rightarrow \mathcal{E}$  a bifunctor.

Suppose that for every  $c \in \mathcal{C}$  there is a limit  $(\lim_{j \in J} F(c, j), \{\lambda_{c,j}\}_{j \in J})$  for  $F(c, -)$ .

Then  $\lim_{j \in J} F(-, j)$  is functorial in the sense there is a unique morphism making the following diagram commute

$$\begin{array}{ccc} \lim_{j \in J} F(c, j) & \xrightarrow{\lim_{j \in J} F(f, j)} & \lim_{j \in J} F(c', j) \\ \downarrow \lambda_{c,j} & & \downarrow \lambda_{c',j'} \\ F(c, j) & \xrightarrow{F(f, \beta)} & F(c', j') \end{array}$$

for all  $f : c \rightarrow c'$  and  $\beta : j \rightarrow j'$ .

*Proof.* Recall that  $(\lim_{j \in J} F(c, j), \{\lambda_{c,j}\}_{j \in J})$  is a terminal object in the category of cones  $(\Delta \downarrow F(c, -))$ . Suppose  $f : c \rightarrow c'$  is a morphism then  $(\lim_{j \in J} F(c, j), \{F(f, 1_j) \circ \lambda_{c,j}\}_{j \in J})$  is an element of  $(\Delta \downarrow F(c', -))$ . Therefore there is a unique morphism into the terminal object of  $(\Delta \downarrow F(c', -))$

$$(\lim_{j \in J} F(c, j), \{F(f, 1_j) \circ \lambda_{c,j}\}_{j \in J}) \rightarrow (\lim_{j \in J} F(c', j), \{\lambda_{c',j}\}_{j \in J})$$

which is exactly what we wanted to prove with  $\beta = 1_j$ . The case of general  $\beta$  then follows from the definition of the cone of the target limit.

The uniqueness shows that  $\lim_{j \in J} F(-, j)$  is indeed functorial.  $\square$

**Proposition 2.6.93** (Pointwise Colimits are Functorial)

Let  $I$  be a small category and  $F : \mathcal{I} \times \mathcal{D} \rightarrow \mathcal{E}$  a bifunctor.

Suppose that for every  $d \in \mathcal{D}$  there is a colimit  $(\text{colim}_{i \in I} F(i, d), \{\mu_{i,d}\}_{i \in I})$  for  $F(-, d)$

Then  $\text{colim}_{i \in I} F(i, -)$  is functorial in the sense there is a unique morphism making the following diagram commute

$$\begin{array}{ccc} F(i, d) & \xrightarrow{F(\alpha, g)} & F(i', d') \\ \downarrow \mu_{i,d} & & \downarrow \mu_{i',d'} \\ \text{colim}_{i \in I} F(i, d) & \xrightarrow{\text{colim}_{i \in I} F(i, g)} & \text{colim}_{i \in I} F(i, d') \end{array}$$

for all  $g : d \rightarrow d'$  and  $\alpha : i \rightarrow i'$ .

**Proposition 2.6.94** (Comparison Map between Limits and Colimits)

Let  $I, J$  be small categories and  $F : I \times J \rightarrow \mathcal{C}$  be a bifunctor.

Suppose that there exists

- a) limits  $i \rightarrow \lim_{j \in J} F(i, j)$  (which is automatically functorial in  $i$  (2.6.92))
- b) colimits  $j \rightarrow \operatorname{colim}_{i \in I} F(i, j)$  (which is automatically functorial in  $j$  (2.6.93))
- c) a colimit  $\operatorname{colim}_{i \in I} \left( \lim_{j \in J} F(\alpha, j) : \lim_{j \in J} F(i, j) \rightarrow \lim_{j \in J} F(i', j) \right)$
- d) a limit  $\lim_{j \in J} \operatorname{colim}_{i \in I} F(i, j)$

(which holds if  $\mathcal{C}$  has limits of shape  $J$  and colimits of shape  $I$ ). Then there exists a unique “comparison morphism”

$$\begin{array}{ccc} \operatorname{colim}_{i \in I} \lim_{j \in J} F(i, j) & \xrightarrow{\quad \Phi \quad} & \lim_{j \in J} \operatorname{colim}_{i \in I} F(i, j) \\ \lambda_{i'} \uparrow & & \downarrow \mu_{j'} \\ \lim_{j \in J} F(i', j) & \xrightarrow{\lambda_{i', j'}} F(i', j') \xrightarrow{\mu_{i', j'}} & \operatorname{colim}_{i \in I} F(i, j') \end{array}$$

making the diagram commute for all  $i' \in I$  and  $j' \in J$ .

*Proof.* The morphisms  $\lambda_{i'}$ ,  $\lambda_{i', j'}$ ,  $\mu_{i', j'}$  and  $\mu_{j'}$  are the usual cone morphisms arising from the limit and colimit constructions. Suppose  $\beta : j' \rightarrow j''$  then we have a commutative diagram, using both functoriality of colimit (2.6.93) and definition of cone  $\lambda_{i', -}$

$$\begin{array}{ccccc} \lim_{j \in J} F(i', j) & \xrightarrow{\lambda_{i', j'}} & F(i', j') & \xrightarrow{\mu_{i', j'}} & \operatorname{colim}_{i \in I} F(i, j') \\ & \searrow \lambda_{i', j''} & \downarrow F(1, \beta) & & \downarrow \operatorname{colim}_{i \in I} F(i, \beta) \\ & & F(i', j'') & \xrightarrow{\mu_{i', j''}} & \operatorname{colim}_{i \in I} F(i, j'') \end{array}$$

By universal property of the (top-right) limit to  $\mu_{i', j'} \circ \lambda_{i', j'}$  we may find for each  $i' \in I$  a unique morphism

$$\Phi_{i'} : \lim_{j \in J} F(i', j) \longrightarrow \lim_{j \in J} \operatorname{colim}_{i \in I} F(i, j)$$

satisfying

$$\mu_{j'} \circ \Phi_{i'} = \mu_{i', j'} \circ \lambda_{i', j'} \quad \forall j' \in J$$

Then by functoriality of limit (2.6.92) and definition of the cocone  $\mu_{-, j'}$  we have a commutative diagram for all  $\alpha : i' \rightarrow i''$

$$\begin{array}{ccccc} \lim_{j \in J} F(i', j) & \xrightarrow{\lambda_{i', j'}} & F(i', j') & \xrightarrow{\mu_{i', j'}} & \operatorname{colim}_{i \in I} F(i, j') \\ \lim_{j \in J} F(\alpha, j) \downarrow & & \downarrow F(\alpha, 1) & & \nearrow \mu_{i'', j''} \\ \lim_{j \in J} F(i'', j) & \xrightarrow{\lambda_{i'', j'}} & F(i'', j'') & & \end{array}$$

so that for all  $j' \in J$

$$\begin{aligned} \mu_{j'} \circ \Phi_{i''} \circ \lim_{j \in J} F(\alpha, j) &= \mu_{i'', j''} \circ \lambda_{i'', j''} \circ \lim_{j \in J} F(\alpha, j) \\ &= \mu_{i', j'} \circ \lambda_{i', j'} \end{aligned}$$

whence by uniqueness

$$\Phi_{i''} \circ \lim_{j \in J} F(\alpha, j) = \Phi_{i'}$$

Therefore by the definition of the (top-left) co-limit there is a unique morphism

$$\Phi : \operatorname{colim}_{i \in I} \lim_{j \in J} F(i, j) \longrightarrow \lim_{j \in J} \operatorname{colim}_{i \in I} F(i, j)$$

such that we have a commutative diagram

$$\begin{array}{ccc}
\operatorname{colim}_{i \in I} \lim_{j \in J} F(i, j) & \xrightarrow{\quad \Phi \quad} & \lim_{j \in J} \operatorname{colim}_{i \in I} F(i, j) \\
\lambda_{i'} \uparrow & \nearrow \Phi_{i'} & \downarrow \mu_{j'} \\
\lim_{j \in J} F(i', j) & \xrightarrow{\lambda_{i', j'}} F(i', j') & \xrightarrow{\mu_{i', j'}} \operatorname{colim}_{i \in I} F(i, j')
\end{array}$$

Then if there is another morphism  $\Phi'$  which makes the square commute define  $\Phi'_{i'} = \Phi' \circ \lambda_{i'}$ , we may use uniqueness to deduce that  $\Phi'_{i'} = \Phi_{i'}$  and  $\Phi = \Phi'$ .  $\square$

**Example 2.6.95** (Direct Limit of an Equaliser)

Let  $\mathbf{2} = \{\emptyset \rightrightarrows 1\}$  be a category with two objects and two arrows, and let  $I$  be a directed poset.

Consider a bifunctor  $F : I \times \mathbf{2} \rightarrow \mathbf{Set}$  then by (...)

$$\lim_{j \in \mathbf{2}} F(i, j) = \operatorname{eq}(F(i, \alpha), F(i, \beta)) = \{y \in F(i, \emptyset) \mid F(i, \alpha)(y) = F(i, \beta)(y)\}$$

and

$$\operatorname{colim}_{i \in I} \lim_{j \in \mathbf{2}} F(i, j) = \{(i, y) \mid y \in F(i, \emptyset) \text{ and } F(i, \alpha)(y) = F(i, \beta)(y)\} / \sim$$

further

$$\operatorname{colim}_{i \in I} F(i, j) = \{(i, y) \mid y \in F(i, j)\} / \sim$$

and

$$\lim_{j \in \mathbf{2}} \operatorname{colim}_{i \in I} F(i, j) = \{[(i, y)] \in \operatorname{colim}_{i \in I} F(i, \emptyset) \mid F(i, \alpha)(y) \sim F(i, \beta)(y)\}$$

In this case we may verify that the comparison map is an isomorphism.

## 2.6.14 Algebraic Categories

**Definition 2.6.96**

Let  $(\mathcal{C}, U)$  be a concrete category. We say it is an algebraic category if  $U$

- a) reflects isomorphisms
- b) creates equalisers, products and direct limits / filtered colimits
- c) has a left adjoint (i.e. free algebras exist)

In particular it also preserves limits and filtered colimits by (2.6.76).

**Proposition 2.6.97**

The following are algebraic categories: **AbGrp**, **A-Mod**, **Alg<sub>k</sub>**.

**Proposition 2.6.98** (Injective  $\iff$  monic)

Let  $\mathcal{C}$  be an algebraic category. Then a morphism  $f : c \rightarrow d$  is a monomorphism if and only if  $U(f)$  is injective.

*Proof.* If  $U(f)$  is injective and  $f \circ g = f \circ h$  then  $U(f) \circ U(g) = U(f) \circ U(h)$  whence  $U(g) = U(h)$  and by faithfulness  $g = h$  as required.

Conversely if  $f : c \rightarrow d$  is monic, consider the left adjoint  $F$  to  $U$ , then by definition we have a natural bijection

$$\Psi_c : \operatorname{Mor}(F(\{\star\}), c) \xrightarrow{\sim} U(c)$$

Suppose  $U(f)(x) = U(f)(y)$  for  $x, y \in U(c)$ . Let  $g, h$  be morphisms corresponding to  $x$  and  $y$  under this bijection, that is  $x = \Psi_c(g)$  and  $y = \Psi_c(h)$ .

Then by naturality of  $\Psi$

$$U(f)(x) = \Psi_d(f \circ g)$$

and

$$U(f)(y) = \Psi_d(f \circ h)$$

so by assumption  $f \circ g = f \circ h$ . As  $f$  is monic we have  $g = h$  whence  $x = y$ . Therefore  $U(f)$  is injective.  $\square$

**Proposition 2.6.99** (Equalisers in Algebraic Categories)

Consider a commutative diagram in an algebraic category  $\mathcal{C}$

$$X \xrightarrow{e} Y \xrightleftharpoons[d_1]{d_0} Z$$

Then  $e$  is an equaliser for  $(d_0, d_1)$  if and only if  $e$  is injective and  $\operatorname{Im}(U(e)) = \{y \in U(Y) \mid U(d_0)(y) = U(d_1)(y)\}$ .

A similar statement applies for kernels.



## Chapter 3

# Topology and Sheaves

### 3.1 Topological Spaces

References :

- General Topology [Kel71, Chap. 1]
- General Topology [Wil70]

Topology is useful in algebraic geometry, but often the natural topologies are usually much coarser so the theory looks rather different.

**Definition 3.1.1** (Topological Space)

A topological space  $(X, \mathcal{T}_X)$  consists of a set  $X$  and family of open sets  $\mathcal{T}_X \subseteq \mathcal{P}(X)$  satisfying the following properties

- $X, \emptyset \in \mathcal{T}_X$
- $U_i \in \mathcal{T}_X \implies \bigcup_{i \in I} U_i \in \mathcal{T}_X$
- $U, V \in \mathcal{T}_X \implies U \cap V \in \mathcal{T}_X$

A subset  $Z \subset X$  is said to be closed iff  $X \setminus Z$  is open. We may equivalently define the topology in terms of closed sets.

**Proposition 3.1.2**

Let  $X$  be a topological space. Both the open sets and closed sets form a *distributive lattice* under inclusion.

**Definition 3.1.3** (Subspace topology)

Let  $Y \subset X$ , then we may define the **subspace topology** on  $Y$  by

$$\mathcal{T}_Y := \{U \cap Y \mid U \in \mathcal{T}_X\}$$

when  $Y$  is open then this is given by

$$\mathcal{T}_Y = \{U \subseteq Y \mid U \in \mathcal{T}_X\}$$

**Definition 3.1.4** (Base)

We say  $\mathcal{B} \subseteq \mathcal{P}(X)$  is a base (of open sets) on  $X$  if

- For every  $x \in X$  there is a  $U \in \mathcal{B}$  such that  $x \in U$
- Suppose  $U, V \in \mathcal{B}$  and  $x \in U \cap V$  then there exists  $W \in \mathcal{B}$  such that  $x \in W \subseteq U \cap V$

**Proposition 3.1.5** (Topology generated by a base)

Let  $\mathcal{B}$  be a base, then the following is a topology on  $X$

$$\mathcal{T}_{\mathcal{B}} := \left\{ \bigcup_{U_i \in I} U_i \mid I \subseteq \mathcal{B} \right\}$$

i.e. the set of arbitrary unions of sets in  $\mathcal{B}$ .

**Proposition 3.1.6** (Base generating topology)

A base  $\mathcal{B}$  satisfies  $\mathcal{T}_{\mathcal{B}} = \mathcal{T}_X$  if and only if

- $\mathcal{B} \subset \mathcal{T}_X$
- or every  $x \in U$  and  $U \in \mathcal{T}_X$  there exists  $V \in \mathcal{B}$  such that  $x \in V \subseteq U$ .

In this case we say  $\mathcal{B}$  is a base for (the topology on)  $X$ .

**Proposition 3.1.7** (Subbase for a topology)

Let  $X$  be a set and  $\mathcal{B}$  be an arbitrary family of subsets. Then there exists a topology  $\mathcal{T}_{\mathcal{B}}$  such that

- $\mathcal{B} \subset \mathcal{T}_{\mathcal{B}}$
- For every topology  $\mathcal{T}'$  containing  $\mathcal{B}$  we have  $\mathcal{T}_{\mathcal{B}} \subset \mathcal{T}'$ .

We say that  $\mathcal{B}$  is a **subbase** for the topology  $\mathcal{T}_{\mathcal{B}}$ . Explicitly  $\mathcal{T}_{\mathcal{B}}$  consists of arbitrary unions of finite intersections of elements of  $\mathcal{B}$ .

**Proposition 3.1.8** (Base for subspace topology)

Let  $(X, \mathcal{T})$  be a topological space, and  $Y \subset X$  has the subspace topology  $\mathcal{T}_Y$  (3.1.3). If  $\mathcal{B}$  is a base for  $(X, \mathcal{T})$  then

$$\mathcal{B}_Y := \{U \cap Y \mid U \in \mathcal{B}\}$$

is a base for  $(Y, \mathcal{T}_Y)$ .

**Definition 3.1.9** (Adherent point)

For  $Y \subset X$  we say  $x$  is an **adherent point** of  $Y$  if every open neighbourhood of  $x$  intersects  $Y$ .

Similarly we say  $x$  is a **limit point** (or **accumulation point**) for  $Y$  if every open neighbourhood of  $x$  intersects  $Y$  at a point other than  $x$ .

**Proposition 3.1.10** (Closed point)

Let  $x \in X$  then TFAE

- $\{x\}$  is closed
- For every  $y \neq x$  there is  $U \ni y$  such that  $x \notin U$ .

**Definition 3.1.11**

For a topological space  $X$  let  $X_0$  denote the subset of closed points.

**Definition 3.1.12** (Open Embedding)

Let  $f : X \rightarrow Y$  be a map of topological spaces. We say it is an **open embedding** if one of the following equivalent conditions holds

- a)  $f(X)$  is open and  $f$  is a homeomorphism onto its image
- b)  $f$  is continuous, open and injective.

### 3.1.1 Axioms of Countability

**Definition 3.1.13** (Neighbourhood)

Let  $X$  be a topological space and  $x \in X$ . We say a set  $V \subset X$  is a **neighbourhood** of  $x$  if  $x \in U \subset V$  for some open set  $U$ . We write  $\mathcal{U}_x$  for the neighbourhoods of  $x$ , and  $\mathcal{O}_x$  for the open neighbourhoods of  $x$ .

We say that a collection of neighbourhoods  $\mathcal{B}_x$  is a **local base** for  $x$  if every neighbourhood  $U \in \mathcal{U}_x$  contains some neighbourhood  $V \in \mathcal{B}_x$ . For example  $\mathcal{O}_x$  is a local base.

**Definition 3.1.14** (First-Countable)

Let  $X$  be a topological space.  $X$  is **first countable** if every point  $x \in X$  has a countable local base.

**Definition 3.1.15** (Second-Countable)

Let  $X$  be a topological space.  $X$  is **second countable** if there exists a countable basis. Evidently this is stronger than first countability.

**Example 3.1.16**

For example a metric space is always first countable as we may consider balls of radius  $1/n$ .

Further  $\mathbb{R}$  is second countable as we may consider open intervals with rational endpoints.

### 3.1.2 Closure

**Definition 3.1.17** (Adherent Point)

Let  $X$  be a topological space and  $Y \subset X$ . We say that  $x \in X$  is an **adherent point** of  $Y$ , if  $U \in \mathcal{U}_x \implies U \cap Y \neq \emptyset$ .

We say that  $x$  is a **sequential limit point** of  $Y$  if there exists a sequence  $x_n \in Y$  such that  $x_n \rightarrow x$ .

**Proposition 3.1.18** (Adherent Point)

Let  $X$  be a topological space. Then every sequential limit point is an adherent point. If  $X$  is first countable the converse holds.

*Proof.* The first implication is straightforward. For the latter, consider a countable local base at  $x$  given by  $U_n$ . We may assume wlog that it is decreasing. Then by assumption we may choose  $x_n \in U_n \cap Y$ . For any  $V \in \mathcal{B}_x$  we have  $U_N \subset V$  for some  $N$ , and therefore  $n \geq N \implies x_n \in U_n \implies x_n \in V$ . Therefore  $x_n \rightarrow x$ .  $\square$

**Proposition 3.1.19** (Topological Closure)

Let  $Y \subset X$  then the following equality holds

$$\bigcap_{\substack{Z \supseteq Y \\ Z \text{ closed}}} Z = \{x \in X \mid x \text{ adherent point of } Y\}$$

We denote this by  $\bar{Y}$  (or  $\text{cl}_X(Y)$  to emphasise the ambient space  $X$ ) and refer to it as the **closure** of  $Y$  in  $X$ . Furthermore the following properties hold

- a)  $Y \subseteq \bar{Y}$  and  $\bar{Y}$  is closed
- b)  $Y = \bar{Y}$  if and only if  $Y$  is closed
- c)  $(Y \cap U \neq \emptyset \iff \bar{Y} \cap U \neq \emptyset)$  for any  $U$  open

When  $X$  is first countable then  $\bar{Y}$  is the set of sequential limit points of  $Y$ .

*Proof.* Suppose  $Z$  is a closed set containing  $Y$  and  $x$  is an adherent point of  $Y$ . Then  $x \notin Z \implies x \in X \setminus Z \implies (X \setminus Z) \cap Y \neq \emptyset$  a contradiction. Conversely assume  $x \notin \bar{Y}$  then there exists  $Z \supseteq Y$  closed such that  $x \notin Z \implies x \in X \setminus Z$ . This means  $x$  is not an adherent point.

- a) An arbitrary intersection of closed sets is closed
- b) This follows because  $\bar{Y}$  is the smallest closed superset.
- c) One implication is clear because  $Y \subseteq \bar{Y}$ . Conversely if  $x \in \bar{Y} \cap U$  then  $x$  must be a limit point of  $Y$  hence  $U \cap Y \neq \emptyset$  as required.

□

### Corollary 3.1.20

Let  $X$  be a first countable topological space. Then  $\text{cl}_X(Y)$  is the set of sequential limit points of  $Y$ .

### Proposition 3.1.21 (Closure in Subspace Topology)

Let  $X$  be a topological space and  $Y$  a subset. For  $Z \subset Y$  we have

$$\text{cl}_Y(Z) = \text{cl}_X(Z) \cap Y$$

*Proof.* By (3.1.19)  $\text{cl}_X(Z)$  is a closed subset of  $X$  and therefore  $\text{cl}_X(Z) \cap Y$  is a closed subset of  $Y$  in the subspace topology. Therefore  $\text{cl}_Y(Z) \subset \text{cl}_X(Z) \cap Y$  by minimality. Similarly  $\text{cl}_Y(Z) = W \cap Y$  for  $W$  closed in  $X$ . By minimality  $\text{cl}_X(Z) \subset W \implies \text{cl}_X(Z) \cap Y \subset W \cap Y = \text{cl}_Y(Z)$ . □

### Proposition 3.1.22

Let  $X$  be a topological space with  $X = \bigcup_{i \in I} U_i$  an open cover. Let  $Y \subset X$  be a subset then  $Y$  is closed in  $X$  if and only if  $Y \cap U_i$  is closed in  $U_i$  for all  $i \in I$ .

*Proof.* One direction is obvious by definition of the subspace topology. Conversely  $U_i \setminus (Y \cap U_i)$  is open in  $U_i$  and therefore in  $X$ . We see that

$$X \setminus Y = \bigcup_{i \in I} U_i \setminus (Y \cap U_i)$$

is open and therefore  $Y$  is closed. □

### Proposition 3.1.23

Let  $Y_1, \dots, Y_n$  be subsets of a topological space  $X$ . Then

$$\overline{Y_1 \cup \dots \cup Y_n} = \bar{Y}_1 \cup \dots \cup \bar{Y}_n$$

*Proof.* By induction it is sufficient to demonstrate the case  $n = 2$ .  $\overline{Y_1 \cup Y_2}$  is a closed set containing both  $Y_1$  and  $Y_2$ . Therefore by definition

$$\bar{Y}_1 \cup \bar{Y}_2 \subset \overline{Y_1 \cup Y_2}$$

On the other hand  $\bar{Y}_1 \cup \bar{Y}_2$  is closed and contains  $Y_1 \cup Y_2$ . Therefore the reverse inclusion follows immediately. □

### Proposition 3.1.24 (Dense subset)

Let  $Y \subset X$  then the following are equivalent

- a)  $\bar{Y} = X$
- b)  $Y \cap U \neq \emptyset$  for any  $U \subset X$  open

In this case we say  $Y$  is **dense**.

*Proof.* 1  $\implies$  2) Follows from (3.1.19).c)

2  $\implies$  1). Suppose  $Y \subseteq \bar{Y} \subsetneq X$  then  $X \setminus \bar{Y}$  is an open set not intersecting  $Y$  a contradiction.  $\square$

### 3.1.3 Continuous Maps

**Proposition 3.1.25** (Continuous at a point)

Let  $f : X \rightarrow Y$  a map of topological spaces and  $x \in X$ . Then the following are equivalent

- a)  $V \in \mathcal{U}_{f(x)} \implies f^{-1}(V) \in \mathcal{U}_x$
- b)  $V \in \mathcal{B}_{f(x)} \implies f^{-1}(V) \in \mathcal{U}_x$
- c)  $V \in \mathcal{U}_{f(x)} \implies \exists U \in \mathcal{U}_x$  s.t.  $f(U) \subseteq V$
- d)  $V \in \mathcal{B}_{f(x)} \implies \exists U \in \mathcal{B}_x$  s.t.  $f(U) \subseteq V$

where  $\mathcal{B}_{f(x)}$  is any local base at  $f(x)$  and  $\mathcal{B}_x$  is any local base at  $x$ . We say that  $f$  is **continuous** at  $x$ .

*Proof.* a)  $\implies$  b) is obvious. Suppose b) holds and  $V \in \mathcal{U}_{f(x)}$  then by definition there exists  $V' \in \mathcal{B}_{f(x)}$  such that  $V' \subset V$ . By hypothesis  $f^{-1}(V') \in \mathcal{U}_x$  from which it follows that  $f^{-1}(V) \in \mathcal{U}_x$ .

a)  $\implies$  c)  $f(f^{-1}(V)) \subseteq V$

c)  $\implies$  d) By definition there exists  $U' \in \mathcal{B}_x$  such that  $U' \subseteq U \implies f(U') \subseteq f(U) \subseteq V$

d)  $\implies$  b) Observe that  $f(U) \subseteq V \implies U \subseteq f^{-1}(V)$  whence  $f^{-1}(V) \in \mathcal{U}_x$  as required.  $\square$

**Proposition 3.1.26**

Let  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$  be maps of topological spaces such that  $f$  is continuous at  $x$  and  $g$  is continuous at  $f(x)$ . Then  $g \circ f$  is continuous at  $x$ .

**Proposition 3.1.27** (Characterisation of Continuous Maps)

Let  $f : X \rightarrow Y$  be a map of topological spaces. Then the following are equivalent

- a)  $f$  is continuous at all points  $x \in X$
- b)  $V \subset Y$  open  $\implies f^{-1}(V) \subset X$  is open
- c)  $Z \subset Y$  closed  $\implies f^{-1}(Z) \subset X$  is closed

Let  $\mathcal{B}_1, \mathcal{B}_2$  be bases of open sets for  $X$  and  $Y$  respectively. Then this is equivalent to the following condition

$$\forall V \in \mathcal{B}_2, \text{ and } x \in f^{-1}(V), \exists U \in \mathcal{B}_1 \text{ s.t. } x \in U \subset f^{-1}(V)$$

**Proposition 3.1.28**

Let  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$  be maps of topological spaces such that  $f$  is continuous and  $g$  is continuous. Then  $g \circ f$  is continuous.

**Proposition 3.1.29**

Let  $f : X \rightarrow Y$  be a map of topological spaces and  $X = \bigcup_{i \in I} U_i$  an open cover. Then  $f$  is continuous iff  $f|_{U_i}$  is continuous for all  $i \in I$ .

*Proof.* Observe that  $f^{-1}(V) = \bigcup_{i \in I} f^{-1}(V) \cap U_i = \bigcup_{i \in I} (f|_{U_i})^{-1}(V)$ .  $\square$

**Proposition 3.1.30**

Let  $f : X \rightarrow Y$  be a map of topological spaces. Then  $f$  is continuous iff

$$f(\bar{Z}) \subset \overline{f(Z)}$$

for all subsets  $Z \subset X$ .

*Proof.* Suppose  $f$  is continuous and  $f(Z) \subset W$  is closed. Then by (3.1.27)  $Z \subset f^{-1}(W)$  is closed. By definition  $\bar{Z} \subset f^{-1}(W) \implies f(\bar{Z}) \subset f(f^{-1}(W)) \subset W$ . As  $W$  was an arbitrary closed set containing  $f(Z)$  we find by definition  $f(\bar{Z}) \subset \overline{f(Z)}$ .

Conversely suppose  $W \subset Y$  is closed and  $Z := f^{-1}(W)$ . Then  $f(\bar{Z}) \subset \overline{f(Z)} = \overline{W \cap \text{Im}(f)} \subset W \implies f^{-1}(f(\bar{Z})) \subset Z \implies \bar{Z} \subset Z \implies \bar{Z} = Z$  and therefore  $Z$  is closed. Therefore  $f$  is continuous by (3.1.27).  $\square$

### 3.1.4 Quasi-Homeomorphism

**Proposition 3.1.31** (Quasi-Homeomorphism)

Let  $f : X \rightarrow Y$  be a map. The following are equivalent

- a) The map  $W \rightarrow f^{-1}(W)$  is a bijection between open sets
- b) The map  $Z \rightarrow f^{-1}(Z)$  is a bijection between closed sets

In this case we say  $f$  is a **quasi-homeomorphism**. Such a map induces a bijection between irreducible sets, respectively irreducible components.

*Proof.* This follows by taking complements and observing  $X \setminus f^{-1}(A) = f^{-1}(Y \setminus A)$ .  $\square$

**Proposition 3.1.32** (Induced Quotient)

Let  $\pi : X \rightarrow Y$  be a surjective map where  $X$  is a topological space. The family

$$\mathcal{T}_Y := \{U \subset Y \mid \pi^{-1}(U) \in \mathcal{T}_X\}$$

is a well-defined topology on  $Y$ . Suppose that  $\pi^{-1}(\pi(V)) \subset V$  for all  $V$  open (resp. closed) subsets of  $X$ . Then  $\pi$  is both open, closed and a quasi-homeomorphism.

*Proof.* As arbitrary intersections and unions commute with inverse image we see that  $\mathcal{T}_Y$  is a well-defined topology.

As  $\pi$  is surjective we have  $\pi(\pi^{-1}(U)) = U$  for all open subsets  $U \subset Y$ . Generically  $V \subset \pi^{-1}(\pi(V))$  and by assumption the reverse inclusion holds. Therefore we see  $\pi$  induces a quasi-homeomorphism. This also shows  $\pi$  is open and therefore closed.  $\square$

### 3.1.5 Kolmogorov Spaces

**Definition 3.1.33**

Let  $X$  be a topological space. We say that  $X$  is **Kolmogorov** or  $T_0$  if for all distinct pairs  $x, y \in X$  there exists an open set  $U$  such that either  $x \in U$  and  $y \notin U$ , or  $x \notin U$  and  $y \in U$ .

**Proposition 3.1.34** (Indistinguishable points)

Let  $X$  be a topological space and  $x, y \in X$ . The following are equivalent

- a)  $\overline{\{x\}} = \overline{\{y\}}$
- b)  $x \in \overline{\{y\}}$  and  $y \in \overline{\{x\}}$
- c) Every closed  $Z \subset X$  contains both or neither of  $x, y$
- d) Every open  $U \subset X$  contains both or neither of  $x, y$

We say  $x, y$  are **topologically indistinguishable**, otherwise we say  $x$  and  $y$  are **distinguishable**.

$X$  is Kolmogorov iff every pair of points are distinguishable.

Being indistinguishable determines an equivalence relation on  $X$ , and we denote by  $X_0 := X / \sim$  the **Kolmogorov Quotient**.

*Proof.* a)  $\implies$  b) Clear

b)  $\implies$  a) By definition of closure  $x \in \overline{\{y\}} \iff \overline{\{x\}} \subset \overline{\{y\}}$ .

c)  $\iff$  d) Follows by taking complements

c)  $\implies$  b) Suppose  $x \in Z$  for  $Z$  closed then by hypothesis  $y \in Z$ . As  $Z$  was arbitrary we see that  $y \in \overline{\{x\}}$ , and symmetrically  $x \in \overline{\{y\}}$ .

b)  $\implies$  c) By assumption  $x \in Z \implies y \in Z$  for  $Z$  closed and symmetrically  $y \in Z \implies x \in Z$ .  $\square$

**Definition 3.1.35** (Kolmogorov Quotient)

Let  $X$  be a topological space. The set  $X_0$  of equivalence classes of topologically indistinguishable points is the **Kolmogorov Quotient**

**Proposition 3.1.36** (Kolmogorov Quotient Properties)

Let  $X$  be a topological space and  $X_0$  be the Kolmogorov Quotient with quotient map  $\pi : X \rightarrow X_0$ . Relative to the induced topology

$$\mathcal{T}_{X_0} := \{W \subset X_0 \mid \pi^{-1}(W) \in \mathcal{T}_X\}$$

$\pi$  is a quasi-homeomorphism and an open and closed map. Further  $X_0$  is Kolmogorov.

*Proof.* By (3.1.32) it is sufficient to show that  $\pi^{-1}(\pi(U)) \subset U$  for all open subsets  $U \subset X$ . Suppose  $x \in \pi^{-1}(\pi(U))$  then  $\pi(x) = \pi(y)$  for some  $y \in U$ . Then by definition of  $\pi$ ,  $x$  and  $y$  are indistinguishable and so  $x \in U$ .

Suppose  $\pi(x) \neq \pi(y)$  then by definition there exists an open set  $U \subset X$  such that  $x \in U$  and  $y \notin U$ . As  $\pi$  is open then  $\pi(U)$  is open. By construction  $\pi(x) \in \pi(U)$  and  $\pi(y) \notin \pi(U)$ . This shows that  $X_0$  is Kolmogorov.  $\square$

**Proposition 3.1.37** (Kolmogorov Quotient Functoriality)

Let  $f : X \rightarrow Y$  be a continuous map, then there is a unique (continuous) map  $f_0 : X_0 \rightarrow Y_0$  such that the following diagram commutes

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow \pi & & \downarrow \pi \\ X_0 & \xrightarrow{f_0} & Y_0 \end{array}$$

*Proof.* Given  $x, x' \in X$  and suppose  $f(x) \not\sim f(x')$ . Then there exists an open set  $U$  such that  $f(x') \in U$  and  $f(x) \notin U$ . Therefore  $x' \in f^{-1}(U)$  and  $x \notin f^{-1}(U)$  and  $x \not\sim x'$ . Equivalently  $x \sim x' \implies f(x) \sim f(x')$ . Therefore the map

$$f_0([x]) = [f(x)]$$

is well-defined.

For  $W_0 \subset Y_0$  we have  $\pi^{-1}(f_0^{-1}(W_0)) = f^{-1}(\pi^{-1}(W_0))$  is open. As  $\pi$  is surjective and open  $f_0^{-1}(W) = \pi(\pi^{-1}(f_0^{-1}(W_0)))$  is open, and therefore  $f_0$  is continuous.

Any such  $f_0$  satisfies  $\{f_0(x_0)\} = \pi(f(\pi^{-1}(\{x_0\})))$  and so is unique.  $\square$

**Proposition 3.1.38**

Let  $X$  be a topological space with open cover  $X = \bigcup_{i \in I} U_i$ . If every  $U_i$  is Kolmogorov with respect to the subspace topology then so is  $X$ .

*Proof.* Consider  $x, y \in X$ . We may assume wlog that  $x, y \in U_i$  for some  $i$  (otherwise we would be done). Then by assumption there exists  $V \subset U_i$  such that  $x \in V$  and  $y \notin V$ , whence  $x \in V$  and  $y \notin V$  as required.  $\square$

### 3.1.6 Symmetric Spaces

**Definition 3.1.39**

Let  $X$  be a topological space. We say that  $x$  is **quasi-closed** if  $y \in \overline{\{x\}} \implies x \in \overline{\{y\}}$ .

**Proposition 3.1.40** (Symmetric Topology)

Let  $X$  be a topological space. Then the following are equivalent

- a) For all  $x \in X$  we have  $\overline{\{x\}} = \{y \in X \mid y \text{ topologically indistinguishable from } x\}$ ,
- b) For all  $x, y \in X$   $x \in \overline{\{y\}} \implies y \in \overline{\{x\}}$  (i.e. every point is **quasi-closed**),
- c) Every open neighbourhood  $U$  of  $x$  contains  $\overline{\{x\}}$ ,
- d)  $X$  is partitioned by sets of the form  $\overline{\{x\}}$ , indexed by  $X_0$ .

In this case we say that  $X$  is **symmetric** or  $R_0$ . Further  $x \in \overline{\{y\}}$  if and only if  $x$  and  $y$  become equal in  $X_0$ .

**Proposition 3.1.41**

Let  $X$  be a symmetric topological space and  $x, y \in X$ . Then the following are equivalent

- a)  $x$  and  $y$  are topologically indistinguishable,
- b)  $x \in \overline{\{y\}}$ ,
- c)  $y \in \overline{\{x\}}$ ,
- d)  $\overline{\{x\}} = \overline{\{y\}}$ ,
- e)  $x$  and  $y$  become equal in the Kolmogorov Quotient  $X_0$

**Proposition 3.1.42** (Open Subspace is Symmetric)

Let  $X$  be a symmetric space and  $U \subset X$  an open subset. Then  $U$  is symmetric and for every  $x \in U$  we have

$$\text{cl}_X(\{x\}) = \text{cl}_U(\{x\})$$

*Proof.* By (3.1.21) we have  $\text{cl}_U(\{x\}) = \text{cl}_X(\{x\}) \cap U$ , so these are equal by (3.1.40).c). We conclude that the sets  $\overline{\{x\}}$  partition  $U$  so that  $U$  is also symmetric.  $\square$

**Proposition 3.1.43** (Symmetric Open Cover  $\implies$  Symmetric)

Let  $X$  be a topological space with open cover  $\bigcup_{i=1}^n U_i$  such that  $U_i$  are symmetric in the subspace topology. Then  $X$  is symmetric.

*Proof.* Suppose  $x \in \text{cl}_X(\{y\})$  then by (3.1.42) we have  $x \in \text{cl}_{U_i}(\{y\})$ , whence  $y \in \text{cl}_{U_i}(\{x\})$  by (3.1.40) and  $y \in \text{cl}_X(\{x\})$ . Therefore by (3.1.40) we have  $X$  is symmetric.  $\square$

**Proposition 3.1.44**

Let  $X$  be a topological space. Then the following are equivalent

- a)  $X$  is both Symmetric and Kolmogorov
- b) Every point  $x \in X$  is closed.

**Corollary 3.1.45**

Let  $X$  be a symmetric topological space. Then every point of  $X_0$  is closed.

### 3.1.7 Hausdorff Spaces

**Definition 3.1.46**

Let  $X$  be a topological space. We say that  $X$  is **Hausdorff** If for all distinct pairs  $x, y \in X$  there exists open neighbourhoods  $U, V$  of  $x$  and  $y$  respectively such that  $U \cap V = \emptyset$ .

Observe that **Hausdorff**  $\implies$  **Kolmogorov**, **Symmetric**, but not conversely (e.g. co-finite topology).

**Proposition 3.1.47**

Let  $X$  be a Hausdorff topological space. Then  $X$  is both Kolmogorov and Symmetric. Further singletons are closed.

### 3.1.8 Convergent Sequences

**Proposition 3.1.48** (Convergent Sequence)

Let  $X$  be a topological space,  $(x_n)$  a sequence in  $X$  and  $x \in X$ . Let  $\mathcal{B}_x$  be a local base for  $x$ . Then the following are equivalent

- a) For every  $V \in \mathcal{U}_x$  there exists  $N$  such that

$$n \geq N \implies a_n \in V$$

- b) For every  $V \in \mathcal{B}_x$  there exists  $N$  such that

$$n \geq N \implies a_n \in V$$

- c) The function

$$\begin{aligned} \mathbb{N} \cup \{\infty\} &\rightarrow X \\ n &\rightarrow x_n \\ \infty &\rightarrow x \end{aligned}$$

is continuous at  $\infty$  with respect to the topology on  $\mathbb{N} \cup \{\infty\}$  given by open sets of the form  $\{N, N+1, \dots\} \cup \{\infty\}$  and arbitrary subsets of  $\mathbb{N}$ .

In this case we say that  $x_n \rightarrow x$ .

**Proposition 3.1.49** (Continuous  $\implies$  Sequentially Continuous)

Let  $f : X \rightarrow Y$  be a continuous map and  $x_n \rightarrow x$  a sequence in  $X$ . Then  $f(x_n) \rightarrow f(x)$ .

*Proof.* If we regard the sequence as a continuous map  $\mathbb{N} \cup \infty$  (3.1.48) then the result follows from (3.1.26).  $\square$

**Proposition 3.1.50** (Limits are Unique in Kolmogorov Spaces)

Let  $X$  be a Kolmogorov topological space. Then limits of sequences are unique.

**Proposition 3.1.51** (Sequentially Continuous  $\implies$  Continuous)

Let  $X, Y$  be topological spaces,  $f : X \rightarrow Y$  a map and  $x \in X$  a point with a countable local base  $\mathcal{B}_x$ . Suppose that for all sequences  $x_n \rightarrow x$  we have  $f(x_n) \rightarrow f(x)$ . Then  $f$  is continuous at  $x$ .

In particular if  $X$  is first countable then  $f : X \rightarrow Y$  is continuous iff it is sequentially continuous.

*Proof.* Consider a point  $x \in X$  and  $y := f(x)$ . By hypothesis  $\mathcal{B}_x = \{U_n\}$  is countable. By replacing with the base  $U'_n := \bigcap_{k=1}^n U_k$  we may suppose that

$$U_1 \supseteq U_2 \supseteq \dots \supseteq U_n \supseteq \dots$$

is decreasing. Suppose  $f$  is not continuous at  $x$  then by (3.1.25).d) there exists  $V \in \mathcal{U}_{f(x)}$  such that  $f(U_n) \setminus V \neq \emptyset$  for all  $n$ . Therefore we may choose  $x_n \in U_n$  such that  $f(x_n) \notin V$ . Consider  $U \in \mathcal{U}_x$  then by definition there is some  $N$  such that  $U_N \subseteq U$ . Furthermore  $n \geq N \implies U_n \subseteq U_N \subseteq U \implies x_n \in U$ . Therefore  $x_n \rightarrow x$ . By construction  $f(x) \in V$  but  $f(x_n) \notin V$  for all  $n$  and in particular  $f(x_n) \not\rightarrow f(x)$ .  $\square$

### 3.1.9 Connected Topological Spaces

**Proposition 3.1.52** (Connected Topological Space)

Let  $X$  be a topological space. Then the following are equivalent

- a)  $X$  cannot be written as the disjoint union  $U \sqcup V$  of two non-empty open subsets  $U, V \subset X$
- b) There are no proper non-empty subsets which are both open and closed

In this case we say that  $X$  is **connected**.

**Proposition 3.1.53** (Connected Subset)

Let  $X$  be a topological space and  $Z \subset X$ . Then the following are equivalent

- a)  $Z$  is connected in the subspace topology
- b)  $Z$  cannot be written as a disjoint  $(Z \cap U) \sqcup (Z \cap V)$  with both  $Z \cap U, Z \cap V \neq \emptyset$

**Proposition 3.1.54**

Let  $X$  be a topological space and  $\{Z_i\}_{i \in I}$  a family of connected subsets such that  $\bigcap_{i \in I} Z_i \neq \emptyset$ . Then  $\bigcup_{i \in I} Z_i$  is connected.

**Definition 3.1.55** (Connected Component)

Let  $X$  be a topological space. We say  $Z \subset X$  is a **connected component** if it is both connected and not properly contained in a connected subset.

**Proposition 3.1.56**

Let  $X$  be a topological space. Then every connected subset  $Z$  is contained in a unique connected component.

*Proof.* Define

$$\overline{Z} := \bigcup \{Z' \supset Z \mid Z' \subset X \text{ connected}\}$$

then by (3.1.54) this is connected. If  $Z'' \supset \overline{Z} \supset Z$  is connected then by construction  $Z'' \subset \overline{Z}$ , so by definition  $\overline{Z}$  is a connected component. Let  $\overline{Z}'$  be another connected component containing  $Z$ , then by the same argument  $\overline{Z}' \subset \overline{Z}$  and by definition  $\overline{Z}' = \overline{Z}$ .  $\square$

**Proposition 3.1.57**

Let  $X$  be a topological space, then the connected components partition  $X$ .

We denote this collection by  $\pi_0(X)$ .

*Proof.* Evidently  $\{x\}$  is connected so that the connected components cover  $X$  by (3.1.56).

Suppose we have two connected components  $Z_1, Z_2$  are not disjoint. Then by (3.1.54)  $Z_1 \cup Z_2$  is connected. By maximality we have  $Z_1 \cup Z_2 = Z_1 = Z_2$  as required.  $\square$

### 3.1.10 Irreducible Topological Spaces

References :

- a) [Bou98a, Chapter II §4.2]
- b) [Sta15, 004U]

**Proposition 3.1.58** (Irreducible space)

Let  $X$  be a topological space. Then the following are equivalent

- a)  $X = Z_1 \cup Z_2$  closed implies either  $Z_1 = X$  or  $Z_2 = X$
- b)  $U, V \neq \emptyset \implies U \cap V \neq \emptyset$  for open sets  $U, V$

c)  $U \neq \emptyset \implies \overline{U} = X$  i.e. every non-empty open set is *dense*

and we say  $X$  is irreducible.

*Proof.* a)  $\implies$  b) Suppose  $U, V$  are open sets such that  $U \cap V = \emptyset$ . Then  $X = (X \setminus U) \cup (X \setminus V)$ . By hypothesis  $X = (X \setminus U)$  or  $X = (X \setminus V)$  whence either  $U$  or  $V$  is empty.

b)  $\implies$  a) Similar.

c)  $\iff$  b) Follows directly from (3.1.24) □

**Proposition 3.1.59** (Irreducible Subset)

Let  $Y \subset X$  be a subset of a topological space. Then the following conditions on  $Y$  are equivalent

- a)  $Y$  is irreducible in the subspace topology.
- b)  $Y \subseteq Z_1 \cup Z_2 \implies Y \subseteq Z_1$  or  $Y \subseteq Z_2$  where  $Z_1, Z_2$  are closed subsets of  $X$
- c)  $U \cap Y \neq \emptyset, V \cap Y \neq \emptyset \implies (U \cap V) \cap Y \neq \emptyset$  for  $U, V$  open

and we say  $Y$  is an **irreducible subset**.

*Proof.* a)  $\implies$  b). Suppose that  $Y$  is irreducible in the subspace topology and  $Y \subseteq Z_1 \cup Z_2$ . This implies  $Y = (Z_1 \cap Y) \cup (Z_2 \cap Y)$  is a decomposition of closed sets. So either  $Z_1 \cap Y = Y$  or  $Z_2 \cap Y = Y \implies Y \subseteq Z_1$  or  $Y \subseteq Z_2$  as required.

b)  $\implies$  a). Suppose that  $Y = (Z_1 \cap Y) \cup (Z_2 \cap Y)$ . Then  $Y \subseteq Z_1 \cup Z_2$ , and for example  $Y \subseteq Z_1$ , which implies  $Z_1 \cap Y = Y$ . □

**Proposition 3.1.60**

Let  $Y \subset X$  be a **closed** subset then the following are equivalent

- a)  $Y$  is an irreducible subset
- b)  $Y = Z_1 \cup Z_2 \implies Y = Z_1$  or  $Y = Z_2$  where  $Z_1, Z_2$  are closed subsets of  $X$
- c)  $Y \subseteq Z_1 \cup Z_2 \implies Y \subseteq Z_1$  or  $Y \subseteq Z_2$  where  $Z_1, Z_2$  are closed subsets of  $X$

In other words in the lattice of closed subsets, the irreducible subsets are precisely the *join-prime* subsets.

*Proof.* a)  $\implies$  b). Clearly  $Z_1, Z_2$  are also closed subsets of  $Y$ , so the result follows by definition.

b)  $\iff$  c). This is (2.4.2).

c)  $\implies$  a). This was already proven in (3.1.59). □

**Remark 3.1.61**

Singletons  $\{x\}$  are always irreducible.

**Definition 3.1.62** (Irreducible Component)

We say that  $Y$  is an irreducible component if it is a maximal irreducible subset.

**Proposition 3.1.63**

Let  $f : X \rightarrow Y$  be a continuous map and  $Z \subset X$  irreducible. Then  $f(Z)$  is irreducible in  $Y$ .

*Proof.* Suppose  $f(Z) \subset Y_1 \cup Y_2$  for  $Y_1, Y_2$  closed. Then  $Z \subset f^{-1}(f(Z)) \subset f^{-1}(Y_1) \cup f^{-1}(Y_2)$ . Then by (3.1.59).b) wlog  $Z \subset f^{-1}(Y_1)$  therefore  $f(Z) \subset f(f^{-1}(Y_1)) \subset Y_1$ . Therefore by the same criterion  $f(Z)$  is irreducible. □

**Proposition 3.1.64**

Let  $Y \subset X$ . Then  $Y$  is irreducible iff  $\overline{Y}$  is.

*Proof.* This follows from (3.1.59).b) and (3.1.19) that  $Y \subset Z \iff \overline{Y} \subset Z$ . □

**Proposition 3.1.65** (Decomposition into Irreducible Components)

A topological space  $X$  may be decomposed into irreducible components. More precisely

- a) Every irreducible component is closed
- b) Every irreducible closed subset is contained in an irreducible component
- c)  $X$  is the union of irreducible components

*Proof.* We prove each in turn

- a) By (3.1.64)  $\bar{Y}$  is irreducible and closed, so by maximality  $Y = \bar{Y}$  is closed.
- b) We may show that the lattice of closed subsets is chain complete so we may use to show that the lattice of irreducible closed subsets is also chain complete (2.4.4). Therefore we may apply (2.1.55) to find an irreducible component.
- c) As  $\{x\}$  is irreducible every element is contained in an irreducible component by b).

□

**Corollary 3.1.66**

For  $x \in X$  the closure  $\overline{\{x\}}$  is an irreducible closed subset.

**Corollary 3.1.67**

$X$  is irreducible if and only if it has a single irreducible component.

**Definition 3.1.68** (Generic Point)

Let  $Z$  be an irreducible closed subset of  $X$ , then we say  $\eta \in X$  is a generic point of  $Z$  if  $Z = \overline{\{\eta\}}$ .

**Proposition 3.1.69**

Let  $X$  be an irreducible space then every open subset  $U$  is irreducible and connected.

*Proof.* By (3.1.58)  $U$  is dense in  $X$ . Therefore by (3.1.64)  $U$  is irreducible.

Suppose that  $U$  is not connected then by (...)  $U = (U \cap V) \sqcup (U \cap W)$  with  $V, W$  open subsets of  $X$  and  $U \cap V, U \cap W$  non-empty. Then as  $U$  is irreducible we have  $U \cap V \cap W \neq \emptyset$ , a contradiction. □

**Proposition 3.1.70**

Let  $X = \bigcup_{i=1}^n X_i$  where  $X_i$  are irreducible closed subsets, and mutually incomparable. Then the  $X_i$  are the irreducible components of  $X$ .

*Proof.* For any irreducible closed subset  $E$  we have by (3.1.60)  $E \subset X_i$  for some  $i$ . In particular every irreducible component is contained in some  $X_i$ . On the other hand every  $X_j$  is contained in an irreducible component by (3.1.65). By incomparability we deduce that each  $X_i$  is an irreducible component, and that the set is complete. □

**Proposition 3.1.71**

Let  $Y \subset X$  such that  $Y = \bigcup_{i=1}^n Y_i$  where  $Y_i$  are the irreducible components of  $Y$ . Then the set of irreducible components of  $\bar{Y}$  is  $\{\bar{Y}_i\}_{i=1 \dots n}$ , and these are distinct.

*Proof.* By (3.1.64)  $\bar{Y}_i$  are irreducible subsets which, by (3.1.23) cover  $\bar{Y}$ . By (3.1.65)  $Y_i$  is closed in  $Y$  and so by (3.1.21)  $\bar{Y}_i \cap Y = Y_i$ . This shows that the  $\bar{Y}_i$  are distinct and also incomparable. Then (3.1.70) shows that these must be the irreducible components. □

**Proposition 3.1.72**

Let  $U \subset X$  be an open subset. There is an order isomorphism

$$\begin{aligned} \{Y \subset U \mid Y \text{ irreducible, closed and non-empty}\} &\longleftrightarrow \{Z \subset X \mid Z \text{ irreducible and closed and } Z \cap U \neq \emptyset\} \\ Y &\rightarrow \text{cl}_X(Y) \\ Z \cap U &\leftarrow Z \end{aligned}$$

which restricts to irreducible components.

*Proof.* By (3.1.64) the first map is well-defined. Conversely  $Z \cap U$  is closed in  $U$  and open in  $Z$ , so irreducible by (3.1.69).

By (3.1.58)  $Z \cap U$  is dense in  $Z$ , that is  $Z = \text{cl}_Z(Z \cap U) \stackrel{(3.1.21)}{=} \text{cl}_X(Z \cap U) \cap Z \implies Z \subset \text{cl}_X(Z \cap U)$ , and so by minimality they are equal.

By (3.1.21)  $\text{cl}_X(Y) \cap U = \text{cl}_U(Y) = Y$ . Therefore the maps are mutually inverse and order preserving. □

**Proposition 3.1.73**

Suppose  $X$  is a topological space with open cover  $\bigcup_{i \in I} U_i$  such that

- a)  $U_i$  is irreducible for all  $i \in I$  and
- b)  $U_i \cap U_j$  is non-empty for all  $i, j \in I$ .

Then  $X$  is irreducible.

*Proof.* Suppose that  $X = Z_1 \cup Z_2$ . Pick some  $i \in I$  then by (3.1.59) wlog  $U_i \cap Z_1 \neq \emptyset$ . Similarly for all  $j \in J$  by assumption  $U_i \cap U_j$  is a non-empty open subset of  $U_j$  and therefore dense (3.1.58). Further  $U_i \cap U_j$  is contained in the closed subset  $Z_1 \cap U_j$ . We conclude that  $U_j \subset Z_1 \cap U_j$ . Therefore  $X \subset Z_1$ . By (3.1.59) we further conclude that  $X$  is irreducible.  $\square$

### 3.1.11 Noetherian Topological Spaces

**Definition 3.1.74** (Noetherian)

A topological space  $X$  is **Noetherian** if the lattice of closed subsets is **Artinian** (i.e. satisfies the descending chain condition).

**Proposition 3.1.75** (Decomposition into Irreducibles)

Let  $X$  be a Noetherian topological space. Then every closed subset  $Y$  may be expressed uniquely as a finite, **incomparable** union of irreducible closed subsets. These are precisely the irreducible components of  $Y$ .

In particular  $X$  has only finitely many irreducible components.

*Proof.* The lattice of closed subsets is distributive and Artinian by definition. Therefore the result follows from (3.1.60) and (2.4.7).  $\square$

**Proposition 3.1.76**

Let  $X$  be a Noetherian topological space and  $Y \subset X$ . Then  $Y$  is Noetherian.

**Proposition 3.1.77**

Let  $X = \bigcup_{i=1}^N X_i$  be topological space such that  $X_i$  is Noetherian under the subspace topology. Then  $X$  is Noetherian.

*Proof.* Let  $\{Z_n\}_{n \in \mathbb{N}}$  be a descending chain of closed subsets of  $X$ . By definition  $\{Z_n \cap X_i\}_{n \in \mathbb{N}}$  terminates for  $n \geq m_i$ . Then for  $n \geq \max(m_1, \dots, m_N)$  the sequence  $\{Z_n\}_{n \in \mathbb{N}}$  also terminates.  $\square$

**Proposition 3.1.78**

Let  $X$  be a Noetherian topological space and  $U \subset X$  an open subset. Then there is a bijection

$$\begin{array}{ccc} \{Y \subset U \mid Y \text{ closed and non-empty}\} & \longleftrightarrow & \{Z \subset X \mid Z \text{ closed and every irreducible component meets } U\} \\ Y & \rightarrow & \text{cl}_X(Y) \\ Z \cap U & \leftarrow & Z \end{array}$$

which is order preserving and restricts to closed irreducible subsets and irreducible components.

*Proof.* By (3.1.76) and (3.1.75)  $Y = Y_1 \cup \dots \cup Y_n$  where  $Y_i$  are the irreducible components of  $Y$ . Then by (3.1.71) the irreducible components of  $\bar{Y}$  are  $\bar{Y}_i$ , which evidently intersect  $U$ . Furthermore by definition  $Z \cap U$  is closed in  $U$  and the maps are well-defined.

By (3.1.72) the maps restricts to closed irreducible subsets (resp. components) and are bijections, in particular for  $Z$  irreducible closed we have  $\overline{Z \cap U} = Z$ .

By (3.1.21)  $\bar{Y} \cap U = Y$ . For the case  $Z$  is closed then as before  $Z = Z_1 \cup \dots \cup Z_n$  is a decomposition into irreducible components. By assumption  $Z_i \cap U \neq \emptyset$  for all  $i = 1 \dots n$ . Therefore by the irreducible case (3.1.72) and (3.1.23)

$$\overline{Z \cap U} = \overline{Z_1 \cap U} \cup \dots \cup \overline{Z_n \cap U} = Z_1 \cup \dots \cup Z_n = Z.$$

Therefore the maps are mutually inverse as required.  $\square$

### 3.1.12 Krull Dimension

References :

- Éléments de géométrie algébrique IV [Gro64, Chap. 0 §14.4.1]
- Some remarks on biequidimensionality of topological spaces and Noetherian schemes [Hei17]

For a Noetherian topological space  $X$  the closed subsets form an Artinian, **distributive lattice**. Furthermore the irreducible subsets are precisely the **join-prime** elements by (3.1.60). Therefore we may use the notions of Krull Lattice developed in Section 2.5.

**Definition 3.1.79** (Chain of irreducibles)

Let  $X$  be a Noetherian topological space. A **chain** of irreducible closed subsets

$$Z_0 \subsetneq Z_1 \subsetneq \dots \subsetneq Z_n$$

is said to have **length**  $n$ . A chain is **saturated** if there is no proper refinement, that is if  $Y$  is irreducible then

$$Z_i \subseteq Y \subseteq Z_{i+1} \implies Y = Z_i \text{ or } Y = Z_{i+1}.$$

If in addition  $Z_n$  (resp.  $Z_0$ ) is maximal (resp. minimal) then the chain is **maximal**.

**Definition 3.1.80** (Krull Lattice of Closed Subsets)

Let  $X$  be a topological space.

- The **Krull dimension**  $\dim X$  of  $X$  is the maximal length of all chains of irreducible closed subsets. Note this may be  $\infty$ .
- The **height** or **codimension** of an irreducible closed subset  $Y \subseteq X$ , denoted  $\text{codim}(Y, X)$ , is the maximal length of chains of irreducible subsets containing  $Y$ .
- When  $Y$  is not irreducible we write

$$\text{codim}(Y, X) = \inf_{\alpha} \text{codim}(Y_{\alpha}, X)$$

where  $Y_{\alpha}$  varies among the irreducible components of  $Y$ .

If  $\dim X < \infty$  then we say  $X$  is **finite-dimensional**. In this case it's clear the closed subsets of a topological space form a **Krull Lattice** where the irreducible closed subsets are precisely the join-prime elements of the lattice.

Note any saturated chain for  $Y \subset X$  must start at  $Y$  and terminate at an irreducible component of  $X$ . In particular if  $X$  is irreducible then a saturated chain must terminate at  $X$ .

**Proposition 3.1.81** (Extending Chains)

Let  $X$  be a finite-dimensional topological space. Then

- Every chain is contained in a saturated chain with the same endpoints
- Every chain is contained in a maximal chain

**Proposition 3.1.82** (Simple properties of (co-)dimension)

Let  $X$  be a topological space. Then the following properties hold

- $\dim(X) = \sup_{\alpha} \dim(X_{\alpha})$  where  $X_{\alpha}$  are the irreducible components of  $X$
- $\text{codim}(Y, X) = \sup_{\alpha} \text{codim}(Y, X_{\alpha})$  where  $X_{\alpha}$  are the irreducible components of  $X$  containing  $Y$
- $\dim Y + \text{codim}(Y, X) \leq \dim X$  for every closed subset  $Y \subset X$
- $\text{codim}(Y, Z) + \text{codim}(Z, T) \leq \text{codim}(Y, T)$  for  $Y \subset Z \subset T$  irreducible closed subsets
- $Y \subsetneq Z$  is a saturated chain of irreducible closed subsets if and only if  $\text{codim}(Y, Z) = 1$ .
- Let  $Y$  be a closed subset of  $X$ . Then  $\text{codim}(Y, X) = 0 \iff$  every irreducible component of  $Y$  is an irreducible component of  $X$ . In particular if  $X$  is irreducible then  $\text{codim}(Y, X) = 0 \iff Y = X \iff \dim X = \dim Y$ .

In particular if  $X$  is finite-dimensional then all codimensions are also finite.

**Proposition 3.1.83**

Let  $X$  be an irreducible topological space and  $Y \subset X$  a closed subset. Then  $\dim Y \leq \dim X$  with equality iff  $Y = X$ .

*Proof.* The inequality follows from (3.1.82).b). If  $\dim Y = \dim X$  then by c)  $\text{codim}(Y, X) = 0$  and so by f) we have  $Y = X$ .  $\square$

**Definition 3.1.84** (Properties)

Let  $X$  be a topological space of finite dimension. Then we say  $X$  is

- **Equidimensional** if all irreducible components of  $X$  have the same dimension
- **Equicodimensional** if  $\text{codim}(Y, X)$  is constant as  $Y$  varies over minimal irreducible subsets of  $X$
- **Biequidimensional** if all **maximal chains** of irreducible subsets have the same length.
- **Quasi-Biequidimensional** if every irreducible component is biequidimensional
- **Catenary** if any two **saturated chains** with the same endpoints, say  $Y$  and  $Z$ , have the same length, namely  $\text{codim}(Y, Z)$

Observe that **quasi-biequidimensional** + **equidimensional**  $\iff$  **biequidimensional** and **irreducible**  $\implies$  **equidimensional**

**Proposition 3.1.85** (Equivalent characterisations of biequidimensional)

Suppose  $X$  is a topological space of finite dimension. Then the following are equivalent

- a)  $X$  is quasi-biequidimensional
- b)  $X$  is catenary and every irreducible component is equidimensional
- c)  $X$  satisfies the codimension formula for  $Z \subset Y$  irreducible

$$\dim Y = \dim Z + \operatorname{codim}(Z, Y)$$

- d)  $X$  satisfies c) in the case  $\operatorname{codim}(Z, Y) = 1$

Furthermore for irreducible subsets  $Z \subset Y$

$$\operatorname{codim} Z = \operatorname{codim}(Z, Y) + \operatorname{codim} Y$$

*Proof.* This is a translation of (2.5.11) to the topological case. □

**Proposition 3.1.86** (Codimension Formula)

Suppose  $X$  is a quasi-biequidimensional topological space. Then for every subset  $Z \subset X$  which is irreducible and closed, and every closed subset  $Y \subset Z$  the following relations hold.

$$\begin{aligned} \dim Z &= \dim Y + \operatorname{codim}(Y, Z) \\ \operatorname{codim} Y &= \operatorname{codim}(Y, Z) + \operatorname{codim} Z \end{aligned}$$

Further every closed subset is also quasi-biequidimensional.

*Proof.* This follows from (2.5.13) and (2.5.15). □

**Proposition 3.1.87**

Suppose  $X = \bigcup_{i \in I} U_i$ . Then  $\dim X = \sup_{i \in I} \dim U_i$ .

*Proof.* By (3.1.72) a chain of irreducible closed subsets of  $U_i$  lifts to a chain of the same length in  $X$ . Therefore  $\dim U_i \leq \dim X$  for all  $i \in I$  whence  $\sup_{i \in I} \dim U_i \leq \dim X$ . Similarly given a chain of irreducible closed subsets

$$X_0 \subsetneq \dots \subsetneq X_n \subseteq X$$

choose  $i$  such that  $X_0 \cap U_i \neq \emptyset$ . Then (3.1.78) shows that this restricts to a chain of the same length in  $U_i$ . Therefore  $\dim X \leq \dim U_i \leq \sup_{i \in I} \dim U_i$ . □

**Proposition 3.1.88**

Let  $X$  be a catenary topological space and  $U \subset X$  an open subset. Then  $U$  is catenary.

*Proof.* Consider  $Z \subset Y \subset U$  irreducible closed subsets. Then for any saturated chain

$$Z = Z_0 \subsetneq \dots \subsetneq Z_n = Y$$

we have by (3.1.72)

$$\overline{Z} = \overline{Z_0} \subsetneq \dots \subsetneq \overline{Z_n} = \overline{Y}$$

a saturated chain in  $X$ . As  $X$  is catenary we find  $n = \operatorname{codim}(\overline{Z}, \overline{Y})$ . As the chain was arbitrary we find  $U$  is catenary. □

**Lemma 3.1.89**

Suppose that  $X$  is a topological space and  $x \in X$  is quasi-closed. Then  $\dim \overline{\{x\}} = 0$ .

Conversely suppose  $Z$  is an irreducible closed set of dimension 0. Then  $Z = \overline{\{z\}}$  for all  $z \in Z$  and  $Z$  consists of quasi-closed elements.

*Proof.* By (3.1.66)  $\overline{\{x\}}$  is irreducible. Suppose  $\emptyset \neq Z \subset \overline{\{x\}}$  is closed then  $y \in Z \implies x \in \overline{\{y\}} \subset Z$  whence  $\overline{\{x\}} = Z$ . This shows that  $\dim \overline{\{x\}} = 0$ .

Conversely if  $\dim Z = 0$  then  $\overline{\{z\}} \subset Z$  is an irreducible subset for all  $z \in Z$ , whence they are equal. This also shows that  $z$  is quasi-closed. □

**Proposition 3.1.90**

Let  $X$  be a Noetherian topological space and  $Z$  a closed subset. Then

$$\dim Z = 0 \iff Z = \overline{\{z_1\}} \cup \dots \cup \overline{\{z_n\}}$$

for some  $z_1, \dots, z_n \in Z$  quasi-closed.

*Proof.* By (3.1.75)  $Z = Z_1 \cup \dots \cup Z_n$  for  $Z_i$  is a decomposition into irreducible components. By (3.1.82).a)  $\dim Z_i = 0$ , so by (3.1.89)  $Z_i = \overline{\{z_i\}}$ .

Conversely  $Z = \overline{\{z_1\}} \cup \dots \cup \overline{\{z_n\}}$  is a decomposition into irreducible closed subsets which are incomparable (3.1.89). By (3.1.70) these are the irreducible components. Therefore

$$\dim Z \stackrel{(3.1.82).a)}{=} \sup_i \dim \overline{\{z_i\}} \stackrel{(3.1.89)}{=} 0$$

□

**Proposition 3.1.91** (Curves have co-finite topology)

Let  $X$  be an irreducible, Noetherian and symmetric topological space such that  $\dim X = 1$ . Then  $X_0$  has the cofinite topology.

*Proof.* We may reduce to the case  $X = X_0$  is Kolmogorov, then by (3.1.44) every point  $x \in X$  is closed.

Suppose  $Z \subset X$  is finite then  $Z = \bigcup_{z \in Z} \{z\}$  is a finite union of closed subsets and therefore closed. Conversely if  $Z \subset X$  is closed and proper then by (3.1.82).f)  $\dim Z = 0$  and therefore  $Z$  is finite by (3.1.90). □

**Proposition 3.1.92**

Let  $X$  be a symmetric topological space. Then there is a correspondence

$$\begin{aligned} X_0 &\longleftrightarrow \{W \subset X \mid \dim W = 0 \text{ closed and irreducible}\} \\ x &\longrightarrow \overline{\{x\}} \end{aligned}$$

**3.1.13 Product Topology****Definition 3.1.93** (Product Topology)

Let  $\{X_i\}_{i \in I}$  be a family of topological spaces. The **product topology** is the topology generated by the base of open sets of the form

$$\prod_{i \in I} U_i$$

where  $U_i \subset X_i$  is open and only finitely many are proper subsets of  $X$ .

**Proposition 3.1.94**

Let  $\{X_i\}_{i \in I}$  be family of topological spaces. The family of sets given in (3.1.93) is a base. The topology generated is the smallest family of subsets for which the projection maps

$$\pi_i : \prod_{i \in I} X_i \rightarrow X_i$$

are continuous. Furthermore if each  $X_i$  is Hausdorff (resp. Kolmogorov), then so is the product.

**Proposition 3.1.95** (Product of Sequences)

Let  $X_1, \dots, X_m$  be a family of topological spaces and  $a_n^i \rightarrow \alpha^i$  for  $i = 1 \dots m$ . Then  $(a_n^1, \dots, a_n^m) \rightarrow (\alpha^1, \dots, \alpha^m)$ .

**Proposition 3.1.96** (Diagonal is Closed)

Let  $X$  be a topological space. Then  $X$  is Hausdorff if and only if  $\{(x, x) \mid x \in X\}$  is closed in  $X \times X$ .

**Proposition 3.1.97**

Let  $f, g : X \rightarrow Y$  be continuous maps and  $Y$  a Hausdorff space. Then

$$\{x \in X \mid f(x) = g(x)\}$$

is closed.

*Proof.* The level set is  $(f \times g)^{-1}(\Delta_Y)$  which by (3.1.96) is closed. □

### 3.1.14 Quasi-Compactness

**Definition 3.1.98** (Cluster Point of a Sequence)

Let  $(x_n)$  be a sequence in a topological space  $X$ . We say that  $x \in X$  is

- a) a **cluster point** (or **accumulation point**) of  $x_n$  if for every  $U \in \mathcal{O}_x$  the set  $\{m \in \mathbb{N} \mid x_m \in U\}$  is infinite
- b) a **subsequential limit** of  $x_n$  if there exists some subsequence  $x_{n_k} \rightarrow x$

In general the latter implies the former, and if  $X$  is first countable the concepts are equivalent as we will see.

**Lemma 3.1.99** (Cluster Point is Subsequential Limit Point)

Let  $X$  be a first countable topological space and  $x$  a cluster point of a sequence  $(x_n)$ . Then there exists some subsequence  $x_{n_k}$  such that  $x_{n_k} \rightarrow x$ .

*Proof.* Let  $(U_n)$  be a local base at  $x$ . By replacing it with  $U'_n := U_1 \cap \dots \cap U_n$  we may assume it is without loss of generality decreasing. Define  $n_0 = 0$  and  $n_k$  inductively by

$$n_k := \min\{m > n_{k-1} \mid x_m \in U_k\}$$

which is well-defined precisely because  $x$  is a cluster point. Then  $x_{n_k} \in U_k$  which shows that  $x_{n_k} \rightarrow x$  as required.  $\square$

**Definition 3.1.100**

Let  $X$  be a topological space. We say it is

- a) **Quasi-Compact** if every open cover of  $X$  has a finite subcover
- b) **Countably Quasi-Compact** if every countable open cover of  $X$  has a finite subcover
- c) **Sequentially Quasi-Compact** if every sequence  $(x_n)$  in  $X$  has a convergent subsequence

If in addition  $X$  is Hausdorff (e.g. a metric space) we say that  $X$  is compact (resp. countably compact, sequentially compact).

**Proposition 3.1.101** (Criteria for Countable Compactness)

Let  $X$  be a topological space. Then the following are equivalent

- a)  $X$  is countably quasi-compact
- b) Let  $\{Z_n\}_{n \in \mathbb{N}}$  be a countable collection of closed subsets such that every finite set has non-empty intersection. Then the intersection  $\bigcap_{n \in \mathbb{N}} Z_n$  is non-empty
- c) For every increasing sequence of proper open subsets

$$U_1 \subseteq U_2 \subseteq \dots \subseteq U_n \subseteq \dots$$

the union  $\bigcup_{n=1}^{\infty} U_n$  is proper.

- d) For every decreasing sequence of non-empty closed subsets

$$Z_1 \supseteq Z_2 \supseteq \dots \supseteq Z_n \supseteq \dots$$

the intersection  $\bigcap_{n=1}^{\infty} Z_n$  is non-empty

*Proof.* a)  $\iff$  b) and c)  $\iff$  d) are simple consequences of De Morgan's laws. For a)  $\implies$  c) suppose  $\bigcup_{n=1}^{\infty} U_n = X$  then by assumption  $U_{n_1} \cup \dots \cup U_{n_k} = X$ , and by the increasing assumption  $U_{n_k} = X$  which is a contradiction.

For c)  $\implies$  a), suppose we have an open cover  $X = \bigcup_{n=1}^{\infty} U_n$  define the increasing open cover

$$U'_n := \bigcup_{i=1}^n U_i$$

by the contrapositive  $U'_n$  is not proper for some  $n$ , which means precisely that there is a finite subcover.  $\square$

**Proposition 3.1.102**

Let  $X$  be a topological space. Then sequentially quasi-compact  $\implies$  countably quasi-compact. Under the assumption of first countability the converse holds.

*Proof.*  $\implies$  ) Consider a countable open cover  $X = \bigcup_{n=1}^{\infty} U_n$  with no finite subcover. Then there exists  $x_n \in X \setminus \bigcup_{i=1}^n U_i$ . By assumption there is a convergent subsequence  $x_{n_k} \rightarrow x$ . We have  $x \in U_N$  for some  $N$  but  $n_k \geq N \implies x_{n_k} \notin U_N$ .

$\impliedby$  ) Consider a sequence  $(x_n)$  and define

$$Z_n := \overline{\{x_k : k > n\}}$$

which is a decreasing sequence of non-empty closed subsets. By the countable compactness criteria (3.1.101)  $Z := \bigcap_{n=1}^{\infty} Z_n$  is non-empty. However we may show this is precisely the set of cluster points of  $(x_n)$ , i.e. there exists a cluster point  $x \in Z$ . By (3.1.99) there exists a subsequence such that  $x_{n_k} \rightarrow x$  as required.  $\square$

**Lemma 3.1.103** (Lindelof Lemma)

Let  $X$  be a second countable topological space. Then every open cover has a countable subcover.

*Proof.* Let  $\mathcal{B}$  be a countable base and  $\{U_\alpha\}_{\alpha \in \mathcal{A}}$  an open cover of  $X$ . Note for every  $U_\alpha$  we have some subfamily  $\mathcal{B}_\alpha \subset \mathcal{B}$  such that  $U_\alpha = \bigcup \mathcal{B}_\alpha$ . Consider  $\mathcal{B}' := \bigcup \{\mathcal{B}_\alpha \mid \alpha \in \mathcal{A}\} \subset \mathcal{B}$ . By definition for every  $V \in \mathcal{B}'$  we have  $V \subset U_\alpha$  for some  $\alpha \in \mathcal{A}$ . In other words there exists a map  $f : \mathcal{B}' \rightarrow \mathcal{A}$  such that  $V \subset U_{f(V)}$ . We may verify that  $\{U_\alpha\}_{\alpha \in \text{Im}(f)}$  is a countable subcover.  $\square$

We summarise the relationships between the concepts

**Proposition 3.1.104**

Let  $X$  be a topological space. Then

- a) Quasi-Compact, Sequentially Quasi-Compact  $\implies$  Countably Quasi-compact
- b) First Countable and (Countably) Quasi-Compact  $\implies$  Sequentially Quasi-Compact
- c) Second Countable and Countably or Sequentially Quasi-Compact  $\implies$  Quasi-Compact

*Proof.* a) The first implication is obvious and the second implication is (3.1.102)

b) This is (3.1.102)

c) This follows from (3.1.103) and a)  $\square$

**Proposition 3.1.105** (Compact subsets of Hausdorff Spaces are Closed)

Let  $X$  be a Hausdorff topological space and  $Y \subset X$  a subset compact with respect to the subspace topology. Then  $Y$  is closed.

*Proof.* Fix  $x \in X \setminus Y$ . Then for every  $y \in Y$  by assumption there exists disjoint open subsets  $x \in U_y$  and  $y \in V_y$ . Then  $\{V_y \cap Y \mid y \in Y\}$  is an open cover for  $Y$ , so there exists a finite set of points  $y_1, \dots, y_n$  such that  $Y \subseteq V_{y_1} \cup \dots \cup V_{y_n}$ . Furthermore

$$U := U_{y_1} \cap \dots \cap U_{y_n}$$

is an open neighbourhood of  $x$  disjoint from each  $V_{y_i}$  and therefore disjoint from  $Y$ . By (3.1.19) we see that  $x \notin \overline{Y}$ , which shows  $Y = \overline{Y}$ , i.e.  $Y$  is closed.  $\square$

**Proposition 3.1.106** (Closed subsets of Compact Spaces are Compact)

Let  $X$  be a quasi-compact topological space and  $Y \subset X$  a closed subset. Then  $Y$  is quasi-compact under the subspace topology.

*Proof.* Let  $\mathcal{U} := \{U_\alpha\}_{\alpha \in \mathcal{A}}$  be an open cover for  $Y$ . Then by definition  $U_\alpha = U'_\alpha \cap Y$  where  $U'_\alpha \subset X$  are open. Evidently

$$\mathcal{U}' : \{U'_\alpha\}_{\alpha \in \mathcal{A}} \cup \{X \setminus Y\}$$

is an open cover of  $X$ . Therefore by assumption it has a finite subcover, which immediately yields a finite subcover of  $Y$  as required.  $\square$

**Proposition 3.1.107** (Image of a Compact Space is Compact)

Let  $f : X \rightarrow Y$  be a continuous map of topological spaces where  $X$  is quasi-compact. Then the image  $f(X)$  is quasi-compact.

Further if  $Y$  is Hausdorff then  $f$  is an open map (i.e.  $f(U)$  is open for every open  $U \subset X$ ). If in addition  $f$  is injective then it is a homeomorphism onto its image.

*Proof.* We may assume without loss of generality that  $f(X) = Y$ . Let  $\{U_\alpha\}_{\alpha \in \mathcal{A}}$  be an open cover of  $Y$ . In other words  $f(X) \subseteq \bigcup_{\alpha \in \mathcal{A}} U_\alpha \implies X \subseteq \bigcup_{\alpha \in \mathcal{A}} f^{-1}(U_\alpha)$ . By assumption we have a finite subcover

$$X \subseteq f^{-1}(U_{\alpha_1}) \cup \dots \cup f^{-1}(U_{\alpha_n})$$

which implies

$$f(X) \subseteq U_{\alpha_1} \cup \dots \cup U_{\alpha_n}$$

since  $f$  is assumed to be surjective, and we see that  $f(X)$  is compact.  $\square$

Analogously we have similar results for sequentially compact spaces, which may be conceptually simpler.

**Proposition 3.1.108**

*Let  $X$  be a first countable Kolmogorov space and  $Y \subset X$  a subset of a sequentially compact with respect to the subspace topology. Then  $Y$  is closed.*

*Proof.* Consider  $x \in \overline{Y}$  then by (3.1.19) there is some sequence  $x_n \rightarrow x$ . Further by assumption there is some subsequence  $x_{n_k} \rightarrow y \in Y$ . By uniqueness of limits (3.1.50) we have  $x = y$ , and in particular  $\overline{Y} = Y$  is closed.  $\square$

**Proposition 3.1.109**

*Let  $X$  be a sequentially compact topological space, and  $Y \subset X$  a closed subset. Then  $Y$  is sequentially compact.*

*Proof.* Let  $x_n \in Y$  be a sequence. By assumption it has a convergent subsequence  $x_{n_k} \rightarrow x$  where  $x \in X$ . By (3.1.18)  $x$  is an adherent point of  $Y$  and therefore lies in  $Y$  by (3.1.19).  $\square$

**Proposition 3.1.110**

*Let  $f : X \rightarrow Y$  be a continuous map of topological spaces where  $X$  is sequentially compact. Then the image  $f(X)$  is sequentially compact.*

*Proof.* Let  $y_n = f(x_n)$  be a sequence in  $f(X)$ . Then by assumption  $x_{n_k} \rightarrow x$ , and therefore by (...)  $y_{n_k} \rightarrow f(x)$ .  $\square$

**Proposition 3.1.111**

*Let  $X$  be a topological space with an open cover  $X = \bigcup_{i=1}^n U_i$  such that  $U_i$  is quasi-compact under the subspace topology. Then  $X$  is quasi-compact.*

*Proof.* Let  $X = \bigcup_{\alpha \in \mathcal{A}} V_\alpha$  then  $U_i = \bigcup_{\alpha \in \mathcal{A}} (V_\alpha \cap U_i)$ . By assumption there is some finite subset  $\mathcal{A}_i \subset \mathcal{A}$  such that  $U_i = \bigcup_{\alpha \in \mathcal{A}_i} (V_\alpha \cap U_i)$ . Then

$$X = \bigcup_{\alpha \in \mathcal{A}_1 \cup \dots \cup \mathcal{A}_n} V_\alpha$$

Therefore  $X$  is quasi-compact.  $\square$

**Proposition 3.1.112** (Noetherian  $\iff$  Hereditarily Quasi-Compact)

*Let  $X$  be a topological space. Then  $X$  is Noetherian if and only if every open subset is quasi-compact.*

*Proof.*  $\implies$  ) By (3.1.76) every open subset is Noetherian, so it is enough to show that  $X$  is quasi-compact. By definition the lattice of open subsets is Noetherian and by (2.1.62) this holds iff every family of open subsets has a maximal element.

Suppose that we have an open cover  $X = \bigcup_{\alpha \in \mathcal{A}} U_\alpha$ . Let  $\mathcal{F}$  be the family of open subsets expressible as a finite union of sets  $U_\alpha$ . By assumption it has a maximal element

$$U := U_{\alpha_1} \cup \dots \cup U_{\alpha_n}$$

Suppose that  $U \subsetneq X$  is proper then choose  $x \in X \setminus U$ . By hypothesis there is some  $\alpha_{n+1}$  such that  $x \in U_{\alpha_{n+1}}$ . Then  $U \subsetneq U \cup U_{\alpha_{n+1}} \in \mathcal{F}$ , contradicting maximality. Therefore  $U = X$  and  $X$  is quasi-compact as required.

$\impliedby$  ) Omitted.  $\square$

**Proposition 3.1.113**

*Let  $X, Y$  be quasi-compact topological spaces. Then  $X \times Y$  is quasi-compact.*

*Proof.* Let  $\{U_\lambda\}_{\lambda \in \Lambda}$  be an open cover of  $X \times Y$ . Therefore for each  $(x, y)$  there is an  $\lambda(x, y)$  such that  $(x, y) \in U_{\lambda(x, y)}$ . By definition of the product topology (3.1.93) we may find open sets  $V_{(x, y)} \subset X$  and  $W_{(x, y)} \subset Y$  such that  $(x, y) \in V_{(x, y)} \times W_{(x, y)} \subset U_{\lambda(x, y)}$ .

Fix  $x$  then the family  $\{W_{(x, y)}\}_{y \in Y}$  is an open cover of  $Y$ . So we may find a finite subset  $J_x \subset Y$  such that  $\{W_{(x, y)}\}_{y \in J_x}$  covers  $Y$ .

Consider the open sets

$$V'_x := \bigcap_{y \in J_x} V_{(x,y)}$$

which evidently cover  $X$ . Then there is a finite subset  $I \subset X$  such that  $\{V'_x\}_{x \in I}$  covers  $X$ .

Then we claim that the family  $\{U_{\lambda(x,y)}\}_{x \in I, y \in J_x}$  is a finite subcover of  $X \times Y$ . For given  $(x', y') \in X \times Y$  there is an  $x \in I$  such that  $x' \in V'_x$ , and  $y \in J_x$  such that  $y' \in W_{(x,y)}$ . Evidently  $x' \in V_{(x,y)}$ , whence  $(x', y') \in U_{\lambda(x,y)}$  as required.  $\square$

### 3.1.15 Connectedness

#### Definition 3.1.114

We say a topological space  $X$  is **disconnected** if  $X = U \cup V$  for two non-empty, disjoint open subsets  $U, V$ . Otherwise we say  $X$  is **connected**.

#### Proposition 3.1.115

Let  $X$  be a topological space. The following are equivalent

- a)  $X$  is connected
- b) The only subsets which are both open and closed are  $X$  and  $\emptyset$

#### Proposition 3.1.116

Let  $f : X \rightarrow Y$  be a continuous map and  $X$  connected. Then  $f(X)$  is also connected.

### 3.1.16 Order Topology

#### Definition 3.1.117 (Order Topology)

Let  $(X, \leq)$  be a totally ordered set. Then the **order topology** is the topology generated by the subbase of half-open intervals

$$\{y \in X \mid y < x\} \text{ and } \{y \in X \mid y > x\}$$

for  $x \in X$ .

#### Proposition 3.1.118 (Base for order topology)

Let  $(X, \leq)$  be a totally ordered set. Then the sets of the form

$$\{y \in X \mid y < x\}, \{y \in X \mid y > x\} \text{ and } \{y \in X \mid x < y < z\}$$

for  $x, z \in X$  constitute a base for the order topology.

### 3.2 Sheaves

Sheaves are typically used to organise sets of functions which are defined by a local differentiability condition (e.g. holomorphic, smooth, etc.). However in some cases it may be convenient to take an abstract category-theoretic approach as the proofs become more conceptual and naturally leads to further generalisations where the topological space is replaced by an arbitrary category. However for the sheafification construction we take (the standard) concrete approach as this is much simpler and more enlightening.

In general we assume that the value category  $\mathcal{C}$  admits arbitrary products and equalisers (and therefore also limits, and in particular terminal objects), as well as filtered colimits / direct limits. In some cases we may also require it to be algebraic.

**Definition 3.2.1** (Abstract Presheaf)

Let  $X$  be a topological space,  $\mathcal{B}$  a base for the topology and  $\mathcal{C}$  a category. A  **$\mathcal{B}$ -presheaf** on  $X$  is a functor

$$\mathcal{F} : (\mathcal{B}, \supseteq) \longrightarrow \mathcal{C}$$

from the poset category of basic open subsets to  $\mathcal{C}$  ordered by reverse inclusion for which

$$\mathcal{F}(\emptyset) = \star \text{ is a terminal object in } \mathcal{C}$$

This means for every pair of open subsets  $U, V \in \mathcal{B}$  such that  $V \subseteq U$  there exists morphisms  $\rho_{UV} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$  which satisfy the following properties

- a)  $\rho_{UU} = 1_{\mathcal{F}(U)}$
- b)  $\rho_{VW} \circ \rho_{UV} = \rho_{UW}$  for any basic open subset  $W \subset U$

A morphism of presheaves is precisely a natural transformation. That is  $\alpha : \mathcal{F} \rightarrow \mathcal{G}$  is a family of morphisms  $\alpha_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$  for which

$$\rho_{UV} \circ \alpha_U = \alpha_V \circ \rho_{UV}$$

This determines a category  $\mathbf{PSh}(\mathcal{B}; \mathcal{C})$ . We may omit the value category  $\mathcal{C}$  if there are no additional hypotheses.

**Definition 3.2.2** (Abstract Sheaf)

Let  $X$  be a topological space,  $\mathcal{B}$  a base for the topology closed under finite intersections and  $\mathcal{F}$  a  $\mathcal{B}$ -presheaf.

For every family of non-empty basic open subsets  $\{U_i\}_{i \in I}$  with  $U = \bigcup_{i \in I} U_i$ , consider the sequence of morphisms

$$\mathcal{F}(U) \xrightarrow{e} \prod_{i \in I} \mathcal{F}(U_i) \xrightleftharpoons[d_1]{d_0} \prod_{(i,j) \in I \times I} \mathcal{F}(U_i \cap U_j)$$

where  $e = (\rho_{UU_i})_{i \in I}$ ,  $d_0 = (\rho_{U_i(U_i \cap U_j)} \circ \pi_i)_{(i,j) \in I \times I}$  and  $d_1 = (\rho_{U_j(U_i \cap U_j)} \circ \pi_j)_{(i,j) \in I \times I}$ .

We say that  $\mathcal{F}$  is a  **$\mathcal{B}$ -sheaf** if  $e$  is an equaliser for  $(d_0, d_1)$  for all such families of open subsets.

This determines a full subcategory  $\mathbf{Sh}(\mathcal{B}; \mathcal{C})$  of  $\mathbf{PSh}(\mathcal{B}; \mathcal{C})$ .

When  $\mathcal{B}$  is just the collection of all open subsets of  $X$  then we say  $\mathcal{F}$  is a **sheaf**, and we denote the category by  $\mathbf{Sh}(X; \mathcal{C})$ .

**Proposition 3.2.3** (Sheaf Conditions for Concrete Categories)

Let  $X$  be a topological space,  $\mathcal{B}$  a base for the topology closed under finite intersections and  $\mathcal{C}$  is an algebraic category (2.6.96). Then a  $\mathcal{B}$ -presheaf  $\mathcal{F}$  is a sheaf if and only if it satisfies the following conditions, for every open cover  $U = \bigcup_{i \in I} U_i$  with  $U_i$  non-empty

- a)  $\sigma, \tau \in \mathcal{F}(U)$  such that  $\rho_{UU_i}(\sigma) = \rho_{UU_i}(\tau)$  for all  $i \in I$  implies  $\sigma = \tau$
- b)  $\sigma_i \in \mathcal{F}(U_i)$  such that  $\rho_{U_i(U_i \cap U_j)}(\sigma_i) = \rho_{U_j(U_i \cap U_j)}(\sigma_j)$  for all  $(i, j) \in I \times I$  implies there exists  $\sigma \in \mathcal{F}(U)$  such that  $\rho_{UU_i}(\sigma) = \sigma_i$ .

*Proof.* Suppose that  $\mathcal{F}$  is a sheaf in the abstract sense. By assumption the forgetful functor  $\mathcal{C} \rightarrow \mathbf{Set}$  preserves limits and so by (2.6.99)  $e$  is injective and

$$\text{Im}(e) = \{(\sigma_i)_{i \in I} \mid \sigma_i|_{U_i \cap U_j} = \sigma_j|_{U_i \cap U_j}\}$$

which gives both concrete sheaf conditions.

Conversely this condition ensures the sequence is an equaliser in  $\mathbf{Set}$ , and so by hypothesis is so in  $\mathcal{C}$ . □

**Proposition 3.2.4**

Let  $\phi : \mathcal{F} \rightarrow \mathcal{G}$  a morphism of sheaves. Then

- a)  $\phi$  is a **monomorphism** if  $\phi_U$  is a monomorphism for all  $U \subset X$
- b)  $\phi$  is an **isomorphism** if  $\phi_U$  is an isomorphism for all  $U \subset X$

If the value category  $\mathcal{C}$  is algebraic (2.6.96) then  $\phi$  is a monomorphism (resp. isomorphism) if and only if  $\phi_U$  is injective (resp. bijective) for all  $U \subset X$  (2.6.98).

**3.2.1 Stalks****Definition 3.2.5** (Stalk of a presheaf)

Let  $X$  be a topological space,  $\mathcal{B}$  a base closed under finite intersections and  $\mathcal{F}$  a  $\mathcal{B}$ -presheaf.

Let  $Z$  be an irreducible subset of  $X$ , then the family of open subsets  $U$  meeting  $Z$  is closed under finite intersections (3.1.59) and therefore forms a directed poset (2.6.77) ordered by reverse inclusion.

Therefore we may define the **stalk** at  $Z$  by the direct limit (2.6.79)

$$\mathcal{F}_Z := \varinjlim_{U \cap Z \neq \emptyset} \mathcal{F}(U)$$

This means there is a family of morphisms  $\{\rho_{UZ} : \mathcal{F}(U) \rightarrow \mathcal{F}_Z\}$  such that

- a) all open subsets  $V \subset U$  intersecting  $Z$  satisfy  $\rho_{VZ} \circ \rho_{UV} = \rho_{UZ}$ ,
- b) any collection morphisms  $\alpha_U : \mathcal{F}(U) \rightarrow \mathcal{A}$  satisfying  $\alpha_U = \alpha_V \circ \rho_{UV}$  factors uniquely through a morphism  $\alpha_Z : \mathcal{F}_Z \rightarrow \mathcal{A}$

When  $Z = \{x\}$  we denote this simply by  $\mathcal{F}_x$ .

**Proposition 3.2.6** (Explicit form of Stalk)

Consider the situation (3.2.5) where the value category  $\mathcal{C}$  is algebraic (2.6.96). Then

$$U(\mathcal{F}_Z) = (U \circ \mathcal{F})_Z$$

and so the stalk  $\mathcal{F}_Z$  may be constructed explicitly as follows

$$\begin{aligned} \mathcal{F}_Z &:= \{(U, \sigma) \mid U \in \mathcal{B}, \sigma \in \mathcal{F}(U)\} / \sim \\ (U, \sigma) \sim (V, \tau) &\iff \exists W \in \mathcal{B} \text{ s.t. } W \cap Z \neq \emptyset \text{ and } \sigma|_W = \tau|_W \end{aligned}$$

with morphisms

$$\begin{aligned} \rho_{UZ} : \mathcal{F}(U) &\rightarrow \mathcal{F}_Z \\ \sigma &\rightarrow [(U, \sigma)] \end{aligned}$$

In particular every element of the stalk  $\sigma_Z \in \mathcal{F}_Z$  is the image of some section  $\sigma \in \mathcal{F}(U)$  under  $\rho_{UZ}$  for some  $U \in \mathcal{B}$ .

*Proof.* By definition the forgetful functor  $U : \mathcal{C} \rightarrow \mathbf{Set}$  creates filtered colimits and so the result follows from (2.6.82).  $\square$

**Proposition 3.2.7** (Functoriality of Stalks)

Let  $X$  be a topological space with base  $\mathcal{B}$  closed under finite intersections  $\phi : \mathcal{F} \rightarrow \mathcal{G}$  a morphism of  $\mathcal{B}$ -presheaves and  $Z \subset X$  an irreducible subset. Then there exists a unique morphism  $\phi_Z : \mathcal{F}_Z \rightarrow \mathcal{G}_Z$  ensuring the following diagram commute for all open subsets  $U$  meeting  $Z$

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\phi_U} & \mathcal{G}(U) \\ \rho_{UZ} \downarrow & & \downarrow \rho_{UZ} \\ \mathcal{F}_Z & \xrightarrow{\phi_Z} & \mathcal{G}_Z \end{array}$$

Furthermore  $(\psi \circ \phi)_Z = \psi_Z \circ \phi_Z$ . In the explicit case (3.2.6) then  $\phi_Z$  is given by

$$[(U, \sigma)] \rightarrow [(U, \phi_U(\sigma))]$$

This also means that for any  $A \in \mathcal{C}$  the bijection

$$\mathrm{Mor}(\mathcal{F}_Z, A) \longrightarrow (\mathcal{F}|_{\mathcal{N}_Z} \downarrow A)$$

is natural in  $\mathcal{F}$ , where  $\mathcal{N}_Z$  is the category of open sets intersecting  $Z$  ordered by reverse inclusion.

**Lemma 3.2.8** (Lifting Stalks)

Suppose we are in the situation of (3.2.6) and  $\sigma \in \mathcal{F}(U)$ ,  $\tau \in \mathcal{F}(V)$  are sections such that  $Z \cap U \cap V \neq \emptyset$ .

- Then  $\sigma_Z = \tau_Z$  if and only if there is a neighbourhood  $W \subseteq U \cap V$  meeting  $Z$  such that  $\sigma|_W = \tau|_W$ .
- If  $\sigma_Z = \tau_Z$  for all  $Z \cap U \cap V \neq \emptyset$ , then there is an open cover  $U \cap V = \bigcup_i U_i$  such that  $\sigma|_{U_i} = \tau|_{U_i}$ .
- If in addition  $\mathcal{F}$  is a sheaf then  $\sigma|_{U \cap V} = \tau|_{U \cap V}$ .

**Proposition 3.2.9** (Stalks when restricting to a base)

Let  $\mathcal{B}_2 \subset \mathcal{B}_1$  be bases for  $X$  which generate the same topology. Suppose  $\mathcal{F}$  is a  $\mathcal{B}_1$ -presheaf and  $Z \subset X$  is an irreducible subset. Then there is a unique morphism making the following diagram commute for all  $U \in \mathcal{B}_2$  intersecting  $Z$

$$\begin{array}{ccc} \mathcal{F}|_{\mathcal{B}_2}(U) & \xlongequal{\quad} & \mathcal{F}(U) \\ \rho_{UZ} \downarrow & & \downarrow \rho_{UZ} \\ (\mathcal{F}|_{\mathcal{B}_2})_Z & \xrightarrow{\sim} & \mathcal{F}_Z \end{array}$$

and it is an isomorphism.

In the situation of (3.2.6) the map is of the obvious form.

*Proof.* Let  $\mathcal{I}_2$  and  $\mathcal{I}_1$  be the diagram of open subsets  $U$  of  $\mathcal{B}_2$  and  $\mathcal{B}_1$  meeting  $Z$ , ordered by reverse inclusion. Then  $\mathcal{I}_2$  is a full subcategory of  $\mathcal{I}_1$  which is cofinal (2.6.80). So the result follows from (2.6.81)

In the explicit case this may be shown more directly. □

**Proposition 3.2.10**

Let  $\mathcal{C}$  be an algebraic category and  $\phi : \mathcal{F} \rightarrow \mathcal{G}$  a morphism of sheaves. Then the following are equivalent

- $\phi$  is a monomorphism
- $\phi_x$  is injective for all  $x \in X$

**Proposition 3.2.11** (Isomorphism of Sheaves)

Let  $\mathcal{C}$  be an algebraic category and  $\phi : \mathcal{F} \rightarrow \mathcal{G}$  a morphism of sheaves. Then the following are equivalent

- $\phi$  is an isomorphism
- $\phi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$  is an isomorphism for all  $x \in X$

*Proof.* a)  $\implies$  b) follows from (3.2.9). We may prove b)  $\implies$  a) explicitly in the case  $\mathcal{C} = \mathbf{Set}$ . Since  $U$  reflects isomorphisms then we are done. □

The following somewhat tedious result shows that morphisms are uniquely determined by morphisms on basic open sets.

**Proposition 3.2.12**

Let  $X$  be a topological space,  $\mathcal{B}$  a base for the topology closed under finite intersections. Then the functor

$$(-)|_{\mathcal{B}} : \mathbf{Sh}(X; \mathcal{C}) \rightarrow \mathbf{Sh}(\mathcal{B}; \mathcal{C})$$

is full and faithful.

*Proof.* Let  $\phi, \psi : \mathcal{F} \rightarrow \mathcal{G}$  be morphisms which agree on basic open subsets  $U \in \mathcal{B}$ . Consider an arbitrary open subset  $U \subset X$  then by definition  $U = \bigcup_{i \in I} U_i$  for  $U_i \in \mathcal{B}$ . By assumption there is a commutative diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{e} & \prod_{i \in I} \mathcal{F}(U_i) \\ \psi_U \downarrow \downarrow \phi_U & & \downarrow \prod_{i \in I} \phi|_{U_i} = \prod_{i \in I} \psi|_{U_i} \\ \mathcal{G}(U) & \xrightarrow{e} & \prod_{i \in I} \mathcal{G}(U_i) \end{array}$$

which shows  $e \circ \phi_U = e \circ \psi_U$ . But  $e$  is a monomorphism (being part of an equaliser) so  $\phi_U = \psi_U$ . Therefore the functor is faithful.

Let  $\phi : \mathcal{F}|_{\mathcal{B}} \rightarrow \mathcal{G}|_{\mathcal{B}}$  be a morphism of  $\mathcal{B}$ -sheaves. For an open set  $U$  choose a basic open cover  $U = \bigcup_{i \in I} U_i$  then we have a commutative diagram

$$\begin{array}{ccccc}
\mathcal{F}(U) & \xrightarrow{e} & \prod_{i \in I} \mathcal{F}(U_i) & \rightrightarrows & \prod_{(i,j) \in I \times I} \mathcal{F}(U_i \cap U_j) \\
\downarrow \tilde{\phi}_U & & \downarrow \prod_{i \in I} \phi_{U_i} & & \downarrow \prod_{(i,j) \in I \times I} \phi_{U_i \cap U_j} \\
\mathcal{G}(U) & \xrightarrow{e} & \prod_{i \in I} \mathcal{G}(U_i) & \rightrightarrows & \prod_{(i,j) \in I \times I} \mathcal{G}(U_i \cap U_j)
\end{array}$$

where for the right hand square we must pick the top arrows or bottom arrows. Chasing the diagram shows that  $(\prod_{i \in I} \phi_{U_i}) \circ e$  is equalised by the bottom two arrows, and so the factorisation exists by the universal property and  $\tilde{\phi}_U$  the unique morphism which makes the left square commute, i.e. such that  $\phi_{U_i} \circ \rho_{UU_i} = \rho_{UU_i} \circ \tilde{\phi}_U$  for all  $i \in I$ .

We claim that for any basic open subset  $W \subset U$  the following diagram commutes

$$\begin{array}{ccccc}
\mathcal{F}(U) & \longrightarrow & \mathcal{F}(W) & \xrightarrow{e} & \prod_{i \in I} \mathcal{F}(W \cap U_i) \\
\downarrow \tilde{\phi}_U & & \downarrow \phi_W & & \downarrow \prod_{i \in I} \phi_{W \cap U_i} \\
\mathcal{G}(U) & \longrightarrow & \mathcal{G}(W) & \xrightarrow{e} & \prod_{i \in I} \mathcal{G}(W \cap U_i)
\end{array}$$

and  $\tilde{\phi}_U$  is the unique morphism such that this commutes for all  $W \in \mathcal{B}$ . The right hand square commutes by assumption. As  $e$  is a monomorphism it is enough to show that

$$\left( \prod_{i \in I} \phi_{W \cap U_i} \right) \circ e \circ \rho_{UW} = e \circ \phi_W \circ \rho_{UW} = e \circ \rho_{UW} \circ \tilde{\phi}_U$$

First observe we have a commutative diagram

$$\begin{array}{ccc}
\mathcal{F}(U) & \xrightarrow{e} & \prod_{i \in I} \mathcal{F}(U_i) \\
\rho_{UW} \downarrow & & \downarrow \prod_{i \in I} \rho_{U_i(W \cap U_i)} \\
\mathcal{F}(W) & \xrightarrow{e} & \prod_{i \in I} \mathcal{F}(W \cap U_i)
\end{array}$$

which follows because both directions are equal to the diagonal  $(\rho_{U(W \cap U_i)})_{i \in I}$ . Therefore we may combine the two diagrams to find

$$\begin{aligned}
e \circ \phi_W \circ \rho_{UW} &= \left( \prod_{i \in I} \phi_{W \cap U_i} \right) \circ e \circ \rho_{UW} \\
&= \left( \prod_{i \in I} \phi_{W \cap U_i} \right) \circ \left( \prod_{i \in I} \rho_{U_i(W \cap U_i)} \right) \circ e \\
&= \left( \prod_{i \in I} \phi_{W \cap U_i} \circ \rho_{U_i(W \cap U_i)} \right) \circ e \\
&= \left( \prod_{i \in I} \rho_{U_i(W \cap U_i)} \circ \phi_{U_i} \right) \circ e \\
&= \left( \prod_{i \in I} \rho_{U_i(W \cap U_i)} \right) \circ \left( \prod_{i \in I} \phi_{U_i} \right) \circ e \\
&= \left( \prod_{i \in I} \rho_{U_i(W \cap U_i)} \right) \circ e \circ \tilde{\phi}_U \\
&= e \circ \rho_{UW} \circ \tilde{\phi}_U
\end{aligned}$$

As  $e$  is a monomorphism (2.6.84) then  $\phi_W \circ \rho_{UW} = \rho_{UW} \circ \tilde{\phi}_U$  as required. Then  $\tilde{\phi}_U$  is the unique morphism such that this holds for all  $W \in \mathcal{B}$  and  $W \subset U$ , because it is already uniquely determined by  $W = U_i$ . This also shows that  $\tilde{\phi}_W = \phi_W$ . Finally for an arbitrary open subset  $V = \bigcup_{i \in I} W_i \subset U$  we wish to show we have a commutative diagram

$$\begin{array}{ccccc}
\mathcal{F}(U) & \longrightarrow & \mathcal{F}(V) & \xrightarrow{e} & \prod_{i \in I} \mathcal{F}(W_i) \\
\downarrow \tilde{\phi}_U & & \downarrow \tilde{\phi}_V & & \downarrow \\
\mathcal{G}(U) & \longrightarrow & \mathcal{G}(V) & \xrightarrow{e} & \prod_{i \in I} \mathcal{G}(W_i)
\end{array}$$

We've already shown that the large square and right square commute. This shows that

$$e \circ \tilde{\phi}_V \circ \rho_{UV} = e \circ \rho_{UV} \circ \tilde{\phi}_U$$

and as  $e$  is a monomorphism (2.6.84) the left square also commutes.  $\square$

### 3.2.2 Sheafification

#### Definition 3.2.13 (Sheafification)

Let  $X$  be a topological space,  $\mathcal{B}$  a basis of open sets for  $X$  closed under finite intersections and  $\mathcal{F}$  a  $\mathcal{B}$ -presheaf (with values in a category  $\mathcal{C}$ ). Then we say that a pair  $(\eta_{\mathcal{F}}, \mathcal{F}^+)$  is a **sheafification** of  $\mathcal{F}$  if

- a)  $\mathcal{F}^+$  is a sheaf (with values in a category  $\mathcal{C}$ )
- b)  $\eta_{\mathcal{F}} : \mathcal{F} \rightarrow \mathcal{F}^+|_{\mathcal{B}}$  is a morphism of  $\mathcal{B}$ -presheaves
- c) for every sheaf  $\mathcal{G}$  there is a bijection

$$\begin{array}{ccc} (-)^{\sharp} : \text{Mor}(\mathcal{F}^+, \mathcal{G}) & \xrightarrow{\sim} & \text{Mor}(\mathcal{F}, \mathcal{G}|_{\mathcal{B}}) \\ \phi & \rightarrow & \phi|_{\mathcal{B}} \circ \eta_{\mathcal{F}} \end{array}$$

which is natural in  $\mathcal{G}$ , with the inverse map denoted by  $(-)^{\flat}$ .

In otherwords  $(\mathcal{F}^+, (-)^{\sharp})$  *represents* the functor  $\mathcal{G} \rightarrow \text{Mor}(\mathcal{F}, \mathcal{G}|_{\mathcal{B}})$ .

In this case we denote by  $\epsilon_{\mathcal{G}} : (\mathcal{G}|_{\mathcal{B}})^+ \rightarrow \mathcal{G}$  the unique morphism which maps to  $1_{\mathcal{G}|_{\mathcal{B}}}$  under the bijection. This must be a natural isomorphism (3.2.12) (2.6.64).

#### Proposition 3.2.14 (Sheafification is Functorial)

Using the notation of (3.2.13) suppose there exists a sheafification  $(\eta_{\mathcal{F}}, \mathcal{F}^+)$  for all pre-sheaves  $\mathcal{F}$ . Then this can be extended to a functor

$$(-)^+ : \text{PSh}(X; \mathcal{B}) \rightarrow \text{Sh}(X)$$

uniquely determined by either the condition  $(-)^+$  is left-adjoint to  $(-)|_{\mathcal{B}}$  or that the following diagram commutes

$$\begin{array}{ccc} \mathcal{F}_1 & \xrightarrow{\phi} & \mathcal{F}_2 \\ \eta_{\mathcal{F}_1} \downarrow & & \downarrow \eta_{\mathcal{F}_2} \\ \mathcal{F}_1^+|_{\mathcal{B}} & \xrightarrow[\phi^+|_{\mathcal{B}}]{} & \mathcal{F}_2^+|_{\mathcal{B}} \end{array}$$

for all morphisms  $\phi : \mathcal{F}_1 \rightarrow \mathcal{F}_2$ . More precisely we have the relationship

$$\phi^+ = (\eta_{\mathcal{F}_2} \circ \phi)^{\flat}$$

In particular  $(\psi \circ \phi)^+ = \psi^+ \circ \phi^+$  for all morphisms  $\psi : \mathcal{F}_2 \rightarrow \mathcal{F}_3$  and  $\eta_{\mathcal{F}}$  is the unit of the adjunction and  $\epsilon_{\mathcal{G}}$  the counit.

*Proof.* See (2.6.66).  $\square$

#### Proposition 3.2.15

Using the notation of (3.2.13) suppose there exists a sheafification  $(\eta_{\mathcal{F}}, \mathcal{F}^+)$  for  $\mathcal{F}$ . Then for every  $x \in X$  there is a unique morphism making the following diagram commute for all  $U \in \mathcal{B}$  containing  $x$

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\eta_U} & \mathcal{F}^+(U) \\ \downarrow & & \downarrow \\ \mathcal{F}_x & \xrightarrow{\quad} & (\mathcal{F}^+)_x \end{array}$$

*Proof.* By (3.2.7) and (3.2.9) there is a commutative diagram for all  $U \in \mathcal{B}$  and  $x \in U$

$$\begin{array}{ccccc} \mathcal{F}(U) & \xrightarrow{\eta_U} & \mathcal{F}^+(U) & \xlongequal{\quad} & \mathcal{F}^+(U) \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{F}_x & \longrightarrow & (\mathcal{F}^+|_{\mathcal{B}})_x & \xrightarrow{\sim} & \mathcal{F}_x^+ \end{array}$$

where the bottom left arrow is uniquely determined. This shows that the required map exists and is uniquely determined according to this condition.  $\square$

**Proposition 3.2.16** (Sheafification)

Let  $\mathcal{C}$  be an algebraic category then the sheafification (3.2.13) exists.

More precisely define

$$\mathcal{F}^+(U) := \left\{ s : U \rightarrow \prod_{x \in U} \mathcal{F}_x \mid U = \bigcup_{i \in I} U_i, \sigma_i \in \mathcal{F}(U_i), s(x) = (\sigma_i)_x \quad \forall x \in U_i \right\}$$

Then this is evidently a sheaf and the adjunction  $\phi \rightarrow \phi^\sharp$  is the unique presheaf morphism such that

$$\phi_U^\sharp(\sigma)_x = \phi((\sigma_x)_{x \in U})_x \quad \forall U \in \mathcal{B}, \sigma \in \mathcal{F}(U), x \in U$$

and the inverse adjunction  $\psi^\flat \leftarrow \psi$  is the unique sheaf morphism such that

$$\psi^\flat(s)(x) = \psi_x(s(x)) \quad \forall x \in U$$

(using (3.2.9) to identify  $(\mathcal{G}|_{\mathcal{B}})_x$  and  $\mathcal{G}_x$ ). The unit of the adjunction is given by

$$(\eta_{\mathcal{F}})_U(\sigma) = (\sigma_x)_{x \in U}$$

and for a morphism of  $\mathcal{B}$ -sheaves  $\phi : \mathcal{F} \rightarrow \mathcal{G}$  the morphism  $\phi^+$  defined in (3.2.14) is simply

$$\phi^+(s)(x) = \phi_x(s(x)).$$

Furthermore  $\eta_{\mathcal{F}}$  is an isomorphism if and only if  $\mathcal{F}$  is a  $\mathcal{B}$ -sheaf.

*Proof.* We first demonstrate this when  $\mathcal{C} = \mathbf{Set}$ . The sheaf conditions are easy to verify. There is an obvious unit morphism

$$\begin{aligned} \eta_{\mathcal{F}}(U) : \mathcal{F}(U) &\rightarrow \mathcal{F}^+(U) \\ \sigma &\rightarrow x \rightarrow \sigma_x \end{aligned}$$

which is a natural transformation by (3.2.7). According to (2.6.66) it is enough to ensure the following map is a bijection and natural in  $\mathcal{G}$

$$\begin{aligned} \text{Mor}(\mathcal{F}^+, \mathcal{G}) &\rightarrow \text{Mor}(\mathcal{F}, \mathcal{G}|_{\mathcal{B}}) \\ \phi &\rightarrow \phi|_{\mathcal{B} \circ \eta_{\mathcal{F}}} \end{aligned}$$

Naturality in  $\mathcal{G}$  follows formally. Suppose  $\phi, \psi : \mathcal{F}^+ \rightarrow \mathcal{G}$  are morphisms such that  $\phi|_{\mathcal{B} \circ \eta_{\mathcal{F}}} = \psi|_{\mathcal{B} \circ \eta_{\mathcal{F}}}$ . For a given  $s \in \mathcal{F}^+(U)$  we wish to show that  $\phi(s) = \psi(s)$ . Suppose that  $s$  is determined by sections  $\sigma_i \in \mathcal{F}(U_i)$ , that is to say  $s|_{U_i} = \eta_{\mathcal{F}}(\sigma_i)$ . Therefore by hypothesis  $\psi(s)|_{U_i} = \phi(s|_{U_i}) = \psi(s|_{U_i}) = \psi(s)|_{U_i}$ . By the concrete sheaf conditions (3.2.3) on  $\mathcal{F}$  we conclude that  $\phi(s) = \psi(s)$ , and so the given map is injective.

Conversely given  $\phi : \mathcal{F} \rightarrow \mathcal{G}|_{\mathcal{B}}$ , we wish to construct a morphism of sheaves  $\phi^\flat : \mathcal{F}^+(U) \rightarrow \mathcal{G}(U)$  such that  $\phi^\flat((\sigma_x)_{x \in U}) = \phi(\sigma)$  for all  $\sigma \in \mathcal{F}(U)$  and  $U \in \mathcal{B}$ . We claim that there exists a family of maps  $\{\phi_U^\flat : \mathcal{F}^+ \rightarrow \mathcal{G}\}$  such that  $\phi_U^\flat(s)_x = \phi_x(s(x))$  for all  $s \in \mathcal{F}^+(U)$ . Such a family is unique by (3.2.8).

To show existence consider a section  $s \in \mathcal{F}^+(U)$ , then by definition  $s(x) = ((\sigma_i)_x)$  for some  $\sigma_i \in \mathcal{F}(U_i)$ ,  $U_i \in \mathcal{B}$  and all  $x \in U_i$ . In particular  $(\sigma_i)_x = (\sigma_j)_x$  for all  $x \in U_i \cap U_j$  so by (3.2.8) we have  $\sigma_i|_{U_i \cap U_j} = \sigma_j|_{U_i \cap U_j}$ .

Therefore the sections  $\tau_i := \phi(\sigma_i)$  also satisfy  $\tau_i|_{U_i \cap U_j} = \tau_j|_{U_i \cap U_j}$ . By (3.2.3) there exists  $\tau \in \mathcal{G}(U)$  such that  $\tau|_{U_i} = \tau_i$ . This yields a map  $\phi_U^\flat$  satisfying  $\phi_U^\flat(s)_x = \tau_x = (\tau_i)_x = \phi(\sigma_i)_x = \phi_x((\sigma_i)_x) = \phi_x(s(x))$  as required.

It remains to show that  $\phi^\flat$  is a morphism of sheaves, that is the following diagram is commutative for all open sets  $V \subset U$

$$\begin{array}{ccc} \mathcal{F}^+(U) & \xrightarrow{\phi_U^\flat} & \mathcal{G}(U) \\ \downarrow \rho_{UV} & & \downarrow \rho_{UV} \\ \mathcal{F}^+(V) & \xrightarrow{\phi_V^\flat} & \mathcal{G}(V) \\ \downarrow & & \downarrow \rho_x \\ \mathcal{F}_x & \xrightarrow{\phi_x} & \mathcal{G}_x \end{array}$$

The large and bottom squares commute by construction. Therefore for  $s \in \mathcal{F}^+(U)$  we have  $(\phi_U^b(s)|_V)_x = \phi_x(s|_V(x)) = \phi_V^b(s|_V)_x$ . As this holds for all  $x \in V$  we find from (3.2.8) that  $\phi_U^b(s)|_V = \phi_V^b(s|_V)$  as required.

As  $(-)^{\sharp}$  is injective it is sufficient by (2.6.40) to show that it has a right inverse, specifically  $(\psi^b)^{\sharp} = \psi$ . However  $\psi(\sigma)_x = \psi^b((\sigma_{x'})_{x' \in U})(x) = (\psi^b)^{\sharp}(\sigma)_x$  by construction so we are done by (3.2.8).

Evidently when  $\eta_{\mathcal{F}}$  is an isomorphism then  $\mathcal{F}$  is a  $\mathcal{B}$ -sheaf. We claim that the converse holds, namely suppose  $\mathcal{F}$  is a  $\mathcal{B}$ -sheaf, then we require to show  $(\eta_{\mathcal{F}})_U$  is injective and surjective. It is injective by (3.2.8). Similarly it is surjective for suppose  $s \in \mathcal{F}^+(U)$  for  $U \in \mathcal{B}$  and  $s(x) = (\sigma_i)_x$  for all  $x \in U_i$  and  $\sigma_i \in \mathcal{F}(U_i)$ . Then by the same result we have  $(\sigma_i)|_{U_i \cap U_j} = (\sigma_j)|_{U_i \cap U_j}$ , whence these lift to a section  $\sigma \in \mathcal{F}(U)$  such that  $\sigma|_{U_i} = \sigma_i$ , and in particular  $\sigma_x = (\sigma_i)_x = s(x)$  for all  $x \in U_i$ . Therefore we conclude the map  $(\eta_{\mathcal{F}})_U$  is surjective and therefore an isomorphism in **Set**, which means  $\eta_{\mathcal{F}}$  is an isomorphism of sheaves.

**TODO** when  $\mathcal{C}$  is an algebraic category. □

### Corollary 3.2.17

Let  $\mathcal{C}$  be an algebraic category then there is an *equivalence of categories*

$$\mathrm{Sh}(X; \mathcal{B}) \xrightleftharpoons[(-)^+]{(-)|_{\mathcal{B}}} \mathrm{Sh}(X)$$

*Proof.* By the last statement of (3.2.16), and (2.6.65) the essential image of  $(-)|_{\mathcal{B}}$  is the full subcategory of  $\mathcal{B}$ -sheaves, and the unit and counit natural transformations induce the equivalence of categories. □

### 3.2.3 Inverse Image Functor

#### Definition 3.2.18 (Inverse Image Sheaf)

Let  $f : X \rightarrow Y$  be a continuous map and  $\mathcal{G}$  a presheaf on  $Y$ . We define as follows the *inverse image sheaf*

$$(f^{-1}\mathcal{G})(U) := \left\{ s : U \rightarrow \prod_{x \in U} \mathcal{G}_{f(x)} \mid \forall x \in U \exists V \in \mathcal{O}_{f(x)} \text{ and } \sigma \in \mathcal{G}(V) \text{ s.t. } s(y) = \sigma_y \text{ for all } y \in f^{-1}(V) \right\}$$

By definition it is a sheaf. Furthermore we see there is a canonical isomorphism

$$\begin{array}{ccc} (f^{-1}\mathcal{G})_x & \xrightarrow{\sim} & \mathcal{G}_{f(x)} \\ (U, s) & \rightarrow & s(x) \end{array}$$

### 3.2.4 Skyscraper Sheaf

#### Lemma 3.2.19

Consider a family of objects in  $\{A_i\}_{i \in I}$  such that  $i \in I_0 \implies A_i$  is terminal. Then there is a canonical isomorphism for which the following diagram commutes for all morphisms  $f$

$$\begin{array}{ccc} B & \xrightarrow{f} & \prod_{i \in I} A_i \\ & \searrow (\pi_i \circ f)_{i \in I_0} & \downarrow \sim \\ & & \prod_{i \in I_0} A_i \end{array}$$

namely that given by  $(\pi_i)_{i \in I_0}$ .

*Proof.* Let  $\phi = (\pi_i)_{i \in I_0}$  and let  $\psi : \prod_{i \in I_0} A_i \rightarrow \prod_{i \in I} A_i$  be the unique morphism such that

$$\pi_j \circ \psi = \begin{cases} \pi_j & j \in I_0 \\ !\prod_{i \in I_0} A_i & j \notin I_0 \end{cases}$$

Then  $\pi_i \circ \psi \circ \phi = \pi_i \circ \phi = \pi_i$  when  $i \in I_0$  and it equals  $!\prod_{i \in I_0} A_i \circ \phi = !\prod_{i \in I} A_i = \pi_i$  when  $i \notin I_0$ , whence  $\psi \circ \phi = 1$ . The reverse is similar. □

#### Proposition 3.2.20

Let  $A \in \mathcal{C}$  be an object,  $\star$  a terminal object and  $x \in X$ . The following assignment

$$i_x(A)(U) := \begin{cases} A & x \in U \\ \star & x \notin U \end{cases}$$

determines a sheaf with restriction maps given by

$$\rho_{UV} = \begin{cases} 1_A & x \in V \\ !_A & x \in U \setminus V \\ 1_\star & x \notin U \end{cases}$$

This is known as the **skyscraper sheaf** at  $x$ .

*Proof.* Suppose that  $U = \bigcup_{i \in I} U_i$  and let  $I_0 = \{i \in I \mid x \in U_i\}$  then we may show there is a commutative diagram

$$\begin{array}{ccccc} \mathcal{F}(U) & \xrightarrow{e} & \prod_{i \in I} \mathcal{F}(U_i) & \xrightarrow[d_1]{d_0} & \prod_{(i,j) \in I \times I} \mathcal{F}(U_i \cap U_j) \\ & \searrow e & \downarrow \sim & & \downarrow \sim \\ & & \prod_{i \in I_0} \mathcal{F}(U_i) & \xrightarrow[d_1]{d_0} & \prod_{(i,j) \in I_0 \times I_0} \mathcal{F}(U_i \cap U_j) \end{array}$$

where  $e = (\rho_{UU_i})$ ,  $d_0 = (\rho_{U_i(U_i \cap U_j)} \circ \pi_i)$  and  $d_1 = (\rho_{U_j(U_i \cap U_j)} \circ \pi_j)$ . Then it's enough to show that the bottom diagram is an equaliser.

We have  $i, j \in I_0$  we have  $\mathcal{F}(U) = \mathcal{F}(U_i) = \mathcal{F}(U_i \cap U_j) = A$ .

Suppose  $f : B \rightarrow \prod_{i \in I_0} \mathcal{F}(U_i)$  such that  $d_0 \circ f = d_1 \circ f$ . Then

$$\begin{aligned} d_0 \circ f = d_1 \circ f &\implies \pi_{(i,j)} \circ d_0 \circ f = \pi_{(i,j)} \circ d_1 \circ f \quad \forall i, j \in I_0 \\ &\implies 1_A \circ \pi_i \circ f = 1_A \circ \pi_j \circ f \quad \forall i, j \in I_0 \\ &\implies \pi_i \circ f = \pi_j \circ f \quad \forall i, j \in I_0 \end{aligned}$$

Therefore define the morphism  $f' : B \rightarrow \mathcal{F}(U)$  to be  $\pi_j \circ f$  for some  $j \in I_0$ . Observing  $e = (1_A)_{i \in I_0}$  we have  $e \circ f' = (f')_{i \in I_0} = (\pi_j \circ f)_{i \in I_0} = (\pi_i \circ f)_{i \in I_0} = f$ .  $\square$

**Proposition 3.2.21** (Stalk functor is left-adjoint to Sky-Scraper Sheaf)

Let  $X$  be a topological space,  $\mathcal{C}$  a category with a terminal object  $\star$  and directed colimits,  $A \in \mathcal{C}$  and  $\mathcal{F}$  a sheaf on  $X$ . Then for all  $x \in X$  there is a bijection

$$\begin{aligned} \text{Mor}(\mathcal{F}_x, A) &\xrightarrow{\sim} \text{Mor}(\mathcal{F}, i_x(A)) \\ \phi &\rightarrow \phi_U^\# := \begin{cases} \phi \circ \rho_{Ux} & x \in U \\ !_\mathcal{F}(U) & x \notin U \end{cases} \end{aligned}$$

which is natural in both  $\mathcal{F}$  and  $A$ .

*Proof.* Let  $\mathcal{N}_x$  be the category of open subsets  $U$  of  $X$  containing  $x$  ordered by reverse inclusion. Then by definition of the stalk (3.2.7) there is a bijection

$$\begin{aligned} \text{Mor}(\mathcal{F}_x, A) &\xrightarrow{\sim} (\mathcal{F}|_{\mathcal{N}_x} \downarrow A) \\ \phi &\rightarrow (\phi \circ \rho_{Ux})_{U \in \mathcal{N}_x} \end{aligned}$$

which is natural in  $\mathcal{F}$  and  $A$ . It then remains to verify that there is a natural bijection

$$\begin{aligned} (\mathcal{F}|_{\mathcal{N}_x} \downarrow A) &\xrightarrow{\sim} \text{Mor}(\mathcal{F}, i_x(A)) \\ (\Phi_U)_{U \ni x} &\rightarrow \begin{cases} \Phi_U & x \in U \\ !_\mathcal{F}(U) & x \notin U \end{cases} \end{aligned}$$

$\square$

### 3.2.5 Equaliser Sheaf

**Proposition 3.2.22** (Equaliser Sheaf may be constructed pointwise)

Let  $\alpha, \beta : \mathcal{F} \rightrightarrows \mathcal{G}$  be morphisms of presheaves (resp. sheaves) with a family of pointwise equalisers  $\{e_U : \text{eq}(\alpha_U, \beta_U) \rightarrow \mathcal{F}(U)\}_{U \subset X}$ . Then the assignment

$$\text{eq}(\alpha, \beta)(U) := \text{eq}(\alpha_U, \beta_U)$$

can be made into a presheaf uniquely such that

$$e : \text{eq}(\alpha, \beta) \rightarrow \mathcal{F}$$

is a morphism of presheaves. Further  $e$  is the equaliser of  $(\alpha, \beta)$  in the category of presheaves (resp. sheaves).

When  $\mathcal{C}$  is an algebraic category this may be constructed in the obvious way

$$\text{eq}(\alpha, \beta)(U) := \{\sigma \in \mathcal{F}(U) \mid \alpha_U(\sigma) = \beta_U(\sigma)\}$$

*Proof.* We require to show that for all open subsets  $V \subset U$  there is a unique restriction morphism  $\rho_{UV} : \text{eq}(\alpha_U, \beta_U) \rightarrow \text{eq}(\alpha_V, \beta_V)$  ensuring the following diagram commutes

$$\begin{array}{ccc} \text{eq}(\alpha_U, \beta_U) & \xrightarrow{e_U} & \mathcal{F}(U) \\ \downarrow \rho_{UV} & & \downarrow \rho_{UV} \\ \text{eq}(\alpha_V, \beta_V) & \xrightarrow{e_V} & \mathcal{F}(V) \end{array}$$

We have that

$$\alpha_V \circ \rho_{UV} \circ e_U = \rho_{UV} \circ \alpha_U \circ e_U = \rho_{UV} \circ \beta_U \circ e_U = \beta_V \circ \rho_{UV} \circ e_U$$

whence by the universal property applied to  $\text{eq}(\alpha_V, \beta_V)$  there exists a unique factorisation of  $\rho_{UV} \circ e_U$  through  $e_V$ . The uniqueness then shows that this satisfies conditions to be a presheaf, and  $e$  is a morphism of presheaves.

We first show that  $e$  is then an equaliser of  $(\alpha, \beta)$  in the category of presheaves. Evidently  $\alpha \circ e = \beta \circ e$ . Suppose that  $\alpha \circ e' = \beta \circ e'$ , then  $\alpha_U \circ e'_U = \beta_U \circ e'_U$  whence there is a factorisation unique factorisation  $i_U$  such that  $e_U \circ i_U = e'_U$ .

$$\begin{array}{ccccc} & & e'_U & & \\ & \nearrow & & \searrow & \\ \mathcal{E}(U) & \xrightarrow{i_U} & \text{eq}(\alpha_U, \beta_U) & \xrightarrow{e_U} & \mathcal{F}(U) \\ \downarrow \rho_{UV} & & \downarrow \rho_{UV} & & \downarrow \rho_{UV} \\ \mathcal{E}(V) & \xrightarrow{i_V} & \text{eq}(\alpha_V, \beta_V) & \xrightarrow{e_V} & \mathcal{F}(V) \\ & \nwarrow & & \nearrow & \\ & & e'_V & & \end{array}$$

The large square and right-hand square commute which we may use to show that

$$e_V \circ \rho_{UV} \circ i_U = e_V \circ i_V \circ \rho_{UV}$$

As  $e_V$  is an equaliser it is a monomorphism (2.6.84) whence  $\rho_{UV} \circ i_U = i_V \circ \rho_{UV}$ , which shows that the left-hand square commutes. This means precisely that  $i$  is a morphism of presheaves.

Finally suppose  $\mathcal{F}$  is a sheaf, then it is sufficient to show that  $\text{eq}(\alpha, \beta)$  is also a sheaf. For  $U = \bigcup_{i \in I} U_i$  consider the commutative diagram

$$\begin{array}{ccccc} \text{eq}(\alpha_U, \beta_U) & \xrightarrow{e_U} & \mathcal{F}(U) & \xrightarrow[\beta_U]{\alpha_U} & \mathcal{G}(U) \\ (\rho_{UU_i})_{i \in I} \downarrow & & \downarrow (\rho_{UU_i})_{i \in I} & & \downarrow (\rho_{UU_i})_{i \in I} \\ \prod_{i \in I} \text{eq}(\alpha_{U_i}, \beta_{U_i}) & \xrightarrow{\prod_{i \in I} e_{U_i}} & \prod_{i \in I} \mathcal{F}(U_i) & \xrightarrow[\prod_{i \in I} \beta_{U_i}]{\prod_{i \in I} \alpha_{U_i}} & \prod_{i \in I} \mathcal{G}(U_i) \\ d_0 \downarrow \downarrow d_1 & & d_0 \downarrow \downarrow d_1 & & \\ \prod_{(i,j) \in I \times I} \text{eq}(\alpha_{U_{ij}}, \beta_{U_{ij}}) & \xrightarrow{\prod_{(i,j) \in I \times I} e_{U_{ij}}} & \prod_{(i,j) \in I \times I} \mathcal{F}(U_{ij}) & & \end{array}$$

where the middle column and first row are equalisers by assumption. We require to prove that the first column is an equaliser.

We may show that the bottom arrow is a monomorphism (...), which then can be used to argue that in the first column  $d_0 \circ (\rho_{UU_i})_{i \in I} = d_1 \circ (\rho_{UU_i})_{i \in I}$ .

Suppose that we have a map  $\psi : \mathcal{E}(U) \rightarrow \prod_{i \in I} \text{eq}(\alpha_{U_i}, \beta_{U_i})$  which satisfies  $d_0 \circ \psi = d_1 \circ \psi$ . Then  $\psi' := (\prod_{i \in I} e_{U_i}) \circ \psi$  also satisfies  $d_0 \circ \psi' = d_1 \circ \psi'$ . Therefore there exists a map  $\psi'' : \mathcal{E}(U) \rightarrow \mathcal{F}(U)$  such that  $(\rho_{UU_i})_{i \in I} \circ \psi'' = \psi'$ , because the middle column is an equaliser.

Furthermore we may chase the diagram to show  $(\rho_{UU_i})_{i \in I} \circ \alpha_U \circ \psi'' = (\rho_{UU_i})_{i \in I} \circ \beta_U \circ \psi''$ . By assumption  $(\rho_{UU_i})_{i \in I}$  is a monomorphism so  $\alpha_U \circ \psi'' = \beta_U \circ \psi''$ . Therefore there exist a map  $\psi''' : \mathcal{E}(U) \rightarrow \text{eq}(\alpha_U, \beta_U)$  such that  $e_U \circ \psi''' = \psi''$ .

Then

$$\left(\prod_{i \in I} e_{U_i}\right) \circ (\rho_{UU_i})_{i \in I} \circ \psi''' = (\rho_{UU_i})_{i \in I} \circ e_U \circ \psi''' = (\rho_{UU_i})_{i \in I} \circ \psi'' = \psi' = \left(\prod_{i \in I} e_{U_i}\right) \circ \psi$$

Again by (...)  $\prod_{i \in I} e_{U_i}$  is a monomorphism so we deduce that  $(\rho_{UU_i})_{i \in I} \circ \psi''' = \psi$ . Finally suppose  $\psi''''$  is another factorisation, then we may use the fact that  $e_U$  and  $(\rho_{UU_i})_{i \in I}$  (from the middle column) are both monomorphisms to deduce the factorisation is unique.

Therefore the first column is an equaliser as required.  $\square$

**Corollary 3.2.23** (Kernel may be constructed pointwise)

Let  $\mathcal{C}$  be a pointed category and  $\alpha : \mathcal{F} \rightarrow \mathcal{G}$  a morphism of presheaves (resp. sheaves). Suppose for all  $U \subset X$  have a kernel  $(i_U, \ker(\alpha_U))$  of  $\alpha_U$ .

Then we may define  $\ker(\alpha)$  as follows

$$\ker(\alpha)(U) := \ker(\alpha_U)$$

and this may be made into a presheaf (resp. sheaf) uniquely such that  $i$  is a morphism of presheaves (resp. sheaves). Furthermore  $(i, \ker(\alpha))$  is a kernel in the category of presheaves (resp. sheaves).

When  $\mathcal{C}$  is an algebraic category this may be constructed in the obvious way namely

$$\ker(\alpha)(U) := \{\sigma \in \mathcal{F}(U) \mid \alpha_U(\sigma) = 0\}$$

### 3.2.6 Cokernel Sheaf

**Proposition 3.2.24**

Let  $\mathcal{C}$  be a pointed category, then  $\mathbf{PSh}(X; \mathcal{C})$  and  $\mathbf{Sh}(X; \mathcal{C})$  are pointed categories in the obvious way.

**Proposition 3.2.25** (Cokernel of presheaves)

Let  $\mathcal{C}$  be a pointed category with coequalisers. Consider  $\alpha : \mathcal{F} \rightarrow \mathcal{G}$  a morphism in  $\mathbf{PSh}(X; \mathcal{C})$  with family of pointwise cokernels  $\{\pi_U : \mathcal{G}(U) \rightarrow \text{coker}(\alpha_U)\}_{U \subset X}$ . Then the assignment

$$\text{coker}^{\text{pre}}(\alpha) : U \rightarrow \text{coker}(\alpha_U)$$

can be made into a presheaf uniquely such

$$\mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\pi^{\text{pre}}} \text{coker}^{\text{pre}}(\alpha)$$

is a morphism of presheaves, and it is in fact a cokernel in  $\mathbf{PSh}(X; \mathcal{C})$ .

**Proposition 3.2.26** (Cokernel of Sheaves)

Let  $\mathcal{C}$  be an algebraic pointed category. Consider  $\alpha : \mathcal{F} \rightarrow \mathcal{G}$  in  $\mathbf{Sh}(X; \mathcal{C})$ . Then

$$\mathcal{F} \rightarrow \mathcal{G} \xrightarrow{\eta \circ \pi^{\text{pre}}} \text{coker}(\alpha) := (\text{coker}^{\text{psh}}(\alpha))^+$$

is a cokernel in the category of sheaves for  $f$ , where  $\eta$  is the counit of the sheafification construction (3.2.16).

*Proof.* We have  $\pi \circ \alpha = \eta \circ \pi^{\text{pre}} \circ \alpha = \eta \circ 0 = 0$ . Suppose that  $\pi' : \mathcal{G} \rightarrow \mathcal{H}$  is a morphism of sheaves and  $\pi' \circ \alpha = 0$  then we have a commutative diagram

$$\begin{array}{ccccc} \mathcal{F} & \xrightarrow{\alpha} & \mathcal{G} & \xrightarrow{\pi^{\text{pre}}} & \text{coker}^{\text{psh}}(\alpha) & \xrightarrow{\eta} & \text{coker}(\alpha) \\ & & & \searrow \pi' & \swarrow h^{\text{pre}} & \downarrow h & \\ & & & & & \mathcal{H} \end{array}$$

then  $h^{\text{pre}}$  exists and is unique by (3.2.25) and  $h = (h^{\text{pre}})^b$  exists and is unique by (3.2.13).  $\square$

**Definition 3.2.27**

Let  $\mathcal{C}$  be an algebraic pointed category and  $i : \mathcal{F} \rightarrow \mathcal{G}$  a monomorphism. Then we write the cokernel (if it exists) as simply  $\mathcal{G}/\mathcal{F}$ .

**Proposition 3.2.28** (Cokernels commute with stalks)

Let  $\mathcal{C}$  be a pointed category with filtered colimits. Consider a diagram in  $\mathbf{Sh}(X; \mathcal{C})$

$$\mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\beta} \mathcal{H}$$

where  $\beta$  is a cokernel of  $\alpha$ . Then for all  $x \in X$  the following diagram

$$\mathcal{F}_x \xrightarrow{\alpha_x} \mathcal{G}_x \xrightarrow{\beta_x} \mathcal{H}_x$$

exhibits  $\beta_x$  as a cokernel of  $\alpha_x$ .

*Proof.* Essentially this follows formally from (2.6.91) because  $(-)_x$  is a left-adjoint to the skyscraper sheaf (3.2.21).  $\square$

### 3.2.7 Constant Sheaves

#### Definition 3.2.29

Let  $A \in \mathcal{C}$  and  $X$  a topological space. Define the **constant presheaf** as follows

$$\mathcal{A}^{\text{pre}}(U) := A$$

with restriction maps the identity.

#### Proposition 3.2.30

Let  $X$  be an irreducible topological space then the constant presheaf  $\mathcal{A}^{\text{pre}}(U) = A$  is a sheaf, which we denote by  $\underline{A}$ .

*Proof.* Suppose that  $U = \bigcup_{i \in I} U_i$  where  $U_i$  are non-empty, then as  $U$  is irreducible (...) by definition every  $U_i \cap U_j \neq \emptyset$ . Therefore for sections on  $U_i$  are only compatible if they are equal, and so there is an obvious extension to  $U$ .  $\square$

#### Proposition 3.2.31 (Quotient of Constant Sheaf)

Let  $\mathcal{F}$  be a subsheaf of the constant sheaf  $\underline{A}$ , define the quotient presheaf as follows

$$(\underline{A}/\mathcal{F})(U) := \left\{ \sigma : U \rightarrow \prod_{x \in U} A/\mathcal{F}_x \mid \sigma(x) \in A/\mathcal{F}_x \right\}$$

### 3.3 Space with Functions

**Definition 3.3.1** (Space with functions)

Let  $X$  be a topological space and  $\mathcal{O}_X$  be a subsheaf of the  $\bar{k}$ -valued functions on  $X$  (which is a sheaf of  $k$ -algebras). Then we say  $(X, \mathcal{O}_X)$  is a **space with functions**. Additionally it's **local** if it satisfies the condition

- a) For every  $f \in \mathcal{O}_X(U)$  we have  $D(f) := \{f \neq 0\}$  is open and  $\frac{1}{f} \in \mathcal{O}_X(D(f))$ .

Note that the sheaf restriction maps  $\rho_{UV} : \mathcal{O}_X(U) \rightarrow \mathcal{O}_X(V)$  correspond simply to restrictions of functions.

In the latter case  $X$  is locally ringed space over  $k$  because  $\mathcal{O}_{X,x}$  is a local  $k$ -algebra with maximal ideal  $\mathfrak{m}_{X,x}$  consisting of stalks vanishing at  $x$ .

A **regular morphism** or **regular map**  $(X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$  is a continuous function  $\phi : X \rightarrow Y$  such that

$$g \in \mathcal{O}_Y(U) \implies g \circ \phi \in \mathcal{O}_X(\phi^{-1}(U))$$

This determines a morphism of sheaves

$$\phi^\# : \mathcal{O}_Y \longrightarrow \phi_* \mathcal{O}_X$$

We say a regular morphism is **dominant** if  $\phi(X)$  is dense in  $Y$ .

An isomorphism is a homeomorphism and induces an isomorphism of sheaves, or equivalently has a two-sided regular inverse.

**Proposition 3.3.2**

The composition of regular maps is well-defined and associative.

#### 3.3.1 Local Rings

**Definition 3.3.3** (Local Ring)

Let  $X$  be a local space with functions and  $W \subset X$  an irreducible closed subset. Then we define the **local ring** at  $W$  to be (see (3.2.5))

$$\mathcal{O}_{X,W} := \varinjlim_{U \cap W \neq \emptyset} \mathcal{O}_X(U)$$

If  $\overline{W} = \overline{W'}$  then  $\mathcal{O}_{X,W} = \mathcal{O}_{X,W'}$ , so for example  $\mathcal{O}_{X,W} = \mathcal{O}_{X,\overline{W}}$ .

It is a local ring with unique maximal ideal

$$\mathfrak{m}_{X,W} := \{[(V, \sigma)] \mid V \cap W \neq \emptyset, \sigma \in \mathcal{O}_X(V), \sigma(x) = 0 \forall x \in V \cap W\}$$

Define the **residue field**

$$k(\mathfrak{m}_{X,W}) := \mathcal{O}_{X,W} / \mathfrak{m}_{X,W}$$

In the case  $W = \{x\}$  we write  $\mathcal{O}_{X,x}$ ,  $\mathfrak{m}_{X,x}$  and  $k(\mathfrak{m}_{X,x})$ .

*Proof.* Consider  $(V, \sigma) \in \mathcal{O}_{X,W} \setminus \mathfrak{m}_{X,W}$ . Then by definition  $D(\sigma) \cap W \neq \emptyset$  and  $(D(\sigma), \sigma^{-1})$  is an explicit inverse.  $\square$

**Corollary 3.3.4** (Field of Rational Functions)

Let  $X$  be an irreducible space with functions. Then  $\mathcal{O}_{X,X} =: k(X)$  is a field, which we call the **field of rational functions**.

**Proposition 3.3.5** (Stalk Maps)

Suppose  $\phi : X \rightarrow Y$  is a regular morphism and  $Z \subset \overline{\phi(W)}$ . Then there exists a unique (local)  $k$ -algebra homomorphism  $\phi_W : \mathcal{O}_{Y,Z} \rightarrow \mathcal{O}_{X,W}$  such that the following diagram commutes

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\rho \circ \phi_V^\#} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,Z} & \dashrightarrow & \mathcal{O}_{X,W} \end{array}$$

for all open sets  $V \subset Y$  for which  $V \cap Z \neq \emptyset$  and some  $U \subset \phi^{-1}(V)$  for which  $U \cap W \neq \emptyset$ . This is functorial in the sense that

$$(\psi \circ \phi)_W = \psi_Z \circ \phi_W$$

*Proof.* If  $V$  meets  $Z$  then by (3.1.19)  $\phi^{-1}(V)$  meets  $W$ . Therefore the vertical maps are well-defined. We define

$$\phi_W((V', \sigma)) := (\phi^{-1}(V'), \sigma \circ \phi)$$

and immediately verify it is well-defined and ensures the diagram commutes.  $\square$

**Proposition 3.3.6** (Stalk Maps II)

Let  $\phi : X \rightarrow Y$  be a regular morphism of local spaces with functions. Suppose  $y \in \overline{\{\phi(x)\}}$ . Then there is a unique local algebra homomorphism  $\phi_x : \mathcal{O}_{Y,\phi(x)} \rightarrow \mathcal{O}_{X,x}$  making the following diagram commute

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi_V^\sharp} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,\phi(x)} & \dashrightarrow & \mathcal{O}_{X,x} \end{array}$$

for all open neighbourhoods  $V$  of  $\phi(x)$  and some open neighbourhood  $U \subset \phi^{-1}(V)$  of  $x$ . This is functorial in the sense that

$$(\psi \circ \phi)_x = \psi_{\phi(x)} \circ \phi_x$$

*Proof.* This follows immediately from (3.3.5). □

**Proposition 3.3.7** (Stalk Maps Function Field)

Let  $X, Y$  be irreducible local spaces with functions and  $\phi : X \rightarrow Y$  a dominant regular morphism. Then there is a unique homomorphism  $\phi_* : k(Y) \hookrightarrow k(X)$  making the following diagram commute

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi_V^\sharp} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ k(Y) & \xhookrightarrow[\phi_*]{} & k(X) \end{array}$$

for all non-empty open subsets  $V \subset Y$  and some non-empty open subset  $U \subset \phi^{-1}(V)$ . This is functorial in the sense that

$$(\phi \circ \psi)_* = \psi_* \circ \phi_*$$

*Proof.* This follows immediately from (3.3.5) with  $W = X$  and  $T = Y$ . □

**Proposition 3.3.8**

Let  $X$  be a local space with functions and  $Z, W$  be irreducible subsets such that  $Z \subset \overline{W}$ . Then there is a canonical homomorphism  $\mathcal{O}_{X,Z} \rightarrow \mathcal{O}_{X,W}$  making the following diagram commute

$$\begin{array}{ccc} \mathcal{O}_X(U) & & \\ \downarrow & \searrow & \\ \mathcal{O}_{X,Z} & \dashrightarrow & \mathcal{O}_{X,W} \end{array}$$

for all open subsets  $U$  of  $X$  meeting  $Z$ .

This holds in particular when  $X$  is irreducible we may take  $W = X$  and  $\mathcal{O}_{X,W} = k(X)$ .

*Proof.* This follows immediately from (3.3.5) by considering the case  $\phi = 1_X$ . □

**Proposition 3.3.9** (Stalk Maps under Specialisation)

Let  $X, Y$  be irreducible local spaces with functions,  $\phi : X \rightarrow Y$  a dominant regular morphism and  $x \in X$  such that  $y \in \overline{\{\phi(x)\}}$ . Then for all neighbourhoods  $V$  of  $\phi(x)$  we have a commutative diagram

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi_V^\sharp} & \mathcal{O}_X(\phi^{-1}(V)) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,y} & \xrightarrow{\phi_x} & \mathcal{O}_{X,x} \\ \downarrow & & \downarrow \\ k(Y) & \xhookrightarrow[\phi_*]{} & k(X) \end{array}$$

**Lemma 3.3.10** (Evaluation of Stalks)

Let  $X$  be a local space of functions and  $x \in X$ . Then there is a unique function

$$\text{ev}_x : k(\mathfrak{m}_{X,x}) \hookrightarrow \bar{k}$$

making the following diagram commute for all open neighbourhoods  $U$  of  $x$

$$\begin{array}{ccccc}
\mathcal{O}_X(U) & \longrightarrow & \mathcal{O}_{X,x} & \longrightarrow & k(\mathfrak{m}_{X,x}) \\
& & & \searrow \text{ev}_x & \downarrow \text{ev}_x \\
& & & & \bar{k}
\end{array}$$

This is functorial in the sense that for a regular morphism  $\phi : X \rightarrow Y$  there is a commutative diagram

$$\begin{array}{ccc}
k(\mathfrak{m}_{Y,\phi(x)}) & \xrightarrow{\phi_x} & k(\mathfrak{m}_{X,x}) \\
\downarrow \text{ev}_{\phi(x)} & & \downarrow \text{ev}_x \\
\bar{k} & \xlongequal{\quad} & \bar{k}
\end{array}$$

### 3.3.2 Open Immersions

#### Definition 3.3.11 (Open Immersion)

Let  $X, Y$  be spaces with functions. Then we say a regular map  $\phi : X \rightarrow Y$  is an **open immersion** if

- a)  $\phi$  is an **open embedding**, and
- b)  $\phi$  induces an isomorphism of sheaves

$$\phi_* (\mathcal{O}_Y|_{\phi(X)}) \xrightarrow{\sim} \mathcal{O}_X$$

#### Proposition 3.3.12 (Open Subspace)

Let  $X$  be a (local) space of functions and  $U \subset X$  an open subset. Then  $(U, \mathcal{O}_X|_U)$  is a (local) space of functions which we call an **open subspace**. We may write  $\mathcal{O}_U := \mathcal{O}_X|_U$ .

The inclusion  $i : U \hookrightarrow X$  is an open immersion and an open immersion  $\phi : X \rightarrow Y$  induces an isomorphism between  $X$  and the open subvariety  $\phi(X)$ .

#### Proposition 3.3.13

Let  $X$  be a space of functions and  $V \subset U$  open subsets. Then  $(V, \mathcal{O}_U|_V) = (V, \mathcal{O}_X|_V)$ .

#### Proposition 3.3.14

Let  $\phi : X \rightarrow Y$  be an open embedding and  $Z \subset Y$  closed. Then  $Z$  is irreducible iff  $\phi^{-1}(Z)$  is irreducible.

In particular if  $Y$  is irreducible then so is  $X$ .

*Proof.* We may reduce to the case  $X = U$  is an open subset of  $Y$ . Then it follows from (3.1.72) and (3.1.69).  $\square$

#### Proposition 3.3.15

Let  $\phi : X \rightarrow Y$  be a regular open embedding. Then the following are equivalent

- a)  $\phi$  is an open immersion
- b) for every irreducible subset  $W \subset X$  the canonical map

$$\mathcal{O}_{Y,Z} \rightarrow \mathcal{O}_{X,W}$$

is an isomorphism for all  $Z$  such that  $\bar{Z} = \overline{\phi(W)}$

- c) b) holds for some  $Z$  such that  $\bar{Z} = \overline{\phi(W)}$

In particular for all open subsets  $V \subset Y$  and  $U \subset \phi^{-1}(V)$  such that  $U \cap W \neq \emptyset$  we have a commutative diagram

$$\begin{array}{ccc}
\mathcal{O}_Y(V) & \xrightarrow{\phi^\#} & \mathcal{O}_X(U) \\
\downarrow & & \downarrow \\
\mathcal{O}_{Y,\phi(W)} & \xrightarrow{\sim} & \mathcal{O}_{X,W}
\end{array}$$

*Proof.* b)  $\implies$  a) By definition we may factor into a regular homeomorphism  $X \xrightarrow{\sim} V \subset Y$ . It is an isomorphism of spaces by (3.2.11).

a)  $\implies$  b) We may reduce to the case that  $X \subset Y$  is an open subset. Because an open subset of  $X$  intersects  $W$  if and only if it intersects  $Z$  we find that the map is bijective.

c)  $\iff$  b) we simply observe that if  $\bar{Z} = \overline{Z'}$  then the canonical maps are the same.  $\square$

**Proposition 3.3.16**

Let  $\phi : X \rightarrow Y$  be a dominant open immersion. If  $Y$  is irreducible, then so is  $X$  and the map

$$\phi_* : k(Y) \hookrightarrow k(X)$$

is an isomorphism.

In particular if  $X$  is irreducible and  $U \subset X$  is a non-empty open subset, then there is a canonical isomorphism

$$k(X) \xrightarrow{\sim} k(U)$$

*Proof.* By (3.3.14)  $X$  is irreducible, so it is a special case of (3.3.15).  $\square$

**Proposition 3.3.17** (Stalk Maps under Open Immersions I)

Let  $X$  be a local space with functions,  $U \subset X$  an open subset and  $W \subset X$  a irreducible closed subset such that  $U \cap W \neq \emptyset$ . Then  $U \cap W$  is an irreducible subset of  $U$  and we have canonical isomorphisms of stalks making the following diagram commute

$$\begin{array}{ccc} \mathcal{O}_X(V) & \longrightarrow & \mathcal{O}_U(U \cap V) \\ \downarrow & & \downarrow \\ \mathcal{O}_{X,W} & \xrightarrow{\sim} & \mathcal{O}_{U,W \cap U} \end{array}$$

for all open  $V \subset X$  for which  $V \cap W \neq \emptyset$ .

*Proof.* The inclusion  $i : U \hookrightarrow X$  is an open immersion, so  $U \cap W$  is irreducible by (3.1.72). This is then a special case of (3.3.15).  $\square$

**Lemma 3.3.18**

Let  $U, Z$  be subsets of a topological space  $X$ , with  $U$  open. Then

$$\overline{Z} \cap U \subset \overline{Z \cap U}$$

*Proof.* Suppose  $x \in \overline{Z} \cap U$  and  $V$  is an open subset containing  $x$ . Then  $U \cap V$  is an open set containing  $x$  and therefore by (3.1.19)  $Z \cap U \cap V \neq \emptyset$ . As  $V$  was an arbitrary neighbourhood of  $x$  we conclude that  $x \in \overline{Z \cap U}$ .  $\square$

**Proposition 3.3.19** (Restricting Stalk Maps to Open Subspaces)

Let  $\phi : X \rightarrow Y$  be a regular morphism of local space with functions. Suppose  $U \subset X$  and  $V \subset Y$  are open subsets such that  $U \subset \phi^{-1}(V)$ , and  $W \subset X$ ,  $Z \subset Y$  are irreducible subsets such that  $Z \subset \phi(W)$  and  $U \cap W \neq \emptyset$ . Then  $\phi(W \cap U) \subset \overline{Z \cap V}$  and there is a commutative cube for every open set  $V' \subset V$  such that  $V' \cap Z \neq \emptyset$  and  $U' \subset \phi^{-1}(V')$  such that  $U' \cap W \neq \emptyset$

$$\begin{array}{ccccc} & & \mathcal{O}_X(U') & \xrightarrow{\quad} & \mathcal{O}_U(U \cap U') \\ & \nearrow \phi^\# & \downarrow & & \nearrow (\phi|_U)^\# \\ \mathcal{O}_Y(V') & \xlongequal{\quad} & \mathcal{O}_V(V') & & \\ \downarrow & & \downarrow & & \downarrow \\ & \nearrow \phi_W & \mathcal{O}_{X,W} & \xrightarrow{\sim} & \mathcal{O}_{U,W \cap U} \\ \mathcal{O}_{Y,Z} & \xrightarrow{\sim} & \mathcal{O}_{V,Z \cap V} & \xrightarrow{(\phi|_U)_{W \cap U}} & \end{array}$$

where the stalk isomorphisms are defined in (3.3.17) and  $(\phi|_U)_{W \cap U}$  is the unique (local) homomorphism making the bottom square (or right side) commute

*Proof.* Observe by (3.3.18) that

$$\phi(W \cap U) \subset \phi(W) \cap V \subset \overline{Z \cap V} \subset \overline{Z \cap V}$$

which means the stalk map of  $\phi|_U$  is well-defined. The sides of the cube are well-defined and commute by (3.3.17) and (3.3.5), and the top by definition. As the horizontal arrows are isomorphisms then we see that the bottom square commutes, and  $(\phi|_U)_{W \cap U}$  is the unique homomorphism for which this is the case.  $\square$

### 3.3.3 Closed Immersions

**Definition 3.3.20** (Closed Immersion)

Let  $\phi : X \rightarrow Y$  be a regular morphism of spaces with functions. We say that  $\phi$  is a **closed immersion** if

- a)  $\phi$  is a homeomorphism onto a closed subset  $\phi(X)$  of  $Y$
- b) For all  $x \in X$  we have the induced map on stalks

$$\mathcal{O}_{Y, \phi(x)} \rightarrow \mathcal{O}_X$$

is surjective.

**Proposition 3.3.21** (Closed Subset Structure Sheaf)

Let  $(X, \mathcal{O}_X)$  be a space of functions and  $Z \subset X$  a closed subset. Then there exists a unique sheaf of functions  $\mathcal{O}_Z$  such that the inclusion  $j : Z \rightarrow X$  is a closed immersion.

More precisely for  $U \subset Z$  an open subset

$$\mathcal{O}_Z(U) := \{f : U \rightarrow \bar{k} \mid f(y) = g_i(y) \forall y \in U \cap U_i \text{ for some } g_i \in \mathcal{O}_X(U_i) \text{ where } Z \cap \bigcup_{i \in I} U_i = U\}$$

*Proof.* Evidently  $j$  is a homeomorphism onto  $Z$ . We first show that  $j$  is regular. Suppose  $g \in \mathcal{O}_X(U')$  then by definition we have  $g \circ j \in \mathcal{O}_Z(j^{-1}(U')) = \mathcal{O}_Z(U' \cap Z)$ .

Given  $(V, f) \in \mathcal{O}_{Z, z}$  by taking a smaller open subset we may suppose that  $f(y) = g(y)$  for all  $y \in V$  for some  $g \in \mathcal{O}_X(U)$  and  $U \cap Z = V$ . Then the image of  $(U, g)$  under the stalk map is  $(U \cap Z, g \circ j)$  which is precisely  $(V, f)$ . Therefore the stalk map  $j_z$  is surjective for all  $z \in Z$ .

Let  $\mathcal{F}$  be a sheaf of functions for which  $j$  is a (regular) closed immersion and suppose  $f \in \mathcal{O}_Z(U)$ . Then  $f|_{U_i \cap Z}(y) = g_i(y)$  for all  $y \in Z \cap U_i$  and  $g_i \in \mathcal{O}_X(U_i)$ . By assumption  $f|_{U_i \cap Z} = g_i \circ j \in \mathcal{F}(U_i \cap Z)$ . By the sheaf condition we have  $f \in \mathcal{F}(U)$  which shows  $\mathcal{O}_Z$  is a subsheaf of  $\mathcal{F}$ .

We need to show that the inclusion of sheaves  $\mathcal{O}_Z \hookrightarrow \mathcal{F}$  is an isomorphism. By (3.2.10) and (3.2.11) it is enough to show that it is surjective on stalks. However this follows by assumption by observing the composition  $\mathcal{O}_{X, j(z)} \rightarrow \mathcal{O}_{Z, z} \rightarrow \mathcal{F}_z$  is surjective.  $\square$

**Remark 3.3.22**

This can be defined formally as  $j^{-1}(\mathcal{O}_X/\mathcal{I})$  where  $\mathcal{I}$  is the presheaf of regular functions which vanish on  $Z$ . We show in Section ?? that this is consistent with the concept of affine subvariety.

**Proposition 3.3.23**

Let  $\phi : X \rightarrow Y$  be a regular morphism of spaces with functions. Then the following are equivalent

- a)  $\phi$  is a closed immersion
- b)  $\phi(X)$  is closed and  $\phi$  determines an isomorphism  $X \xrightarrow{\sim} \phi(X)$  of spaces with functions using the structure defined in (3.3.21)

*Proof.* a)  $\implies$  b) By definition  $\phi(X)$  is closed and  $\phi$  determines a homeomorphism onto  $\phi(X)$ . Let  $Z := \phi(X)$  and  $\mathcal{O}_Z$  denote the canonical structure sheaf. There is a well-defined map

$$\mathcal{O}_Z(U) \rightarrow \mathcal{O}_X(\phi^{-1}(U))$$

which is injective, and therefore injective on stalks (3.2.10). By (3.2.11) we need only show that the map  $\mathcal{O}_{Z, \phi(x)} \rightarrow \mathcal{O}_{X, x}$  is surjective. However by hypothesis the composite map

$$\mathcal{O}_{Y, \phi(x)} \rightarrow \mathcal{O}_{Z, \phi(x)} \rightarrow \mathcal{O}_x$$

is surjective from which the result follows.

b)  $\implies$  a) Evidently  $\phi$  is a homeomorphism onto  $\phi(X)$ . Further we have a sequence of stalk maps

$$\mathcal{O}_{Y, \phi(x)} \xrightarrow{\sim} \mathcal{O}_{\phi(X), \phi(x)} \rightarrow \mathcal{O}_{X, x}$$

where the first is surjective by (3.3.21) and the second is an isomorphism by assumption. Therefore the composite is surjective and  $\phi$  is a closed immersion.  $\square$

### 3.3.4 Locally Closed Subspace

**Lemma 3.3.24** (Locally Closed Structure Sheaf)

Let  $(X, \mathcal{O}_X)$  be a space with functions,  $(Z, \mathcal{O}_Z)$  a closed subset with canonical structure defined in (3.3.21) and  $(U, \mathcal{O}_U)$  an open subset with canonical structure (3.3.12). Then

$$\mathcal{O}_Z|_{Z \cap U} = \mathcal{O}_U|_{Z \cap U}$$

### 3.3.5 Glueing

**Proposition 3.3.25** (Patching Topologies)

Let  $X = \bigcup_{i=1}^n U_i$  be a set with topologies given for each  $U_i$ . Suppose that for all  $i, j = 1 \dots n$  the following properties hold

- $U_i \cap U_j$  is open in  $U_i$ , and
- the identity map  $1 : U_i \cap U_j \rightarrow U_i \cap U_j$  is continuous with respect to the subspace topologies on  $U_i$  and  $U_j$  respectively.

Then there is a unique topology on  $X$  such that

- the  $U_i$  are open in  $X$ ,
- the topology on  $U_i$  coincides with the subspace topology from  $X$

More precisely it is given by

$$\{U \subset X \mid U \cap U_i \text{ open in } U_i \text{ for all } i = 1 \dots n\}$$

**Proposition 3.3.26** (Glueing Topologies)

Let  $X = \bigcup_{i=1}^n U_i$  be a set together with bijections  $\theta_i : U_i \xrightarrow{\sim} W_i$  where  $W_i$  are topological spaces. Suppose that for all  $i, j = 1 \dots n$  the following properties hold

- $\theta_i(U_i \cap U_j)$  is open in  $W_i$ ,
- the map  $\theta_{ij} := \theta_j \circ \theta_i^{-1} : \theta_i(U_i \cap U_j) \rightarrow \theta_j(U_i \cap U_j)$  is continuous.

Then there is a unique topology on  $X$  such that

- $U_i$  are open in  $X$ ,
- $\theta_i$  is a homeomorphism with respect to the subspace topology on  $U_i$  for all  $i = 1 \dots n$ .

More precisely the topology is given by

$$\{U \subset X \mid \theta_i(U \cap U_i) \text{ is open in } W_i \text{ for all } i = 1 \dots n\}$$

**Proposition 3.3.27** (Glueing Regular Maps)

Let  $X, Y$  be spaces with functions and  $X = \bigcup_{i=1}^n U_i$  an open cover. Suppose there exists regular maps

$$\phi_i : (U_i, \mathcal{O}_X|_{U_i}) \rightarrow (Y, \mathcal{O}_Y)$$

which satisfy  $\phi_i|_{U_i \cap U_j} = \phi_j|_{U_i \cap U_j}$  for all  $i, j = 1 \dots n$ . Then there exists a unique map  $\phi : X \rightarrow Y$  such that  $\phi|_{U_i} = \phi_i$ , which is regular.

*Proof.* As a function  $\phi$  is clearly well-defined and unique. It is continuous by (3.1.29). Suppose  $g \in \mathcal{O}_Y(V)$  then by definition  $(g \circ \phi)|_{\phi^{-1}(V) \cap U_i} = g \circ \phi_i \in \mathcal{O}_{U_i}(\phi^{-1}(V) \cap U_i) = \mathcal{O}_X(\phi^{-1}(V) \cap U_i)$ . As  $\mathcal{O}_X$  is a sheaf this shows that  $g \circ \phi$  is regular as required.  $\square$

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